NEET UG PHYSICS

MASTER TEXTBOOK

NEET UG 2027 Master Reference Series · Complete Authoritative Edition

NEET UG PHYSICS

MASTER TEXTBOOK

Complete Master Theory, Formulas & Experimental Compendium

Strictly Aligned with NEET UG 2027 NMC Syllabus

Authored by

Satya Prakash

Strictly Calibrated to the Revised NMC Syllabus & Rationalized NCERT Textbooks

Target Examination: National Eligibility cum Entrance Test (NEET UG 2027)

Published for Independent Medical Aspirants, Educators & Academic Study Circles

Comprehensive Academic Edition · Global Print-on-Demand

NEET UG PHYSICS MASTER TEXTBOOK

Comprehensive Theory, Rigorous Derivations, Experimental Skills & Formula Compendium

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Source Repository & Official Errata:
`https://github.com/spdedsec/neet-study-materials`

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First Edition: October 2026
Publishing Imprint: Independent Academic Publishing / Satya Prakash Editorial Imprint
Target Cycle: NEET UG 2027 (Undergraduate Medical Admissions Entrance Examination)
ISBN-13 (Paperback): 979-8-1626-5967-9
ISBN-13 (Hardcover): 979-8-1626-5967-9 (Reference Library Edition)

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Typesetting & Typography: Typeset in LaTeX 2_ε with the Tectonic XeTeX engine using Latin Modern typefaces. Mathematical typesetting follows ISO 80000-2 standards. Produced and bound under rigorous academic publishing standards.

Printed in the United States of America / United Kingdom / India.

To my parents, whose quiet sacrifices and endless belief built the foundation of my life;

To the teachers and mentors who showed me that physics is not a collection of arbitrary formulas,
but the poetry of how nature actually works;

And to you, the medical aspirant reading this in the quiet hours of the night—
may every solved problem and mastered derivation bring you one step closer
to the stethoscope, the white coat, and a lifetime of healing.

Acknowledgements

Creating a complete, mathematically uncompromising textbook calibrated to a competitive examination as fiercely contested as NEET UG is never a solitary effort. I owe an immense debt of gratitude to several people and institutions whose influence echoes through these chapters.

First and foremost, I thank my family for their patience, understanding, and unwavering encouragement during the countless late nights and weekends spent drafting, proofreading, and verifying derivations.

I express deep gratitude to the mentors and physics educators whose classroom discussions shaped my conviction that physics must be taught from fundamental principles rather than mechanical memorization. Their insistence on clear vector diagrams, rigorous free-body analysis, and boundary condition checks inspired the structure of every chapter in this book.

I am especially grateful to the students and peer study groups whose probing questions, skepticism, and frequent exam mistakes illuminated the exact cognitive hurdles aspirants face. The "NEET Trap" boxes in this book were born directly from real student doubts and mock exam error logs.

Finally, I acknowledge the vibrant open-source scientific typesetting community, specifically the developers of the LaTeX 2_ε ecosystem and the Tectonic compiler, which made it possible to produce a publication of this typographical caliber.

Satya Prakash
October 2026

Author's Preface: Straight Talk Before We Begin

Let's be completely frank with each other. If you are preparing for NEET UG, you already know that physics is where ranks are decided. Biology gives you the foundation, Chemistry keeps you in the game, but Physics is what separates a student struggling for a cutoff from one who walks into a top government medical college.

Yet for most aspirants, physics feels like a daily war. You sit down with coaching modules running over 800 pages, and half the material is either filled with fluff or contains obsolete topics that the National Medical Commission (NMC) eliminated years ago. You waste precious study hours memorizing Carnot refrigerator formulas, potentiometer wire balancing equations, and Earth magnetism tables that will not appear on your exam paper. Meanwhile, when the actual exam twists a rolling friction problem on an inclined plane or asks an Assertion-Reasoning question on Gauss's law with non-symmetric charges, formulas from memory suddenly fail.

I wrote this book because I wanted to hand you the exact companion I wished I had when studying: honest, rigorous, beautifully organized, and talking to you straight without any fake "miracle tricks" or empty motivation.

Here is how this book is built, and how you should use it:

1. **Bedrock Principles Over Rote Memorization:** Every major chapter starts from first principles. When we look at work-energy, fluid dynamics, or electromagnetic induction, you will see the complete mathematical derivation. Why? Because when you understand where a formula comes from and what assumptions were made to reach it, you become immune to the traps exam setters lay for unsuspecting students.
2. **The Five Tactical Callout Systems:** Throughout every chapter, you will encounter five distinct color-coded boxes. Treat them like field markers:
   - **Core Principle Boxes (`conceptbox`):** The foundational laws and definitions. Pay close attention to these—NTA draws multi-statement and factual questions directly from these definitions.
   - **Master Formula Boxes (`formulabox`):** Every critical governing equation with explicit variable definitions, limits of integration, and SI units.
   - **Application Regime Boxes (`usecasebox`):** The real-world conditions where formulas work, where they break down, and shortcuts for common numerical situations.
   - **NEET Trap Boxes (`trapbox`):** The exact spots where thousands of students lose 5 marks (-1 penalty plus the 4 marks they missed). These

cover sign convention mistakes, hidden assumptions, and misleading exceptions.

- **Direct NCERT Lines (`ncertbox`):** Verbatim textual extractions and table data that frequently form the basis of Assertion-Reasoning items.

3. **Strict NMC Syllabus Calibration:** Everything in this book conforms strictly to the revised NMC syllabus for NEET UG 2027. Obsolete topics like Carnot engines, free and damped oscillations, potentiometers, and transistor amplifiers have been flagged and excised from the core theory. You will not waste a single hour on deleted material.
4. **Experimental Skills (Unit 20) Integration:** The NMC added explicit weightage to practical physics. Chapter 6 is dedicated entirely to the 18 prescribed experimental skills—from Vernier calipers and screw gauges to sonometer wires, focal length determination, and meter bridges.

A personal piece of advice: do not read physics like a novel. If you are sitting on your bed flipping pages without a pen in your hand, you are not studying physics. Pick up a notebook, derive the equations alongside the text, draw every free-body diagram yourself, and work through the problem archetypes with the solutions covered.

Mistakes are going to happen. When you get a question wrong in a practice test, do not get demoralized. An error made in practice is a gift: it reveals a blind spot today so you don't repeat it on exam day.

You have the capability and the time to master this subject. Stay consistent, stay disciplined, and keep your focus on the prize. I'll see you on the other side with that white coat.

Satya Prakash
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October 2026

Syllabus Blueprint & NMC Rationalization Guide

The National Eligibility cum Entrance Test (NEET UG 2027) follows the curriculum prescribed by the Under Graduate Medical Education Board (UGMEB) of the National Medical Commission (NMC). This section outlines the structure of the examination, subject weightage, and critical syllabus deletions.

Official Physics Examination Architecture

Component	Compulsory Qs	Marks	Pace Budget	Question Typology
Physics	45 (All Compulsory)	180	45–50 min	Single Choice MCQs, Numerical Calculations,
Exam Total	180 Compulsory	720	180 min (3h)	Assertion-Reason, Statement I/II, Matrix Match

Table 1: Reformed NEET UG Physics Examination Scheme (+4 for correct response, −1 for incorrect response, 0 for unattempted. Optional Section B discontinued by NTA).

Official 20-Unit Physics Syllabus Structure

1. **Unit 1: Physics and Measurement** (Units, dimensions, error propagation, significant figures, Vernier calipers, screw gauge).
2. **Unit 2: Kinematics** (Frame of reference, rectilinear motion, relative velocity, 2D projectile motion, uniform circular motion).
3. **Unit 3: Laws of Motion** (Newton's laws, momentum conservation, equilibrium, friction dynamics, circular dynamics and banking).
4. **Unit 4: Work, Energy and Power** (Work-energy theorem, conservative forces, potential energy of springs, vertical circular motion, collisions).
5. **Unit 5: Rotational Motion** (Centre of mass, moment of inertia, parallel and perpendicular axis theorems, torque, angular momentum, pure rolling).
6. **Unit 6: Gravitation** (Kepler's laws, universal gravitation, acceleration due to gravity variation, gravitational potential and energy, orbital velocity, escape speed).

7. **Unit 7: Properties of Solids and Liquids** (Elasticity, Hooke's law, Young's/bulk/shear moduli, Pascal's law, Archimedes principle, surface tension, capillarity, viscosity, Stokes' law, Bernoulli's theorem).
8. **Unit 8: Thermodynamics** (Thermal equilibrium, zeroth law, heat, work, first law of thermodynamics, isothermal, adiabatic, isobaric, isochoric processes, second law).
9. **Unit 9: Kinetic Theory of Gases** (Equation of state of perfect gas, kinetic postulates, pressure formulation, kinetic interpretation of temperature, rms speed, degrees of freedom, equipartition of energy, mean free path).
10. **Unit 10: Oscillations and Waves** (Periodic motion, SHM kinematics and dynamics, spring-mass systems, simple pendulum, wave motion, longitudinal and transverse waves, superposition, standing waves in strings and pipes, beats).
11. **Unit 11: Electrostatics** (Coulomb's law, electric field, electric dipole, Gauss's law and applications, electric potential, equipotential surfaces, capacitors, dielectrics).
12. **Unit 12: Current Electricity** (Electric current, drift velocity, Ohm's law, resistivity, temperature dependence, Kirchhoff's laws, Wheatstone bridge, internal resistance).
13. **Unit 13: Magnetic Effects of Current and Magnetism** (Biot-Savart law, Ampère's circuital law, force on moving charges and current-carrying conductors, magnetic dipole, moving coil galvanometer).
14. **Unit 14: Electromagnetic Induction and Alternating Currents** (Faraday's laws, Lenz's law, eddy currents, self and mutual inductance, AC circuits, impedance, LCR series circuits, resonance, power in AC, transformers).
15. **Unit 15: Electromagnetic Waves** (Displacement current, Maxwell's equations characteristics, electromagnetic spectrum from radio waves to gamma rays).
16. **Unit 16: Optics** (Ray optics: reflection, refraction, total internal reflection, lens maker's formula, prisms, microscopes, telescopes; Wave optics: Huygens' principle, interference, Young's double slit, diffraction at single slit).
17. **Unit 17: Dual Nature of Matter and Radiation** (Photoelectric effect, Hertz and Lenard's observations, Einstein's photoelectric equation, de Broglie relation).
18. **Unit 18: Atoms and Nuclei** (Alpha-particle scattering, Rutherford model, Bohr model, energy levels, hydrogen spectrum, atomic nuclei, mass defect, binding energy per nucleon).
19. **Unit 19: Electronic Devices** (Semiconductors, energy bands, intrinsic and extrinsic semiconductors, p-n junction diode, I-V characteristics, diode as rectifier, logic gates: OR, AND, NOT, NAND, NOR).
20. **Unit 20: Experimental Skills** (18 prescribed experiments including Vernier calipers, screw gauge, simple pendulum, Metre Scale principle of

moments, Young's modulus by Searle's method, surface tension by capillary rise, viscosity of liquid, sonometer, resonance tube, Ohm's law, meter bridge, p-n junction diode characteristics).

Zero-Tolerance NMC Exclusions: Topics Strictly Deleted from Physics

Candidates must note that legacy textbooks and coaching modules contain several topics that have been **completely eliminated** by the NMC:

- **Carnot Engine, Carnot Cycle & Refrigerators** (Completely eliminated from Thermodynamics).
- **Free, Damped & Forced Oscillations, Resonance** (Eliminated from Oscillations).
- **Doppler Effect in Sound** (Eliminated from Waves).
- **Potentiometer and its applications** (Replaced in practicals by Meter Bridge and Multimeter).
- **Earth's Magnetism and Magnetic Elements** (Eliminated from Magnetism).
- **Cyclotron** (Eliminated from Moving Charges).
- **Radioactive Decay Law, Alpha/Beta/Gamma Decay, Half-Life** (Eliminated from Nuclear Physics).
- **Transistor and its characteristics, Transistor as Amplifier/Oscillator** (Eliminated from Semiconductors).

Preparation Directive: Do not waste valuable study time practicing numericals or derivations on these excluded topics.

Empirical Question Weightage Matrix

Physics Domain	Typical Qs	Marks Weightage
Classical Mechanics (Kinematics, Dynamics, Energy, Rotation, Gravity, Fluids)	16–18	64–72 Marks
Thermal Physics, Thermodynamics, Kinetic Theory & Oscillations	6–8	24–32 Marks
Electrodynamics, Magnetism, EMI & AC Circuits	11–13	44–52 Marks
Optics (Ray & Wave Optics)	4–5	16–20 Marks
Modern Physics & Semiconductors (Dual Nature, Atoms, Nuclei, Diodes/Gates)	6–8	24–32 Marks
Experimental Skills (Unit 20 Laboratory Practicals)	2–3	8–12 Marks
Total Physics Paper	**45 Qs (Compulsory)**	**180 Marks**

Table 2: Empirical Question Distribution in Recent NEET UG Papers.

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PART I

Classical Mechanics & Dynamics

CHAPTER 1

Foundations of Classical Mechanics: Kinematics, Dynamics, Work, Energy & Collisions

This chapter establishes the core mechanics principles for NEET UG. We begin with the mathematical foundation of physical measurements, dimensional homogeneity, and error calculus, progressing into rectilinear and two-dimensional kinematics, Newton's laws of motion with constrained multi-block dynamics, microscopic friction mechanisms, vertical and banked circular motion, and the calculus of work, energy, and collision impulse.

1.1 Physical Measurement, Dimensional Analysis & Calculus of Errors

1.1.1 Foundational Standards of Measurement & Dimensional Homogeneity

Physical quantities are classified into base (fundamental) and derived quantities. The International System of Units (SI) defines seven base quantities: Mass [M], Length [L], Time [T], Electric Current [I] or [A], Thermodynamic Temperature [K], Amount of Substance [mol], and Luminous Intensity [cd]. All mechanical, thermal, and electrodynamic variables in the NEET curriculum derive from power combinations of these fundamental dimensions.

Principle of Dimensional Homogeneity: An equation representing any physical law of the form $A + B = C$ is physically valid if and only if each additive term possesses identical dimensions:

$$[A] = [B] = [C] \tag{1.1}$$

Furthermore, arguments of transcendental functions (such as trigonometric arguments $\sin\theta$, exponents in e^x, and logarithms $\ln x$) must be strictly dimensionless numbers: $[x] = \mathrm{M}^0\mathrm{L}^0\mathrm{T}^0 = 1$.

Capabilities and Inherent Limitations of Dimensional Analysis: Dimensional analysis provides a rapid method for checking algebraic consistency, deducing power-law dependencies among independent variables, and converting numerical magnitudes between unit systems (e.g., CGS and SI). However, it possesses fundamental limitations:

- It cannot determine dimensionless proportionality constants (such as $1/2$ in kinetic energy or 2π in simple pendulum period).
- It fails when a quantity depends on more variables than the available independent fundamental equations (typically three in mechanics: M, L, T).
- It cannot deduce relations involving additive sums of terms or trigonometric functions.

Calculus of Error Propagation & Uncertainty Formulations

Let a derived quantity Z depend on independently measured physical variables A, B, C via the functional relation $Z = f(A, B, C)$. By taking the total differential:

$$dZ = \left(\frac{\partial f}{\partial A}\right) dA + \left(\frac{\partial f}{\partial B}\right) dB + \left(\frac{\partial f}{\partial C}\right) dC \tag{1.2}$$

For experimental uncertainty analysis, maximum absolute error ΔZ is determined by summing the absolute magnitudes of the partial deviations:

- **Sum / Difference ($Z = A \pm B$):**

$$\Delta Z = \Delta A + \Delta B$$

- **Product / Quotient ($Z = \dfrac{A \cdot B}{C}$):**

$$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$$

- **Generalized Power Formulation ($Z = \dfrac{A^p B^q}{C^r}$):**

$$\frac{\Delta Z}{Z} = p\left(\frac{\Delta A}{A}\right) + q\left(\frac{\Delta B}{B}\right) + r\left(\frac{\Delta C}{C}\right)$$

The percentage fractional uncertainty is given by:

$$\%\Delta Z = \left[p\left(\frac{\Delta A}{A}\right) + q\left(\frac{\Delta B}{B}\right) + r\left(\frac{\Delta C}{C}\right)\right] \times 100\% \tag{1.3}$$

Experimental Least Count & Boundary Application Rules

- **Vernier Calipers Least Count (LC):** Given 1 MSD (Main Scale Division) and n Vernier divisions coinciding with $(n-1)$ MSD:

$$\text{LC} = 1\,\text{MSD} - 1\,\text{VSD} = \frac{1\,\text{MSD}}{n} \tag{1.4}$$

- **Screw Gauge Least Count:** For pitch p and total circular scale divisions N:

$$\text{LC} = \frac{\text{Pitch } p}{N} \tag{1.5}$$

$$\text{True Reading} = \text{MSR} + (\text{CSR} \times \text{LC}) - (\pm\text{Zero Error}) \tag{1.6}$$

- **Rule for Significant Figures in Calculus:** For multiplication and division, the final computed value retains the same number of significant figures as the least precise factor. For addition and subtraction, the result retains the least number of decimal places.

NEET Trap: Sign of Zero Error & Subtraction of Uncertainty

- **Zero Error Subtraction:** True Reading = Observed Reading − (Zero Error with appropriate algebraic sign). If a screw gauge has a positive zero error of $+0.04\,\text{mm}$, that quantity must be **subtracted**. If it has a negative zero error of $-0.03\,\text{mm}$, it is added: True = Observed $-(-0.03)$ = Observed $+\,0.03\,\text{mm}$.
- **Never Subtract Errors:** In fractional error propagation for $Z = A/B$ or $Z = A-B$, errors never cancel. Even though variables subtract, their individual experimental uncertainties **additively compound**: $\Delta Z = \Delta A + \Delta B$.

NCERT Critical Extraction: Systematic vs Random Errors

NCERT Highlight: Systematic errors are unidirectional (consistently positive or consistently negative) and arise from instrumental imperfections, zero errors, or experimental methodology flaws. They can be eliminated by calibration. Random errors arise from uncontrollable fluctuations in experimental conditions and follow Gaussian distributions; their influence is minimized by taking the arithmetic mean of N repetitive measurements, reducing the random error magnitude by a factor of $1/\sqrt{N}$.

1.2 Rectilinear Kinematics (Motion in One Dimension)

Calculus Formulations of Rectilinear Motion

Consider a point particle moving along a coordinate axis $x(t)$.

1. **Position and Displacement:** Position vector $\mathbf{r}(t) = x(t)\hat{\mathbf{i}}$. Displacement over time interval $[t_1, t_2]$ is $\Delta x = x(t_2) - x(t_1)$, which is path-independent. Total distance s is path-dependent: $s = \int_{t_1}^{t_2} |v(t)|\,dt \geq |\Delta x|$.
2. **Differential Velocity and Acceleration:**

$$v(t) = \frac{dx}{dt}, \qquad a(t) = \frac{dv}{dt} = \frac{d^2x}{dt^2} = v\frac{dv}{dx} \tag{1.7}$$

3. **Integral Kinematic Relations (Variable Acceleration):**

$$\Delta x = \int_{t_1}^{t_2} v(t)\,dt, \qquad \Delta v = \int_{t_1}^{t_2} a(t)\,dt, \qquad \int_{v_1}^{v_2} v\,dv = \int_{x_1}^{x_2} a(x)\,dx \quad (1.8)$$

Rigorous Calculus Derivation of Uniformly Accelerated Kinematic Equations

For constant acceleration $a =$ constant:

1. **First Kinematic Relation ($v - t$):**

$$a = \frac{dv}{dt} \implies \int_u^v dv = a\int_0^t dt \implies [v]_u^v = a[t]_0^t \implies v = u + at \quad (1.9)$$

2. **Second Kinematic Relation ($x - t$):**

$$v = \frac{dx}{dt} = u + at \implies \int_0^s dx = \int_0^t (u + at)\,dt \implies s = ut + \frac{1}{2}at^2 \quad (1.10)$$

3. **Third Kinematic Relation ($v - x$):**

$$a = v\frac{dv}{dx} \implies \int_u^v v\,dv = a\int_0^s dx \implies \left[\frac{v^2}{2}\right]_u^v = as \implies v^2 = u^2 + 2as \quad (1.11)$$

4. **Displacement in the nth Second (s_n):**

$$\begin{aligned} s_n = s(n) - s(n-1) &= \left[un + \frac{1}{2}an^2\right] - \left[u(n-1) + \frac{1}{2}a(n-1)^2\right] \\ &= u[n - (n-1)] + \frac{1}{2}a[n^2 - (n^2 - 2n + 1)] = u + \frac{a}{2}(2n-1) \end{aligned} \quad (1.12)$$

Motion Under Gravity: Symmetries & Galileo's Odd Integer Theorem

For vertical throw under uniform gravitational acceleration $a = -g$ (taking upward as positive):

- **Maximum Height Attained:** Setting $v = 0$ in $v^2 = u^2 - 2gH \implies H = \frac{u^2}{2g}$.
- **Time of Ascent and Descent:** $t_{\text{ascent}} = \frac{u}{g}$, Total flight time $T = \frac{2u}{g}$. The speed of the particle returning to the release plane equals its launch speed: $v_{\text{return}} = -u$.
- **Galileo's Law of Odd Numbers:** For a body released from rest ($u = 0$), the distances traversed in successive equal time intervals Δt follow the ratio of odd integers:

$$s_1 : s_2 : s_3 : \cdots : s_n = 1 : 3 : 5 : \cdots : (2n-1) \quad (1.13)$$

- **Stopping Distance & Reaction Time:** If driver reacts after time t_r and braking deceleration is a:

$$d_{\text{stop}} = d_{\text{reaction}} + d_{\text{braking}} = u \cdot t_r + \frac{u^2}{2a} \tag{1.14}$$

NEET Trap: Distance vs Displacement in Turning Point Trajectories

When computing the total distance traveled by an object undergoing deceleration (e.g., $u > 0, a < 0$), the kinematic formula $s = ut + \frac{1}{2}at^2$ yields the **net displacement**, which can be zero or negative.

- **Verification Step:** First solve $v(t^*) = u + at^* = 0 \implies t^* = -u/a$.
- If the specified evaluation time $t > t^*$, the particle has reversed direction. The total path distance is:

$$\text{Distance } = |x(t^*) - x(0)| + |x(t) - x(t^*)| = \frac{u^2}{2|a|} + \frac{1}{2}|a|(t - t^*)^2 \tag{1.15}$$

Using $s = ut + \frac{1}{2}at^2$ directly will cause a catastrophic error in NEET problems!

NCERT High-Yield Extraction: Geometric Meanings of Motion Graphs

NCERT Highlight:

- In a Position–Time $(x - t)$ graph: Slope of the tangent at time t gives instantaneous velocity $v = \frac{dx}{dt}$. An inflection point represents zero acceleration.
- In a Velocity–Time $(v-t)$ graph: Slope gives instantaneous acceleration $a = \frac{dv}{dt}$. The signed definite integral (area between curve and time axis) represents net displacement: $\Delta x = \int v\, dt$. Total area with positive magnitudes represents total distance: $s = \int |v|\, dt$.
- In an Acceleration–Time $(a - t)$ graph: Area under the curve yields change in velocity: $\Delta v = \int a\, dt$.

1.3 Two-Dimensional Kinematics, Projectile Motion & Relative Velocity

Orthogonal Vector Decomposition in Planar Motion

Planar motion under constant acceleration vector $\mathbf{a} = a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}}$ decomposes into two completely independent, uncoupled rectilinear motions along perpendicular Cartesian axes:

$$x(t) = x_0 + u_x t + \frac{1}{2}a_x t^2, \qquad y(t) = y_0 + u_y t + \frac{1}{2}a_y t^2 \tag{1.16}$$

The instantaneous velocity vector is $\mathbf{v}(t) = (u_x + a_x t)\hat{\mathbf{i}} + (u_y + a_y t)\hat{\mathbf{j}}$, with speed $|\mathbf{v}| = \sqrt{v_x^2 + v_y^2}$ and trajectory inclination angle $\tan\phi(t) = \frac{v_y(t)}{v_x(t)}$.

Complete Mathematical Derivation of Oblique Projectile Dynamics

Consider a projectile launched with speed u at angle θ to the horizontal over level ground ($a_x = 0, a_y = -g$):

$$u_x = u\cos\theta, \qquad v_x(t) = u\cos\theta, \qquad x(t) = (u\cos\theta)t \tag{1.17}$$

$$u_y = u\sin\theta, \qquad v_y(t) = u\sin\theta - gt, \qquad y(t) = (u\sin\theta)t - \frac{1}{2}gt^2 \tag{1.18}$$

1. Time of Flight (T): Setting vertical displacement $y(T) = 0$:

$$T\left(u\sin\theta - \frac{1}{2}gT\right) = 0 \implies T = \frac{2u\sin\theta}{g} \tag{1.19}$$

2. Maximum Height Attained (H): At the trajectory apex, $v_y = 0 \implies t_{\text{apex}} = \frac{u\sin\theta}{g}$:

$$H = y(t_{\text{apex}}) = u\sin\theta\left(\frac{u\sin\theta}{g}\right) - \frac{1}{2}g\left(\frac{u\sin\theta}{g}\right)^2 = \frac{u^2\sin^2\theta}{2g} \tag{1.20}$$

3. Horizontal Range (R): Evaluating horizontal coordinate at $t = T$:

$$R = x(T) = (u\cos\theta)\left(\frac{2u\sin\theta}{g}\right) = \frac{u^2(2\sin\theta\cos\theta)}{g} = \frac{u^2\sin 2\theta}{g} \tag{1.21}$$

4. Cartesian Equation of Trajectory: Eliminating $t = \frac{x}{u\cos\theta}$:

$$\begin{aligned} y(x) &= u\sin\theta\left(\frac{x}{u\cos\theta}\right) - \frac{g}{2}\left(\frac{x}{u\cos\theta}\right)^2 \\ &= x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} = x\tan\theta\left(1 - \frac{x}{R}\right) \end{aligned} \tag{1.22}$$

High-Yield Projectile Properties & Incline Formulas

- **Complementary Angles of Projection:** For a given initial velocity u, the range is identical for launch angles θ and $(90^\circ - \theta)$:

$$R(\theta) = R(90^\circ - \theta) = \frac{u^2\sin 2\theta}{g} \tag{1.23}$$

  The corresponding maximum heights H_1, H_2 satisfy the master identity:

$$R = 4\sqrt{H_1 H_2}, \qquad H_1 + H_2 = \frac{u^2}{2g} \tag{1.24}$$

- **Horizontal Projectile from Cliff of Height h:** Launched horizontally with speed u:

$$t_{\text{flight}} = \sqrt{\frac{2h}{g}}, \qquad R = u\sqrt{\frac{2h}{g}}, \qquad v_{\text{strike}} = \sqrt{u^2 + 2gh} \tag{1.25}$$

- **Projectile Motion on an Inclined Plane:** For a plane inclined at angle α to horizontal, with launch angle β relative to the inclined plane:

$$\text{Time of flight: } T = \frac{2u\sin\beta}{g\cos\alpha} \tag{1.26}$$

$$\text{Range up the incline: } R_{\text{up}} = \frac{u^2}{g\cos^2\alpha}\left[\sin(2\beta+\alpha) - \sin\alpha\right] \tag{1.27}$$

$$\text{Condition for Maximum Range: } \beta = \frac{\pi}{4} - \frac{\alpha}{2} \implies R_{\text{max,up}} = \frac{u^2}{g(1+\sin\alpha)} \tag{1.28}$$

Relative Velocity Formulations in 2D: River-Boat & Rain-Man

Relative velocity of body A with respect to body B is defined by:

$$\mathbf{v}_{AB} = \mathbf{v}_A - \mathbf{v}_B \tag{1.29}$$

1. River-Boat Problem: Let river width be d, river flow speed be $v_r\hat{\mathbf{i}}$, and boat velocity relative to water be $\mathbf{v}_{br}$:

- **Shortest Time Crossing:** Steer directly across ($\theta = 90^\circ$ to river flow):

$$t_{\min} = \frac{d}{v_{br}}, \qquad \text{Drift } x = v_r \cdot t_{\min} = \frac{v_r d}{v_{br}} \tag{1.30}$$

- **Shortest Path Crossing (Zero Drift):** Feasible only when $v_{br} > v_r$. Steer at angle θ upstream such that horizontal component cancels river flow:

$$\sin\theta = \frac{v_r}{v_{br}}, \qquad v_{\text{resultant}} = \sqrt{v_{br}^2 - v_r^2}, \qquad t = \frac{d}{\sqrt{v_{br}^2 - v_r^2}} \tag{1.31}$$

2. Rain-Man Problem: Rain falling vertically with velocity $\mathbf{v}_r = -v_r\hat{\mathbf{j}}$, observer walking with $\mathbf{v}_m = v_m\hat{\mathbf{i}}$:

$$\mathbf{v}_{r,m} = \mathbf{v}_r - \mathbf{v}_m = -v_m\hat{\mathbf{i}} - v_r\hat{\mathbf{j}}, \qquad \tan\theta = \frac{|\mathbf{v}_m|}{|\mathbf{v}_r|} = \frac{v_m}{v_r} \tag{1.32}$$

The umbrella must be oriented tilted forward at angle $\theta = \tan^{-1}(v_m/v_r)$ relative to vertical.

NEET Trap: Minimum Separation in 2D Interception Problems

When two bodies A and B move with constant velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ from initial positions $\mathbf{r}_A$ and $\mathbf{r}_B$:

- **Wrong Approach:** Differentiating scalar distance formula algebraically leads to enormous computation time.
- **Correct Vector Method:** Freeze body B at the origin of a relative frame.

Body A moves along a straight line with constant relative velocity $\mathbf{v}_{AB} = \mathbf{v}_A - \mathbf{v}_B$ starting from relative position $\mathbf{r}_{AB} = \mathbf{r}_A - \mathbf{r}_B$.

- The shortest distance is simply the length of the perpendicular from B to the relative velocity line:

$$d_{\min} = \frac{|\mathbf{r}_{AB} \times \mathbf{v}_{AB}|}{|\mathbf{v}_{AB}|}, \qquad t_{\min} = -\frac{\mathbf{r}_{AB} \cdot \mathbf{v}_{AB}}{|\mathbf{v}_{AB}|^2} \tag{1.33}$$

1.4 Newton's Laws of Motion, Constrained Dynamics & Pseudo-Forces

Foundations of Newtonian Mechanics & Frame Relativity

1. **Newton's First Law (Law of Inertia):** In the absence of an unbalanced net external force, a body maintains its state of rest or uniform rectilinear motion. This establishes the existence of **Inertial Reference Frames**.
2. **Newton's Second Law:** The rate of change of linear momentum $\mathbf{p} = m\mathbf{v}$ is proportional to and in the direction of the applied net force:

$$\mathbf{F}_{\text{net}} = \frac{d\mathbf{p}}{dt} = m\frac{d\mathbf{v}}{dt} + \mathbf{v}\frac{dm}{dt} = m\mathbf{a} \quad \text{(for constant mass system)} \tag{1.34}$$

3. **Newton's Third Law:** Forces occur strictly in pairs. When body A exerts force $\mathbf{F}_{BA}$ on body B, body B exerts an instantaneous, collinear, equal and opposite reaction force on A: $\mathbf{F}_{AB} = -\mathbf{F}_{BA}$. Action and reaction act on **different bodies** and never cancel each other.
4. **Non-Inertial Reference Frames & Pseudo-Force:** In a frame accelerating with acceleration $\mathbf{a}_f$, Newton's second law is restored by introducing the inertial pseudo-force:

$$\mathbf{F}_{\text{pseudo}} = -m\mathbf{a}_f \tag{1.35}$$

Equilibrium, Lami's Theorem & String/Wedge Constraints

1. Concurrent Force Equilibrium (Lami's Theorem): If three coplanar forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3$ acting on a particle maintain equilibrium ($\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \mathbf{0}$):

$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma} \tag{1.36}$$

where α, β, γ are the angles opposite to forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3$ respectively.

2. Virtual Work Principle for String Constraints: For an inextensible, taut string passing over ideal frictionless massless pulleys, the total internal

work done by string tension across the system is identically zero:

$$\sum_i \mathbf{T}_i \cdot \mathbf{x}_i = 0 \implies \sum_i \mathbf{T}_i \cdot \mathbf{v}_i = 0 \implies \sum_i \mathbf{T}_i \cdot \mathbf{a}_i = 0 \tag{1.37}$$

3. Wedge Constraints: Two bodies maintaining continuous contact without penetration must have identical velocity and acceleration components along the **common normal** to their contact interface:

$$(\mathbf{v}_1 \cdot \hat{\mathbf{n}}) = (\mathbf{v}_2 \cdot \hat{\mathbf{n}}), \qquad (\mathbf{a}_1 \cdot \hat{\mathbf{n}}) = (\mathbf{a}_2 \cdot \hat{\mathbf{n}}) \tag{1.38}$$

Atwood Machine Derivations & High-Yield Accelerations

- **Standard Simple Atwood Machine:** Masses $m_1 > m_2$ connected over an ideal pulley:

$$a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) g, \qquad T = \left(\frac{2m_1 m_2}{m_1 + m_2}\right) g \tag{1.39}$$

$$F_p = 2T = \frac{4m_1 m_2 g}{m_1 + m_2} \quad \text{[Force on Pulley Support]} \tag{1.40}$$

- **Movable Pulley System (Mechanical Advantage):** Mass m_1 on smooth horizontal table connected to movable pulley holding hanging mass m_2. Since $x_2 = x_1/2$, constraint yields $a_1 = 2a_2$:

$$a_2 = \frac{m_2 g}{4m_1 + m_2}, \qquad a_1 = \frac{2m_2 g}{4m_1 + m_2} \tag{1.41}$$

$$T = \frac{2m_1 m_2 g}{4m_1 + m_2} \tag{1.42}$$

- **Apparent Weight in an Accelerating Elevator:**

$$\text{Accelerating upward with } a : N = m(g + a) \tag{1.43}$$

$$\text{Accelerating downward with } a : N = m(g - a) \tag{1.44}$$

$$\text{Free fall } (a = g) : N = 0 \quad \text{(Weightlessness)} \tag{1.45}$$

NEET Trap: Spring Balance Reading During Transients & Cut Strings

- **Spring vs String Cut Difference:** A light spring exerts force $F = kx$. The displacement x cannot change discontinuously. Therefore, immediately after cutting an adjacent string/support at $t = 0^+$, the spring force **remains completely unchanged** ($F_{\text{spring}}(0^+) = F_{\text{spring}}(0^-)$).
- In contrast, string tension is an instantaneous constraint force that drops to zero or readjusts instantaneously ($T(0^+) \neq T(0^-)$).

1.5 Friction: Microscopic Mechanism, Slopes & Multi-Block Dynamics

1.5.1 Microscopic Origin and Laws of Dry Friction (Amontons–Coulomb Model)

Friction is an electromagnetic contact force arising from microscopic adhesive bonding (cold welding of microscopic surface asperities) and mechanical interlocking between contacting surfaces. When two rough solid surfaces are pressed together under normal load N, the actual microscopic contact area is merely a minute fraction (10^{-4} to 10^{-2}) of the apparent macroscopic surface area.

1. Static Friction (f_s): Static friction is a self-adjusting reaction force that opposes the impending tendency of relative tangential motion between two contacting interfaces. It automatically balances the net applied tangential force F_{applied} up to a definite limiting maximum value $f_{s,\max}$:

$$0 \leq f_s \leq f_{s,\max} = \mu_s N \tag{1.46}$$

where μ_s is the dimensionless coefficient of static friction, and N is the normal contact force. As long as $F_{\text{applied}} < \mu_s N$, the acceleration is zero and $f_s = F_{\text{applied}}$.

2. Kinetic Friction (f_k): Once the applied tangential load overcomes the limiting static barrier ($F_{\text{applied}} > \mu_s N$), relative sliding commences. The microscopic cold welds shear continuously, and the opposing kinetic friction force becomes virtually independent of both the sliding velocity (at moderate speeds) and the apparent contact area:

$$f_k = \mu_k N \qquad (\text{with } \mu_k < \mu_s) \tag{1.47}$$

3. Angle of Friction (λ) and Angle of Repose (θ_r): The angle of friction λ is defined as the angle that the resultant contact force $\mathbf{R} = \mathbf{N} + \mathbf{f}_{s,\max}$ makes with the normal vector $\mathbf{N}$ at limiting equilibrium:

$$\tan\lambda = \frac{f_{s,\max}}{N} = \frac{\mu_s N}{N} = \mu_s \implies \lambda = \tan^{-1}\mu_s \tag{1.48}$$

Similarly, the angle of repose θ_r is the maximum angle of inclination of a rough plane with the horizontal such that a block placed upon it remains at rest without slipping down. By resolving forces parallel and perpendicular to the inclined plane:

$$N = mg\cos\theta \tag{1.49}$$

$$f_s = mg\sin\theta \tag{1.50}$$

At limiting equilibrium when slipping is imminent, $f_s = \mu_s N = \mu_s mg\cos\theta$. Equating the two yields:

$$mg\sin\theta_r = \mu_s mg\cos\theta_r \implies \tan\theta_r = \mu_s \implies \theta_r = \lambda = \tan^{-1}\mu_s \tag{1.51}$$

Thus, a body placed on an inclined plane remains at rest if $\theta \leq \theta_r$, slides with constant velocity if $\theta = \theta_r$ (with $\mu_k \approx \mu_s$), and accelerates down the plane if $\theta > \theta_r$.

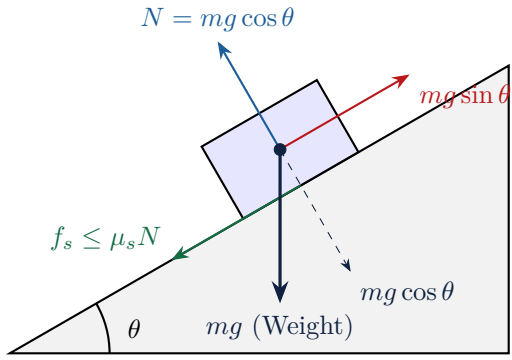


Figure 1.1: Free-Body Force Resolution for a Block of Mass m on a Rough Inclined Plane of Angle θ.

Dynamics on Rough Inclined Planes & Stacked Block Systems

1. Acceleration on Rough Incline:

$$\text{Sliding down incline } (\theta > \theta_r)\text{: } a_{\text{down}} = g(\sin\theta - \mu_k\cos\theta) \tag{1.52}$$

$$\text{Projected upward along incline: } a_{\text{up}} = -g(\sin\theta + \mu_k\cos\theta) \tag{1.53}$$

Ratio of time taken to slide down rough incline vs smooth incline of identical length:

$$t_{\text{rough}} = n \cdot t_{\text{smooth}} \implies \mu_k = \tan\theta\left(1 - \frac{1}{n^2}\right) \tag{1.54}$$

2. Minimum Force to Drag a Block on a Rough Horizontal Surface: Pulling at an upward angle θ to the horizontal reduces normal force $N = mg - F\sin\theta$:

$$F\cos\theta = \mu_s(mg - F\sin\theta) \implies F(\theta) = \frac{\mu_s mg}{\cos\theta + \mu_s\sin\theta} \tag{1.55}$$

Minimizing $F(\theta)$ by setting $\frac{d}{d\theta}(\cos\theta + \mu_s\sin\theta) = 0 \implies \tan\theta = \mu_s$:

$$F_{\min} = \frac{\mu_s mg}{\sqrt{1+\mu_s^2}} \quad \text{at angle } \theta = \tan^{-1}\mu_s \tag{1.56}$$

Two-Block Stacked Friction Problem Archetype

Consider block A of mass m_1 resting atop block B of mass m_2, situated on a frictionless horizontal floor. Friction coefficient between A and B is $\mu_s = \mu_k = \mu$.

- **Case I: External Force F Applied to Lower Block B (m_2):** The only force driving block A forward is static friction $f_s \leq \mu m_1 g$. Maximum allowable acceleration without relative slip:

$$a_{\max} = \frac{f_{s,\max}}{m_1} = \frac{\mu m_1 g}{m_1} = \mu g \tag{1.57}$$

  Threshold force for no slipping:

$$F_{\text{threshold}} = (m_1 + m_2) a_{\max} = \mu (m_1 + m_2) g \tag{1.58}$$

  If $F \leq F_{\text{threshold}}$, both blocks accelerate together at $a = \frac{F}{m_1+m_2}$. If $F > F_{\text{threshold}}$, slipping occurs:

$$a_1 = \mu g, \qquad a_2 = \frac{F - \mu m_1 g}{m_2} \tag{1.59}$$

- **Case II: External Force F Applied to Upper Block A (m_1):** Here friction on B drives B forward: $f_s \leq \mu m_1 g \implies a_{\max} = \frac{\mu m_1 g}{m_2}$.

$$F_{\text{threshold}} = (m_1 + m_2) a_{\max} = \mu m_1 g \left(1 + \frac{m_1}{m_2}\right) \tag{1.60}$$

NEET Trap: Direction of Static Friction on Walking & Driving Wheels

- **Human Walking:** The foot pushes the ground backward. By Newton's third law, static friction exerted by the ground on the foot acts **forward**, providing propulsive acceleration!
- **Automobile / Bicycle Wheels:** For the rear (driving) wheel of a bicycle, engine/pedal torque pushes the ground backward, so ground friction acts **forward**. For the front (driven) wheel, ground friction acts **backward** to generate forward angular rotation.

1.6 Dynamics of Circular Motion, Banking & Conical Pendulum

Kinematics & Force Mechanics in Curvilinear Trajectories

In circular motion of radius r, the acceleration vector decomposes along radial (centripetal) and tangential directions:

$$\mathbf{a} = \mathbf{a}_c + \mathbf{a}_t = -\left(\frac{v^2}{r}\right)\hat{\mathbf{r}} + \left(\frac{dv}{dt}\right)\hat{\boldsymbol{\theta}} = -(\omega^2 r)\hat{\mathbf{r}} + (\alpha r)\hat{\boldsymbol{\theta}} \tag{1.61}$$

1. **Centripetal Force:** A net real force directed toward the instantaneous center of curvature required to continuously bend the velocity vector:

$$F_c = ma_c = \frac{mv^2}{r} = m\omega^2 r \tag{1.62}$$

2. **Centrifugal Force:** In the rotating reference frame attached to the orbiting particle, a radially outward pseudo-force of magnitude $F_{\text{cf}} = m\omega^2 r$ is applied.

Complete Derivation of Banking Dynamics on Curved Tracks

Consider a vehicle negotiating a curve of radius r on a road banked at angle θ with static friction coefficient μ_s:

1. **Unbanked Level Curved Road:** Friction alone provides centripetal acceleration:

$$f_s \leq \mu_s N = \mu_s mg \implies \frac{mv^2}{r} \leq \mu_s mg \implies v_{\max} = \sqrt{\mu_s g r} \tag{1.63}$$

2. **Frictionless Banked Road (Optimum Speed v_0):** Normal force component $N\sin\theta$ provides centripetal acceleration, while $N\cos\theta$ balances gravity:

$$N\sin\theta = \frac{mv_0^2}{r}, \quad N\cos\theta = mg \implies \tan\theta = \frac{v_0^2}{rg} \implies v_0 = \sqrt{rg\tan\theta} \tag{1.64}$$

3. **Rough Banked Road (Maximum Safe Velocity $v_{\max}$):** Vehicle tends to skid outward up the incline; limiting friction $f_s = \mu_s N$ acts downward along the incline:

$$N\cos\theta - f_s\sin\theta = mg \implies N(\cos\theta - \mu_s\sin\theta) = mg \tag{1.65}$$

$$N\sin\theta + f_s\cos\theta = \frac{mv^2}{r} \implies N(\sin\theta + \mu_s\cos\theta) = \frac{mv^2}{r} \tag{1.66}$$

Dividing the two equations yields the exact NEET master formula:

$$v_{\max} = \sqrt{rg\left(\frac{\tan\theta + \mu_s}{1 - \mu_s\tan\theta}\right)} \tag{1.67}$$

4. **Minimum Safe Velocity (Preventing Inward Slide $v_{\min}$):** Limiting friction acts upward along the incline:

$$v_{\min} = \sqrt{rg\left(\frac{\tan\theta - \mu_s}{1 + \mu_s\tan\theta}\right)} \quad (\text{if } \tan\theta > \mu_s) \tag{1.68}$$

Conical Pendulum Dynamics & Period Derivation

A bob of mass m attached to a string of length L traces a horizontal circle of radius $r = L\sin\theta$ at constant angular velocity ω, making semi-vertical angle θ:

$$\text{Vertical force balance: } T\cos\theta = mg \implies T = \frac{mg}{\cos\theta} \tag{1.69}$$

$$\text{Radial centripetal equation: } T\sin\theta = m\omega^2 r = m\omega^2(L\sin\theta) \implies T = m\omega^2 L \tag{1.70}$$

Equating expressions for string tension T:

$$\frac{mg}{\cos\theta} = m\omega^2 L \implies \omega = \sqrt{\frac{g}{L\cos\theta}} \tag{1.71}$$

The time period of the conical pendulum is:

$$T_{\text{period}} = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L\cos\theta}{g}} = 2\pi\sqrt{\frac{h}{g}} \tag{1.72}$$

where $h = L\cos\theta$ is the vertical height of the apex above the circular plane of motion.

NEET Trap: Death Well / Rotor Minimum Angular Velocity

In a cylindrical rotor (death well) of radius R, a person rests against the vertical wall without a floor.

- The normal force $N = m\omega^2 R$ acts **horizontally** providing centripetal acceleration.
- Static friction acts **vertically upward** to balance weight: $f_s = mg$.
- Condition to prevent sliding down:

$$f_s \le \mu_s N \implies mg \le \mu_s(m\omega^2 R) \implies \omega_{\min} = \sqrt{\frac{g}{\mu_s R}}, \quad v_{\min} = \sqrt{\frac{gR}{\mu_s}} \tag{1.73}$$

Notice that μ_s appears in the denominator, in stark contrast to level road turns!

1.7 Work, Energy, Power & Vertical Circular Dynamics

Calculus Formulations of Work & Potential Energy Functions

1. **Work Done by Variable Force:** For a particle moving along path C from $\mathbf{r}_i$ to $\mathbf{r}_f$:

$$W = \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F}\cdot d\mathbf{r} = \int_{x_i}^{x_f} F_x\,dx + \int_{y_i}^{y_f} F_y\,dy + \int_{z_i}^{z_f} F_z\,dz \tag{1.74}$$

2. **Conservative Force Criterion:** A force is conservative if the work done around any closed loop vanishes identically: $\oint \mathbf{F}\cdot d\mathbf{r} = 0 \iff \boldsymbol{\nabla}\times\mathbf{F} = \mathbf{0}$.

3. **Potential Energy Field (U):** For conservative forces, potential energy is defined such that:

$$\Delta U = U_f - U_i = -W_{\text{int}} = -\int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F} \cdot d\mathbf{r} \tag{1.75}$$

$$\implies \mathbf{F} = -\boldsymbol{\nabla} U = -\left(\frac{\partial U}{\partial x}\hat{\mathbf{i}} + \frac{\partial U}{\partial y}\hat{\mathbf{j}} + \frac{\partial U}{\partial z}\hat{\mathbf{k}}\right) \tag{1.76}$$

4. **Equilibrium States & Stability Calculus:**
   - **Equilibrium Condition:** $\dfrac{dU}{dx} = 0 \implies F = 0$
   - **Stable Equilibrium:** $\dfrac{d^2U}{dx^2} > 0$ (Local potential minimum, restoring force)
   - **Unstable Equilibrium:** $\dfrac{d^2U}{dx^2} < 0$ (Local potential maximum, destabilizing)
   - **Neutral Equilibrium:** $\dfrac{d^2U}{dx^2} = 0$ (Flat potential valley)

Work-Energy Theorem Rigorous Proof & Spring Potential Energy

1. Work-Energy Theorem for Variable Force in 1D:

$$W = \int_{x_i}^{x_f} F\,dx = \int_{x_i}^{x_f} m\left(\frac{dv}{dt}\right)dx = \int_{x_i}^{x_f} mv\left(\frac{dv}{dx}\right)dx = \int_{v_i}^{v_f} mv\,dv \tag{1.77}$$

Integrating yields the fundamental Work-Energy Theorem:

$$W_{\text{net}} = \left[\frac{1}{2}mv^2\right]_{v_i}^{v_f} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = \Delta K \tag{1.78}$$

2. Elastic Potential Energy of Ideal Spring: According to Hooke's Law, restoring spring force is $F_s = -kx$. Work done by external agent stretching from 0 to x:

$$W_{\text{ext}} = \int_0^x kx\,dx = \frac{1}{2}kx^2 \implies U_s(x) = \frac{1}{2}kx^2 \tag{1.79}$$

Complete Dynamics of Vertical Circular Motion (String vs Rod)

Let a bob of mass m on a string of length l be projected horizontally at the lowest point (Point L) with velocity u. At angle θ relative to the downward vertical:

Conservation of Mechanical Energy: $\dfrac{1}{2}mu^2 = \dfrac{1}{2}mv^2(\theta) + mgl(1 - \cos\theta)$

$$\implies v^2(\theta) = u^2 - 2gl(1 - \cos\theta) \quad (1.80)$$

Radial Equation for String Tension: $T(\theta) - mg\cos\theta = \dfrac{mv^2(\theta)}{l}$

$$\implies T(\theta) = \frac{mu^2}{l} - mg(2 - 3\cos\theta) \quad (1.81)$$

Key Critical Boundary Values for Complete Vertical Loop (String):

1. **At Highest Point ($\theta = \pi$):** $\cos\pi = -1 \implies v_{\text{top}}^2 = u^2 - 4gl$. To maintain tautness, $T_{\text{top}} \geq 0$:

$$T_{\text{top}} = \frac{mv_{\text{top}}^2}{l} - mg \geq 0 \implies v_{\text{top}} \geq \sqrt{gl} \quad (1.82)$$

2. **Critical Minimum Launch Speed at Bottom ($u_{\min}$):**

$$u^2 - 4gl \geq gl \implies u_{\min} = \sqrt{5gl} \quad (1.83)$$

3. **Speed at Horizontal Position ($\theta = \pi/2$):**

$$v_{\text{horiz}} = \sqrt{u^2 - 2gl} \geq \sqrt{3gl} \quad (1.84)$$

4. **Tension Difference Between Bottom and Top:**

$$T_{\text{bottom}} = \frac{mu^2}{l} + mg, \qquad T_{\text{top}} = \frac{mu^2}{l} - 5mg \implies T_{\text{bottom}} - T_{\text{top}} = 6mg \quad (1.85)$$

5. **Looping in a Light Rigid Rod:** A rod can support compressive stress. Looping requires only $v_{\text{top}} \geq 0$:

$$v_{\text{top}} \geq 0 \implies u_{\min,\text{rod}} = \sqrt{4gl} = 2\sqrt{gl} \quad (1.86)$$

Regimes of Motion in Vertical Circle Based on Launch Velocity u

- **Regime I ($0 < u \leq \sqrt{2gl}$): Oscillation Regime.** Velocity vanishes ($v = 0$) before or at horizontal ($\theta \leq 90°$). Tension remains strictly positive ($T > 0$). The particle oscillates as a simple pendulum.
- **Regime II ($\sqrt{2gl} < u < \sqrt{5gl}$): String Slackening Regime.** Tension vanishes ($T = 0$) before velocity vanishes, at an angle θ in the upper hemisphere ($90° < \theta < 180°$):

$$\cos\theta = \frac{2gl - u^2}{3gl} \quad \text{(negative value)} \quad (1.87)$$

  The particle leaves the circular path and executes parabolic projectile motion under gravity!
- **Regime III ($u \geq \sqrt{5gl}$): Complete Looping Regime.** String never slacks ($T \geq 0$ throughout).

NEET Trap: Instantaneous Power vs Average Power

Power is the rate of doing work:

$$P_{\text{avg}} = \frac{\Delta W}{\Delta t}, \qquad P_{\text{inst}} = \frac{dW}{dt} = \mathbf{F} \cdot \mathbf{v} = Fv\cos\phi \tag{1.88}$$

- In uniform circular motion, $\mathbf{F} \perp \mathbf{v}$ at every instant ($\cos 90^\circ = 0$). Thus, instantaneous power delivered by centripetal force is **identically zero** ($P_{\text{inst}} = 0$), and net work done over any interval is zero.
- For an engine delivering constant power P, velocity scales as $v \propto t^{1/2}$ and displacement scales as $x \propto t^{3/2}$.

1.8 Conservation of Linear Momentum, Collisions & Variable Mass Systems

Conservation of Linear Momentum & Coefficient of Restitution

1. **Principle of Conservation of Linear Momentum:** For an isolated system of particles with zero net external force ($\sum \mathbf{F}_{\text{ext}} = \mathbf{0}$):

$$\frac{d\mathbf{P}_{\text{total}}}{dt} = \mathbf{0} \implies \mathbf{P}_{\text{total}} = \sum m_i \mathbf{v}_i = \text{constant} \tag{1.89}$$

2. **Impulse-Momentum Theorem:** The time integral of net force equals the total change in linear momentum:

$$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F}\, dt = \Delta \mathbf{p} = \mathbf{p}_2 - \mathbf{p}_1 \tag{1.90}$$

3. **Newton's Law of Restitution:** The coefficient of restitution e quantifies the elasticity along the line of impact (common normal):

$$e = \frac{\text{Velocity of Separation}}{\text{Velocity of Approach}} = \frac{v_2 - v_1}{u_1 - u_2} \tag{1.91}$$

   - $e = 1$: Perfectly elastic collision (kinetic energy conserved).
   - $0 < e < 1$: Inelastic collision (partial loss of kinetic energy to heat/deformation).
   - $e = 0$: Perfectly inelastic collision (bodies stick together, maximum kinetic energy loss).

Complete Derivation of Post-Collision Velocities & Kinetic Energy Loss

Consider two masses m_1, m_2 moving with initial velocities u_1, u_2 undergoing a 1D head-on collision with coefficient of restitution e:

$$\text{Conservation of Momentum: } m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \tag{1.92}$$

$$\text{Restitution Equation: } v_2 - v_1 = e(u_1 - u_2) \implies v_2 = v_1 + e(u_1 - u_2) \tag{1.93}$$

Substituting v_2 into the momentum equation:

$$\begin{aligned} m_1u_1 + m_2u_2 &= m_1v_1 + m_2[v_1 + e(u_1 - u_2)] \\ &= (m_1 + m_2)v_1 + em_2(u_1 - u_2) \\ \implies (m_1 + m_2)v_1 &= (m_1 - em_2)u_1 + (1 + e)m_2u_2 \end{aligned} \tag{1.94}$$

This yields the universal post-collision velocity formulas:

$$v_1 = \left(\frac{m_1 - em_2}{m_1 + m_2}\right) u_1 + \left(\frac{(1 + e)m_2}{m_1 + m_2}\right) u_2 \tag{1.95}$$

$$v_2 = \left(\frac{(1 + e)m_1}{m_1 + m_2}\right) u_1 + \left(\frac{m_2 - em_1}{m_1 + m_2}\right) u_2 \tag{1.96}$$

Kinetic Energy Loss in Inelastic Collisions (ΔK_{loss}):

$$\Delta K_{\text{loss}} = K_i - K_f = \frac{1}{2}\left(\frac{m_1m_2}{m_1 + m_2}\right)(u_1 - u_2)^2(1 - e^2) = \frac{1}{2}\mu(u_{\text{rel}})^2(1 - e^2) \tag{1.97}$$

where $\mu = \frac{m_1m_2}{m_1+m_2}$ is the reduced mass of the two-body colliding system.

High-Yield Collision Archetypes & Variable Mass Systems

- **Equal Masses Elastic Collision ($m_1 = m_2, e = 1$):** Velocities swap completely:
$$v_1 = u_2, \qquad v_2 = u_1 \tag{1.98}$$
- **Massive Target at Rest ($m_2 \gg m_1, u_2 = 0, e = 1$):**
$$v_1 = -u_1, \qquad v_2 \approx 0 \quad \text{(Projectile rebounds with inverted speed)} \tag{1.99}$$
- **Massive Projectile on Light Target ($m_1 \gg m_2, u_2 = 0, e = 1$):**
$$v_1 \approx u_1, \qquad v_2 \approx 2u_1 \quad \text{(Target kicked at double projectile velocity)} \tag{1.100}$$
- **Successive Bounces of Ball Dropped from Height h_0:**
  - **Height after n^{th} bounce:** $h_n = e^{2n}h_0$
  - **Total distance traveled until rest:**
$$H_{\text{total}} = h_0 + 2\sum_{n=1}^{\infty} h_n = h_0\left(\frac{1 + e^2}{1 - e^2}\right)$$
  - **Total time elapsed until rest:**
$$T_{\text{total}} = \sqrt{\frac{2h_0}{g}} + 2\sum_{n=1}^{\infty}\sqrt{\frac{2h_n}{g}} = \sqrt{\frac{2h_0}{g}}\left(\frac{1 + e}{1 - e}\right)$$

- **Rocket Propulsion (Tsiolkovsky Equation):** For variable mass system with exhaust velocity u_{rel}:

$$v(t) = v_0 + u_{\text{rel}} \ln\left(\frac{m_0}{m(t)}\right) - gt \tag{1.101}$$

NEET Trap: Oblique Collisions & Line of Impact Conservation

In two-dimensional (oblique) collisions between smooth spheres:

- Forces during collision act **solely along the line of impact** (the common normal connecting the sphere centers at the instant of contact).
- Consequently, momentum is conserved along the line of impact, and Newton's restitution formula $e = -\frac{v_{2n}-v_{1n}}{u_{2n}-u_{1n}}$ applies **only** to normal velocity components.
- Tangential velocity components parallel to the contact plane experience zero impulse and remain **strictly unchanged**:

$$v_{1t} = u_{1t}, \qquad v_{2t} = u_{2t} \tag{1.102}$$

Applying restitution to total velocity vectors without normal projection is an immediate trap!

NCERT Critical Extraction: Collision Classifications & Nuclear Moderation

NCERT Highlight:

- In a nuclear reactor, fast neutrons ($m \approx 1\,\text{u}$) produced by uranium fission must be slowed down to thermal speeds to sustain a chain reaction. Moderators containing light nuclei such as protons or carbon ($m_2 \approx m_1$) are used.
- In an elastic head-on collision with a target of comparable mass ($m_1 \approx m_2$) at rest, the incoming neutron transfers virtually all of its kinetic energy (100%) because $v_1 \approx 0$ and $v_2 \approx u_1$. Heavy stationary nuclei ($m_2 \gg m_1$) merely rebound the neutron elastically without absorbing its kinetic energy.

PART II

Rotational Mechanics, Gravitation & Continuum Mechanics

CHAPTER 2

Rotational Dynamics, Universal Gravitation & Continuum Mechanics

In this chapter, we bridge translational physics with rotational mechanics, celestial gravitation, and the mechanics of deformable media. Topics include the calculus of continuous mass distributions, moment of inertia theorems, angular momentum conservation, pure rolling on rough inclines, Kepler's laws, gravitational field/potential energy formulations, elastic moduli, surface tension, capillarity, and Bernoulli's hydrodynamic theorem.

2.1 Centre of Mass, Continuous Mass Distributions & System Dynamics

Theoretical Postulates of Centre of Mass & Invariant Trajectories

The Centre of Mass (CM) of a system of particles represents a unique point where the entire mass of the system can be considered concentrated for translational dynamic analysis.

1. **Discrete Particle Formulations:** For a collection of N point masses $m_1, m_2, \ldots, m_N$ with position vectors $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N$:

$$\mathbf{r}_{\text{cm}} = \frac{\sum_{i=1}^{N} m_i \mathbf{r}_i}{\sum_{i=1}^{N} m_i} = \frac{1}{M} \sum_{i=1}^{N} m_i \mathbf{r}_i \tag{2.1}$$

2. **Continuous Mass Distributions (Calculus Formulation):**

$$\mathbf{r}_{\text{cm}} = \frac{1}{M} \int \mathbf{r}\, dm, \qquad x_{\text{cm}} = \frac{1}{M} \int x\, dm, \qquad y_{\text{cm}} = \frac{1}{M} \int y\, dm \tag{2.2}$$

3. **System Equations of Motion & Conservation of Linear Momentum:**

$$\mathbf{P}_{\text{total}} = M \mathbf{v}_{\text{cm}}, \qquad \mathbf{F}_{\text{ext}} = \frac{d\mathbf{P}_{\text{total}}}{dt} = M \mathbf{a}_{\text{cm}} \tag{2.3}$$

   Since internal mutual forces between system particles occur in collinear, equal and opposite action-reaction pairs by Newton's third law, they cancel identically in vector summation: $\sum \mathbf{F}_{\text{int}} = \mathbf{0}$. If $\mathbf{F}_{\text{ext}} = \mathbf{0}$, the centre of mass maintains constant velocity ($\mathbf{v}_{\text{cm}} = \text{constant}$), irrespective of internal explosions or collisions.

Master Coordinates of Centre of Mass for Continuous Rigid Bodies

Coordinates are defined from the geometric center or vertex as reference origin:

$$\text{Uniform Thin Rod of Length } L : x_{\text{cm}} = \frac{L}{2} \tag{2.4}$$

$$\text{Uniform Semicircular Ring of Radius } R : y_{\text{cm}} = \frac{2R}{\pi} \approx 0.637R \tag{2.5}$$

$$\text{Uniform Semicircular Disc of Radius } R : y_{\text{cm}} = \frac{4R}{3\pi} \approx 0.424R \tag{2.6}$$

$$\text{Uniform Hemispherical Shell of Radius } R : y_{\text{cm}} = \frac{R}{2} = 0.500R \tag{2.7}$$

$$\text{Uniform Solid Hemisphere of Radius } R : y_{\text{cm}} = \frac{3R}{8} = 0.375R \tag{2.8}$$

$$\text{Uniform Solid Cone of Height } h : y_{\text{cm}} = \frac{h}{4} \quad \text{(from base)} \tag{2.9}$$

$$\text{Uniform Hollow Cone of Height } h : y_{\text{cm}} = \frac{h}{3} \quad \text{(from base)} \tag{2.10}$$

Cavity Problems & Mid-Air Projectile Explosions

- **Negative Mass Technique for Cavities:** When a circular or spherical cavity of mass m_2 with centroid $\mathbf{r}_2$ is excised from an intact parent body of mass m_1 with centroid $\mathbf{r}_1$:

$$\mathbf{r}_{\text{cm, remaining}} = \frac{m_1\mathbf{r}_1 - m_2\mathbf{r}_2}{m_1 - m_2} = \frac{A_1\mathbf{r}_1 - A_2\mathbf{r}_2}{A_1 - A_2} \quad \text{(for laminar 2D bodies)} \tag{2.11}$$

- **Projectile Explosion Invariance:** If a projectile explodes in flight due to internal chemical forces, $\mathbf{F}_{\text{ext}} = M\mathbf{g}$ remains unchanged. Hence, the centre of mass of all fragments continues along the identical parabolic trajectory until the first fragment strikes the ground.

NEET Trap: Shift of Centre of Mass on Frictionless Floors

Consider a man of mass m walking a distance x relative to a boat of mass M floating on calm, frictionless water.

- Since no external horizontal force acts on the (man + boat) system, the centre of mass cannot shift horizontally: $\Delta x_{\text{cm}} = 0$.
- If the boat displaces by distance X_{boat} in the opposite direction:

$$m(x - X_{\text{boat}}) - MX_{\text{boat}} = 0 \implies X_{\text{boat}} = \left(\frac{m}{M+m}\right) x \tag{2.12}$$

Assuming the boat moves distance x is a common pitfall. The boat displacement is strictly proportional to reduced system fractions!

2.2 Rigid Body Rotation, Moment of Inertia & Parallel/Perpendicular Theorems

Rotational Kinematics & Physical Meaning of Moment of Inertia

A rigid body maintains constant mutual distances between all constituent particles.

1. **Rotational Kinematics (Constant Angular Acceleration α):**

$$\omega = \omega_0 + \alpha t, \qquad \theta = \omega_0 t + \frac{1}{2}\alpha t^2 \tag{2.13}$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta, \qquad \theta_n = \omega_0 + \frac{\alpha}{2}(2n-1) \tag{2.14}$$

2. **Moment of Inertia (I) & Radius of Gyration (k):** The quantitative measure of rotational inertia about a specified axis of rotation:

$$I = \sum_{i=1}^{N} m_i r_i^2 = \int r^2\, dm, \qquad I = Mk^2 \implies k = \sqrt{\frac{I}{M}} \tag{2.15}$$

The radius of gyration k represents the effective radial distance from the rotational axis at which the total mass M could be concentrated without altering its moment of inertia.

Proofs of Fundamental Moment of Inertia Theorems

1. Parallel Axis Theorem (Steiner's Theorem): Let I_{cm} be the moment of inertia about an axis passing through the centre of mass, and I be the moment of inertia about a parallel axis located at perpendicular distance d. Let the origin be at the centre of mass ($\int x\, dm = 0, \int y\, dm = 0$). Then:

$$\begin{aligned} I &= \int r^2\, dm = \int \left[(x+d)^2 + y^2\right] dm \\ &= \int (x^2+y^2) dm + 2d \int x\, dm + d^2 \int dm \\ &= I_{\text{cm}} + 0 + Md^2 \implies I = I_{\text{cm}} + Md^2 \end{aligned} \tag{2.16}$$

Domain of Validity: Applies to any rigid body of arbitrary three-dimensional shape.

2. Perpendicular Axis Theorem: For a flat two-dimensional planar lamina situated in the $x-y$ plane ($z=0$):

$$I_x = \int y^2\, dm, \qquad I_y = \int x^2\, dm, \qquad I_z = \int (x^2+y^2) dm = I_x + I_y \tag{2.17}$$

Domain of Validity: Strictly restricted to 2D laminar planar geometries. Never apply to 3D bodies!

Calculus Derivations of Moment of Inertia for Standard Geometries

1. Thin Uniform Rod of Mass M and Length L: Linear mass density $\lambda = M/L$. Elemental mass $dm = \lambda\, dx$:

- **Transverse axis through center ($x = 0$):**

$$I_{\text{cm}} = \int_{-L/2}^{L/2} x^2 \left(\frac{M}{L}\right) dx = \frac{M}{L}\left[\frac{x^3}{3}\right]_{-L/2}^{L/2}$$
$$= \frac{M}{3L}\left[\frac{L^3}{8} - \left(-\frac{L^3}{8}\right)\right] = \frac{1}{12}ML^2 \tag{2.18}$$

- **Transverse axis through end ($x = 0$ to L):** By parallel axis theorem:

$$I_{\text{end}} = I_{\text{cm}} + M\left(\frac{L}{2}\right)^2 = \frac{1}{12}ML^2 + \frac{1}{4}ML^2 = \frac{1}{3}ML^2 \tag{2.19}$$

2. Uniform Circular Disc of Mass M and Radius R: Surface mass density $\sigma = \frac{M}{\pi R^2}$. Consider a thin concentric elemental ring of radius r and radial width dr, mass $dm = \sigma(2\pi r dr) = \frac{2M}{R^2} r\, dr$:

$$I_z = \int_0^R r^2\, dm = \frac{2M}{R^2}\int_0^R r^3\, dr = \frac{2M}{R^2}\left[\frac{R^4}{4}\right] = \frac{1}{2}MR^2 \tag{2.20}$$

By symmetry and perpendicular axis theorem ($I_z = I_x + I_y = 2I_{\text{dia}}$):

$$I_{\text{diameter}} = \frac{1}{2}I_z = \frac{1}{4}MR^2 \tag{2.21}$$

3. Uniform Solid Sphere of Mass M and Radius R: Volume mass density $\rho = \frac{3M}{4\pi R^3}$. Slice the sphere into elemental discs perpendicular to the x-axis at position x, with radius $y = \sqrt{R^2 - x^2}$ and thickness dx. Mass of elemental disc $dm = \rho(\pi y^2 dx) = \frac{3M}{4R^3}(R^2 - x^2)dx$. The moment of inertia of this elemental disc about the x-axis is:

$$dI = \frac{1}{2}y^2 dm = \frac{1}{2}(R^2 - x^2)\left[\frac{3M}{4R^3}(R^2 - x^2)dx\right] = \frac{3M}{8R^3}(R^2 - x^2)^2 dx \tag{2.22}$$

Integrating from $x = -R$ to $+R$:

$$I_{\text{dia}} = \frac{3M}{8R^3}\int_{-R}^{R}(R^4 - 2R^2x^2 + x^4)\, dx$$
$$= \frac{3M}{4R^3}\left[R^5 - \frac{2R^5}{3} + \frac{R^5}{5}\right] = \frac{3M}{4R^3}\left[\frac{8}{15}R^5\right] = \frac{2}{5}MR^2 \tag{2.23}$$

About a tangential axis: $I_{\text{tangent}} = I_{\text{cm}} + MR^2 = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$.

4. Uniform Hollow Spherical Shell of Mass M and Radius R: Surface mass density $\sigma = \frac{M}{4\pi R^2}$. Elemental ring at polar angle θ has radius $r = R\sin\theta$, width $R\,d\theta$, mass $dm = \frac{M}{4\pi R^2}(2\pi R\sin\theta \cdot R\,d\theta) = \frac{1}{2}M\sin\theta\,d\theta$:

$$I_{\text{dia}} = \int_0^{\pi} (R\sin\theta)^2 dm = \frac{1}{2}MR^2 \int_0^{\pi} \sin^3\theta\,d\theta = \frac{1}{2}MR^2\left(\frac{4}{3}\right) = \frac{2}{3}MR^2 \quad (2.24)$$

About a tangential axis: $I_{\text{tangent}} = \frac{2}{3}MR^2 + MR^2 = \frac{5}{3}MR^2$.

Moment of Inertia Master Reference Table

Rigid Body Geometry	Axis of Rotation	Moment of Inertia (I)	Radius of Gyration (k^2/R^2)
Thin Circular Ring	Central perpendicular	MR^2	1.0
Thin Circular Ring	Diametrical axis	$\frac{1}{2}MR^2$	0.5
Thin Circular Disc	Central perpendicular	$\frac{1}{2}MR^2$	0.5
Thin Circular Disc	Diametrical axis	$\frac{1}{4}MR^2$	0.25
Thin Hollow Cylinder	Longitudinal central	MR^2	1.0
Solid Cylinder	Longitudinal central	$\frac{1}{2}MR^2$	0.5
Solid Sphere	Diametrical axis	$\frac{2}{5}MR^2$	0.4
Solid Sphere	Tangential axis	$\frac{7}{5}MR^2$	1.4
Hollow Spherical Shell	Diametrical axis	$\frac{2}{3}MR^2$	0.67
Hollow Spherical Shell	Tangential axis	$\frac{5}{3}MR^2$	1.67

2.3 Rotational Dynamics, Angular Momentum & Pure Rolling on Inclines

Torque, Angular Momentum & Rolling Kinematics

1. **Torque Vector & Rotational Work:**

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = I\boldsymbol{\alpha}, \qquad W_{\text{rot}} = \int \tau \, d\theta, \qquad P_{\text{rot}} = \boldsymbol{\tau} \cdot \boldsymbol{\omega} \tag{2.25}$$

2. **Angular Momentum Vector (L):** For a particle $\mathbf{L} = \mathbf{r} \times \mathbf{p}$. For a rigid body rotating about a fixed principal symmetry axis:

$$\mathbf{L} = I\boldsymbol{\omega}, \qquad \boldsymbol{\tau}_{\text{ext}} = \frac{d\mathbf{L}}{dt} = I\frac{d\boldsymbol{\omega}}{dt} = I\boldsymbol{\alpha} \tag{2.26}$$

   If net external torque vanishes ($\sum \boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$), total angular momentum is strictly conserved: $I_1\omega_1 = I_2\omega_2$.

3. **Kinematics of Pure Rolling (Rolling Without Slipping):** A circular body of radius R rolling on a stationary surface:

$$v_{\text{cm}} = \omega R, \qquad a_{\text{cm}} = \alpha R \tag{2.27}$$

   The contact point P is instantaneously at rest: $\mathbf{v}_P = \mathbf{v}_{\text{cm}} - \omega R\hat{\mathbf{i}} = \mathbf{0}$. The highest point moves at $\mathbf{v}_{\text{top}} = 2\mathbf{v}_{\text{cm}}$.

4. **Total Kinetic Energy in Pure Rolling:**

$$K_{\text{total}} = K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2 = \frac{1}{2}Mv_{\text{cm}}^2\left(1 + \frac{k^2}{R^2}\right) \tag{2.28}$$

Rigorous Derivation of Acceleration, Velocity & Friction on an Incline

Consider a symmetrical rigid body of mass M, radius R, and radius of gyration k rolling without slipping down an inclined plane of inclination angle θ and vertical height h:

- **Translational Equation along Incline:**

$$Mg\sin\theta - f_s = Ma_{\text{cm}}$$

- **Rotational Equation about Centre of Mass:**

$$\tau = f_s R = I_{\text{cm}}\alpha = (Mk^2)\left(\frac{a_{\text{cm}}}{R}\right) \implies f_s = M\left(\frac{k^2}{R^2}\right)a_{\text{cm}}$$

Substituting the friction expression into the translational equation:

$$Mg\sin\theta = Ma_{\text{cm}} + M\left(\frac{k^2}{R^2}\right)a_{\text{cm}} = Ma_{\text{cm}}\left(1 + \frac{k^2}{R^2}\right) \tag{2.29}$$

This yields the foundational rolling acceleration formula:

$$a_{\text{cm}} = \frac{g\sin\theta}{1 + \frac{k^2}{R^2}} \tag{2.30}$$

1. Linear Velocity at the Base of the Incline (v): Applying conservation of mechanical energy ($Mgh = K_{\text{total}}$):

$$Mgh = \frac{1}{2}Mv^2\left(1 + \frac{k^2}{R^2}\right) \implies v = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}} \tag{2.31}$$

2. Necessary Static Friction Force (f_s):

$$f_s = M\left(\frac{k^2}{R^2}\right)\left[\frac{g\sin\theta}{1 + \frac{k^2}{R^2}}\right] = \frac{Mg\sin\theta}{1 + \frac{R^2}{k^2}} \tag{2.32}$$

3. Minimum Friction Coefficient to Prevent Slipping ($\mu_{\min}$): To sustain pure rolling without sliding: $f_s \leq \mu_s N = \mu_s Mg\cos\theta$:

$$\frac{Mg\sin\theta}{1 + \frac{R^2}{k^2}} \leq \mu_s Mg\cos\theta \implies \mu_s \geq \frac{\tan\theta}{1 + \frac{R^2}{k^2}} \tag{2.33}$$

Race Down an Incline: Comparison of Standard Bodies

From the acceleration formula $a \propto \frac{1}{1+k^2/R^2}$, bodies with **smaller** k^2/R^2 possess greater acceleration and reach the bottom first:

$$\left(\frac{k^2}{R^2}\right)_{\text{Solid Sphere}} (0.4) < \left(\frac{k^2}{R^2}\right)_{\text{Disc}} (0.5) < \left(\frac{k^2}{R^2}\right)_{\text{Hollow Sphere}} (0.67) < \left(\frac{k^2}{R^2}\right)_{\text{Ring}} (1.0) \tag{2.34}$$

- **Order of Acceleration and Striking Velocity:**

$$a_{\text{Solid Sphere}} > a_{\text{Disc/Cylinder}} > a_{\text{Hollow Sphere}} > a_{\text{Ring/Hollow Cylinder}} \tag{2.35}$$

- **Order of Time of Descent ($t = \sqrt{2L/a}$):**

$$t_{\text{Solid Sphere}} < t_{\text{Disc/Cylinder}} < t_{\text{Hollow Sphere}} < t_{\text{Ring/Hollow Cylinder}} \tag{2.36}$$

The solid sphere always wins the race, irrespective of mass or radius!

NEET Trap: Work Done by Static Friction in Pure Rolling

In pure rolling motion on a stationary surface:

- The instantaneous velocity of the contact point is zero ($\mathbf{v}_{\text{contact}} = \mathbf{0}$).
- Consequently, the power delivered by static friction is $P = \mathbf{f}_s \cdot \mathbf{v}_{\text{contact}} = 0$.
- **Static friction does zero net work** ($W_{f_s} = 0$) during pure rolling. It acts purely as a kinematic constraint that transfers energy reversibly between translational and rotational modes without dissipating mechanical energy as

thermal heat.

2.4 Universal Gravitation, Kepler's Laws & Field / Potential Calculus

Newton's Law of Gravitation & Kepler's Three Planetary Laws

1. **Newton's Universal Gravitation Law:** Between two point masses M and m separated by distance r:

$$\mathbf{F}_{12} = -\frac{GMm}{r^2}\hat{\mathbf{r}}_{12}, \qquad G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}} \tag{2.37}$$

2. **Kepler's First Law (Law of Orbits):** All planets move in elliptical orbits with the Sun located at one of the two foci.
3. **Kepler's Second Law (Law of Areas):** The radius vector connecting the Sun to a planet sweeps out equal areas in equal intervals of time.
4. **Kepler's Third Law (Law of Periods):** The square of the orbital period T is proportional to the cube of the semi-major axis a: $T^2 \propto a^3$.

Proofs of Kepler's Second and Third Laws

1. Proof of Kepler's Second Law from Angular Momentum Conservation: Gravitational force is purely central: $\mathbf{F} = F(r)\hat{\mathbf{r}}$. Net external torque on the planet about the Sun:

$$\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F} = \mathbf{r}\times[F(r)\hat{\mathbf{r}}] = \mathbf{0} \implies \frac{d\mathbf{L}}{dt} = \mathbf{0} \implies \mathbf{L} = m(\mathbf{r}\times\mathbf{v}) = \text{constant} \tag{2.38}$$

In time dt, the radius vector sweeps out triangular area $d\mathbf{A} = \frac{1}{2}(\mathbf{r} \times d\mathbf{r})$:

$$\frac{d\mathbf{A}}{dt} = \frac{1}{2}\left(\mathbf{r} \times \frac{d\mathbf{r}}{dt}\right) = \frac{1}{2}(\mathbf{r} \times \mathbf{v}) = \frac{\mathbf{L}}{2m} = \text{constant} \tag{2.39}$$

Areal velocity is constant because gravity produces zero torque!

2. Proof of Kepler's Third Law for Circular Orbit of Radius r: Gravitational force provides the necessary centripetal acceleration:

$$\frac{GMm}{r^2} = m\omega^2 r = m\left(\frac{2\pi}{T}\right)^2 r \implies T^2 = \left(\frac{4\pi^2}{GM}\right) r^3 \tag{2.40}$$

Gravitational Field & Potential Formulations (Solid Sphere vs Spherical Shell)

Gravitational field $\mathbf{E} = -\boldsymbol{\nabla} V$. The potential $V(\mathbf{r}) = -\int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{r}'$.

1. **Uniform Solid Sphere of Mass M and Radius R:**

$$\text{Outside } (r \geq R)\text{: } E_{\text{out}}(r) = \frac{GM}{r^2}, \qquad V_{\text{out}}(r) = -\frac{GM}{r} \tag{2.41}$$

$$\text{Inside } (r < R)\text{: } E_{\text{in}}(r) = \frac{GMr}{R^3}, \qquad V_{\text{in}}(r) = -\frac{GM}{2R^3}\left(3R^2 - r^2\right) \tag{2.42}$$

At the center of the sphere ($r = 0$):

$$V_{\text{center}} = -\frac{3GM}{2R} = \frac{3}{2}V_{\text{surface}} \tag{2.43}$$

2. **Uniform Spherical Shell of Mass M and Radius R:**

$$\text{Outside } (r \geq R)\text{: } E_{\text{out}}(r) = \frac{GM}{r^2}, \qquad V_{\text{out}}(r) = -\frac{GM}{r} \tag{2.44}$$

$$\text{Inside } (r < R)\text{: } E_{\text{in}}(r) = 0, \qquad V_{\text{in}}(r) = -\frac{GM}{R} = \text{constant} \tag{2.45}$$

The interior of a spherical shell is an equipotential field-free cavity.

Variation of g, Escape Speed & Satellite Orbital Dynamics

- **Altitude Variation:** At height h above Earth's surface:

$$g_h = g\left(1 + \frac{h}{R}\right)^{-2} \approx g\left(1 - \frac{2h}{R}\right) \quad (\text{for } h \ll R) \tag{2.46}$$

- **Depth Variation:** At depth d below Earth's surface:

$$g_d = g\left(1 - \frac{d}{R}\right) \quad \text{(Linear decrease to zero at Earth's center)} \tag{2.47}$$

- **Latitude Variation Due to Earth's Rotation ($\omega = 7.29 \times 10^{-5}\,\text{rad}\,\text{s}^{-1}$):**

$$g' = g - \omega^2 R\cos^2\lambda \implies g_{\text{equator}} = g - \omega^2 R, \qquad g_{\text{pole}} = g \tag{2.48}$$

- **Escape Velocity (v_e):** Minimum projection speed required to escape Earth's gravitational well:

$$\frac{1}{2}mv_e^2 - \frac{GMm}{R} = 0 \implies v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2\,\text{km}\,\text{s}^{-1} \tag{2.49}$$

- **Satellite Dynamics & Energy Partition:** For orbit radius $r = R + h$:
  - **Orbital Velocity:** $v_o = \sqrt{\dfrac{GM}{r}} = \dfrac{v_e}{\sqrt{2}}$
  - **Total Mechanical Energy:**

$$E = K + U = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r} = -K = \frac{U}{2}$$

NEET Trap: Independence of Escape Velocity from Angle & Drag Paradox

- **Projection Angle Independence:** Escape speed depends solely on gravitational potential at the launch site: $v_e = \sqrt{-2V_{\text{launch}}} = \sqrt{2GM/R}$. It is completely independent of launch direction θ (as long as it does not intercept Earth's surface) and independent of projectile mass m.
- **Satellite Atmospheric Drag Paradox:** When satellite orbits encounter atmospheric resistance, total energy decreases (E becomes more negative $\implies$ r decreases). Because $v_o = \sqrt{GM/r}$, orbital velocity **increases** as the orbit spirals downward!

2.5 Mechanical Properties of Solids: Stress, Strain & Elastic Moduli

Foundations of Elasticity & Stress-Strain Curve

Elasticity is the property by virtue of which a material body regains its original shape and dimensions when deforming external forces are removed.

1. **Stress ($\sigma = F_{\text{int}}/A$):** Internal restoring force per unit area. SI Unit: N m^{-2} or Pa.
2. **Strain ($\epsilon = \Delta L/L$):** Fractional deformation. Strictly dimensionless.
3. **Hooke's Law:** Within the proportional limit, stress is directly proportional to strain: $\sigma = E \cdot \epsilon$, where E is the modulus of elasticity.
4. **Characteristic Points on the Tensile Stress-Strain Diagram:**
   - **Proportional Limit:** Region of linear Hookean behavior ($\sigma \propto \epsilon$).
   - **Elastic Limit (Yield Point σ_y):** Maximum stress beyond which permanent plastic deformation (permanent set) occurs.
   - **Ultimate Tensile Strength (σ_u):** Maximum stress the material can sustain before necking.
   - **Fracture Point:** Separation point. If fracture occurs immediately past yield, material is **brittle** (e.g., glass, cast iron). If wide plastic region exists, material is **ductile** (e.g., copper, aluminium).

Elastic Moduli Formulations & Potential Energy Density

- **Young's Modulus:**

$$Y = \frac{\text{Tensile Stress}}{\text{Longitudinal Strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$$

- **Bulk Modulus & Compressibility:**

$$B = -\frac{\Delta P}{\Delta V/V} = -V\frac{\Delta P}{\Delta V}, \qquad K = \frac{1}{B}$$

- **Shear Modulus (Rigidity):** $\eta = \dfrac{\text{Shearing Stress}}{\text{Shear Strain}} = \dfrac{F/A}{\theta}$
- **Poisson's Ratio:** $\sigma_p = -\dfrac{\text{Lateral Strain}}{\text{Longitudinal Strain}} = -\dfrac{\Delta d/d}{\Delta L/L} \quad (-1 \leq \sigma_p \leq 0.5)$

Elastic Potential Energy Stored in Stretched Wire:

$$W = \int_0^{\Delta L} F(x)\,dx = \int_0^{\Delta L} \left(\frac{YA}{L}x\right) dx = \frac{1}{2}\left(\frac{YA}{L}\right)(\Delta L)^2 = \frac{1}{2}F\Delta L \quad (2.50)$$

Energy density (energy per unit volume u):

$$u = \frac{W}{AL} = \frac{1}{2}\left(\frac{F}{A}\right)\left(\frac{\Delta L}{L}\right) = \frac{1}{2} \times \text{Stress} \times \text{Strain} = \frac{1}{2}Y\epsilon^2 = \frac{\sigma^2}{2Y} \quad (2.51)$$

2.6 Fluid Statics, Pascal's Principle, Surface Tension & Capillary Rise

Hydrostatic Pressure, Pascal's Law & Archimedes' Principle

1. **Hydrostatic Pressure Variation with Depth:** In an incompressible fluid of uniform density ρ:

$$\frac{dP}{dz} = -\rho g \implies P(h) = P_0 + \rho g h \quad (2.52)$$

2. **Pascal's Law:** Pressure exerted anywhere upon a confined incompressible fluid is transmitted undiminished equally in all directions throughout the fluid:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \implies F_2 = F_1\left(\frac{A_2}{A_1}\right) \quad \text{(Hydraulic Lift Principle)} \quad (2.53)$$

3. **Archimedes' Principle & Law of Floatation:** Upward buoyant force equals the weight of displaced fluid:

$$F_B = \rho_{\text{fluid}} V_{\text{submerged}} g, \qquad \frac{V_{\text{submerged}}}{V_{\text{total}}} = \frac{\rho_{\text{body}}}{\rho_{\text{fluid}}} \quad (2.54)$$

Surface Tension & Young-Laplace Excess Pressure Derivations

Surface tension $T = \frac{F}{L} = \frac{dW}{dA}$ represents surface free energy per unit area.

1. **Excess Pressure Inside a Spherical Liquid Drop of Radius R:** A drop has a **single** liquid-gas interface. Increasing radius from R to $R + dR$:

$$dW_{\text{surface}} = T \cdot d(4\pi R^2) = T(8\pi R\, dR) \tag{2.55}$$

   Work done by excess pressure ΔP:

$$dW_{\text{pressure}} = \Delta P(4\pi R^2)dR = T(8\pi R\, dR) \implies \Delta P_{\text{drop}} = \frac{2T}{R} \tag{2.56}$$

2. **Excess Pressure Inside a Spherical Soap Bubble of Radius R:** A bubble possesses **two** liquid-gas interfaces (interior and exterior surfaces):

$$dW_{\text{surface}} = 2 \times T(8\pi R\, dR) = 16\pi RT\, dR \implies \Delta P_{\text{bubble}} = \frac{4T}{R} \tag{2.57}$$

Rigorous Calculus Derivation of Capillary Rise (Jurin's Law)

Consider a narrow capillary tube of radius r dipped vertically into a wetting liquid of density ρ and contact angle θ_c. The curved meniscus forms a spherical cap of radius of curvature $R_{\text{meniscus}} = \frac{r}{\cos\theta_c}$. By the Young-Laplace relation, the pressure just below the concave meniscus is reduced relative to atmospheric pressure P_0 by:

$$\Delta P = \frac{2T}{R_{\text{meniscus}}} = \frac{2T\cos\theta_c}{r} \tag{2.58}$$

To equilibrate with hydrostatic pressure at the flat external liquid surface at depth h:

$$\rho g h = \frac{2T\cos\theta_c}{r} \implies h = \frac{2T\cos\theta_c}{r\rho g} \tag{2.59}$$

Insufficient Capillary Length & Droplet Coalescence Thermics

- **Capillary Tube of Insufficient Length $l < h$:** Liquid rises to the top lip but **never overflows**. The meniscus flattens, increasing its radius of curvature R' such that:

$$hr = lR'\cos\theta_c \implies hR = lR' = \text{constant} \tag{2.60}$$

- **Coalescence of n Identical Droplets:** Volume conservation: $n\left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi R^3 \implies R = n^{1/3}r$. Change in surface area: $\Delta A = 4\pi R^2 - n(4\pi r^2) = 4\pi R^2(1 - n^{1/3}) < 0$. Surface energy released as heat:

$$\Delta E = T|\Delta A| = 4\pi TR^3\left(\frac{1}{r} - \frac{1}{R}\right) = 3VT\left(\frac{1}{r} - \frac{1}{R}\right) \tag{2.61}$$

  Resultant temperature increase: $\Delta T = \frac{3T}{\rho s}\left(\frac{1}{r} - \frac{1}{R}\right)$, where s is specific heat.

2.7 Fluid Dynamics, Bernoulli's Theorem, Viscosity & Stokes' Drag

Continuity, Bernoulli Theorem & Viscous Shear Dynamics

1. **Equation of Continuity:** For steady, incompressible, irrotational fluid flow:
$$A_1 v_1 = A_2 v_2 \implies Av = \text{constant} \tag{2.62}$$
2. **Bernoulli's Principle:** In streamline flow of an ideal (incompressible, non-viscous) fluid, total mechanical energy per unit volume is constant along any streamline:
$$P + \frac{1}{2}\rho v^2 + \rho g y = \text{constant} \tag{2.63}$$
3. **Newton's Law of Viscosity:** Viscous drag force between adjacent fluid layers undergoing laminar shear:
$$F = -\eta A \left(\frac{dv}{dz}\right) \tag{2.64}$$
where η is coefficient of viscosity (SI Unit: $\mathrm{Pa\,s} = \mathrm{N\,s\,m^{-2}} = 10\,\text{Poise}$).
4. **Reynolds Number (Re):** Dimensionless criterion for flow turbulence:
$$Re = \frac{\rho v d}{\eta} \quad (Re < 1000 \implies \text{Laminar}; \quad Re > 2000 \implies \text{Turbulent}) \tag{2.65}$$

Work-Energy Derivation of Bernoulli's Theorem

Consider a fluid element of volume ΔV and mass $\Delta m = \rho \Delta V$ moving along a streamline pipe between stations 1 and 2:

- **Work Done by Pressure Forces:**
$$W_P = P_1 A_1 \Delta x_1 - P_2 A_2 \Delta x_2 = (P_1 - P_2)\Delta V$$
- **Change in Kinetic Energy:**
$$\Delta K = \frac{1}{2}\Delta m (v_2^2 - v_1^2) = \frac{1}{2}\rho \Delta V (v_2^2 - v_1^2)$$
- **Change in Gravitational Potential Energy:**
$$\Delta U = \Delta m g (y_2 - y_1) = \rho \Delta V g (y_2 - y_1)$$

Applying the Work-Energy Theorem ($W_P = \Delta K + \Delta U$):

$$(P_1 - P_2)\Delta V = \frac{1}{2}\rho \Delta V(v_2^2 - v_1^2) + \rho \Delta V g(y_2 - y_1) \tag{2.66}$$

Dividing by ΔV and rearranging terms:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2 \implies P + \frac{1}{2}\rho v^2 + \rho g y = \text{constant} \tag{2.67}$$

Venturimeter Flow Rate & Torricelli's Law of Efflux

1. Venturimeter Flow Rate Derivation: A horizontal pipe ($y_1 = y_2$) with inlet area A_1 and throat area A_2. By continuity $v_2 = (A_1/A_2)v_1$:

$$P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2) = \frac{1}{2}\rho v_1^2 \left(\frac{A_1^2}{A_2^2} - 1\right) = \rho_m g h \tag{2.68}$$

The volume flow rate Q is given by:

$$Q = A_1 v_1 = A_1 A_2 \sqrt{\frac{2(P_1 - P_2)}{\rho(A_1^2 - A_2^2)}} = A_1 A_2 \sqrt{\frac{2\rho_m g h}{\rho(A_1^2 - A_2^2)}} \tag{2.69}$$

2. Torricelli's Law of Efflux: For an open tank filled to total depth H with an orifice at depth h below the liquid surface:

$$\text{Velocity of Efflux: } v = \sqrt{2gh} \tag{2.70}$$

$$\text{Horizontal Range of Jet: } R = v \cdot t = \sqrt{2gh}\sqrt{\frac{2(H-h)}{g}} = 2\sqrt{h(H-h)} \tag{2.71}$$

$$\text{Maximum Horizontal Range: } R_{\max} = H \quad \text{when orifice is at depth } h = \frac{H}{2} \tag{2.72}$$

Orifices at complementary depths h and $(H - h)$ produce identical horizontal ranges R.

Stokes' Law, Terminal Velocity & Poiseuille's Pipe Flow Equation

1. Stokes' Law for Spherical Particle of Radius r:

$$F_v = 6\pi\eta r v \tag{2.73}$$

2. Rigorous Derivation of Terminal Velocity (v_t): For a sphere of density ρ falling in fluid of density σ. At terminal equilibrium:

$$\text{Gravity } F_g = \text{Buoyant Force } F_B + \text{Viscous Drag } F_v \tag{2.74}$$

$$\frac{4}{3}\pi r^3\rho g = \frac{4}{3}\pi r^3\sigma g + 6\pi\eta r v_t \implies 6\pi\eta r v_t = \frac{4}{3}\pi r^3(\rho-\sigma)g \tag{2.75}$$

Solving for terminal velocity v_t:

$$v_t = \frac{2}{9}\frac{r^2(\rho-\sigma)g}{\eta} \tag{2.76}$$

3. Poiseuille's Volume Flow Rate Equation for Cylindrical Pipe: For laminar flow of fluid through a capillary tube of internal radius r and length L under pressure differential ΔP:

$$Q = \frac{\pi\Delta P r^4}{8\eta L}, \qquad \text{Fluid Resistance } R_f = \frac{\Delta P}{Q} = \frac{8\eta L}{\pi r^4} \tag{2.77}$$

NEET Trap: Terminal Velocity Sign for Bubbles & Poiseuille Radius Scaling

- **Air Bubble Rising in Liquid:** Since $\rho_{\text{air}} < \sigma_{\text{liquid}}$, the factor $(\rho-\sigma)$ is **negative**. Terminal velocity is negative, meaning the bubble ascends upward at $v_t = \frac{2}{9}\frac{r^2(\sigma-\rho)g}{\eta}$.
- **Sensitivity of Flow Rate to Radius (Poiseuille r^4 Law):** Flow rate scales as $Q \propto r^4$. If a blood vessel or pipe radius is reduced by a mere 10% ($r' = 0.9r$), the flow rate drops by $1-(0.9)^4 \approx 34.4\%$. To maintain the original flow rate, pressure gradient must increase by $(1/0.9)^4 \approx 1.52$ (52% increase)!

NCERT Critical Extraction: Dynamic Aerodynamic Lift & Magnus Effect

NCERT Highlight:

- **Magnus Effect:** When a spinning cylinder or tennis ball travels through air, the boundary layer drags air around it. On the side where surface motion coincides with flow velocity, air speed is higher, resulting in lower pressure by Bernoulli's principle. On the opposite side, air speed is retarded, producing higher pressure. The resulting net transverse pressure differential generates dynamic curve deflection.
- **Aerofoil Design:** The upper surface of an airplane wing is contoured with greater curvature than the lower surface. Streamlines bunch together above the wing, increasing velocity ($v_{\text{top}} > v_{\text{bottom}}$), creating a vertical upward pressure gradient that generates the aerodynamic lift required for flight.

PART III

Thermal Physics, Thermodynamics & Waves

CHAPTER 3

Thermal Physics, Thermodynamics, Kinetic Theory & Waves

This chapter covers the thermal and vibrational behavior of matter. We explore thermal expansion, calorimetry, radiation laws, the first and second laws of thermodynamics with explicit NMC boundary exclusions, the molecular kinetic theory of gases with degrees of freedom and equipartition of energy, simple harmonic motion kinematics and spring-pendulum dynamics, and wave motion in strings and organ pipes.

A Personal Note from Satya: Why Sleep is Non-Negotiable for Memory Consolidation

Hey, let's take a quick breath before we dive into Thermodynamics and Waves.

If you've been grinding Mechanics for four hours straight and your eyes are burning, close this book right now. I'm serious. Go take a 20-minute nap, or if it's late at night, turn off the study lamp and sleep.

Here is the cold, biological truth that coaching institutes rarely tell you: *your brain is an organic organ, not a silicon hard drive.* When you solve complex physics problems—free-body diagrams, torque balances, fluid pressure integrals—your brain builds temporary synaptic patterns in the hippocampus. But those memories are volatile.

During Slow-Wave Sleep (SWS) and REM sleep, your brain literally replays those firing patterns, transferring the concepts from your fragile short-term memory into the stable neocortex for permanent consolidation. When you pull an all-nighter or cut your sleep down to four hours, two devastating things happen:

1. **Synaptic Fatigue & Working Memory Collapse:** Your prefrontal processing speed drops by nearly 30%. On paper, you think you're working hard; in reality, you stare at a simple Mayer's relation ($C_p - C_v = R$) or spring-pendulum phase constant for ten minutes, unable to connect two basic equations.
2. **Fatal Sign Slips in Exam Conditions:** Sleep deprivation specifically impairs attention-to-detail circuits. You miss minus signs, you invert ratios, and you misread "isobaric" as "isochoric." That's where you lose 5 marks per question.

Aim for 7 to 8 hours of uninterrupted sleep every single night. Sleeping is not "wasting time"—it is the active biological process that turns the physics you practiced today into permanent, instinctive recall on exam day. Rest well, hydrate, and when you wake up fresh, this chapter on thermodynamics will feel ten times clearer.

3.1 Thermal Properties of Matter, Thermal Expansion & Calorimetry

Foundations of Heat, Temperature & Thermometry

1. Heat vs. Temperature:

- **Heat (Q):** Energy in transit across a boundary solely driven by a macroscopic temperature gradient. Heat is a path-dependent boundary phenomenon, not an intrinsic property stored within a system ($[Q] = \text{J}$ or cal, where $1\,\text{cal} = 4.186\,\text{J}$).
- **Temperature (T):** Macroscopic measure of the average translational kinetic energy per molecule of a substance. It dictates the direction of spontaneous thermal energy transfer (from higher T to lower T).

2. Thermometric Scales & Interconversions: Any linear thermometric property $X(T)$ (e.g., liquid column length, platinum wire resistance, gas pressure at constant volume) satisfies:

$$\frac{T - \text{LFP}}{\text{UFP} - \text{LFP}} = \text{constant} \implies \frac{C - 0}{100} = \frac{F - 32}{180} = \frac{K - 273.15}{100} = \frac{R - 0}{80} \tag{3.1}$$

where LFP is the Lower Fixed Point (ice point at 1 atm) and UFP is the Upper Fixed Point (steam point at 1 atm).

$$\frac{C}{5} = \frac{F - 32}{9} = \frac{K - 273.15}{5} \tag{3.2}$$

Key Coincidence Points:

- $C = F = -40^\circ \implies -40^\circ\text{C} = -40^\circ\text{F}$.
- $F = K = 574.25 \implies 574.25^\circ\text{F} = 574.25\,\text{K}$.
- Incremental temperature step equivalence: $\Delta T = 1\,\text{K} \equiv 1^\circ\text{C} \equiv 1.8^\circ\text{F}$.

Thermal Expansion in Solids & Liquids: Master Governing Equations

1. Anisotropic and Isotropic Thermal Expansion in Solids:

- **Linear Expansion:**

$$L(T) = L_0(1 + \alpha\Delta T) \implies \Delta L = L_0\alpha\Delta T \tag{3.3}$$
$$\alpha = \frac{1}{L_0}\frac{dL}{dT} \quad [\text{K}^{-1} \text{ or } {}^\circ\text{C}^{-1}]$$

- **Superficial (Areal) Expansion:**

$$A(T) = A_0(1 + \beta\Delta T) \implies \Delta A = A_0\beta\Delta T, \quad \text{where } \beta \approx 2\alpha \tag{3.4}$$

- **Volumetric (Cubical) Expansion:**

$$V(T) = V_0(1 + \gamma\Delta T) \implies \Delta V = V_0\gamma\Delta T, \quad \text{where } \gamma \approx 3\alpha \tag{3.5}$$

- **Isotropic Ratio:** $\alpha : \beta : \gamma = 1 : 2 : 3 \iff \alpha = \frac{\beta}{2} = \frac{\gamma}{3}$.

2. Proof of $\gamma = 3\alpha$ for Isotropic Solids: Consider a cube of initial edge length L_0 and initial volume $V_0 = L_0^3$. When temperature rises by ΔT:

$$V = L^3 = [L_0(1 + \alpha\Delta T)]^3 = L_0^3(1 + 3\alpha\Delta T + 3\alpha^2\Delta T^2 + \alpha^3\Delta T^3) \tag{3.6}$$

Since $\alpha \sim 10^{-5}\,\mathrm{K}^{-1}$, higher-order terms $\alpha^2\Delta T^2 \ll 1$ and $\alpha^3\Delta T^3 \ll 1$:

$$V \approx V_0(1 + 3\alpha\Delta T) \equiv V_0(1 + \gamma\Delta T) \implies \gamma = \mathbf{3}\alpha \tag{3.7}$$

3. Apparent vs. Real Expansion of Liquids in Vessels:

$$\gamma_{\text{real}} = \gamma_{\text{apparent}} + \gamma_{\text{vessel}} \tag{3.8}$$

- If $\gamma_{\text{real}} > \gamma_{\text{vessel}}$, liquid level rises upon heating.
- If $\gamma_{\text{real}} < \gamma_{\text{vessel}}$, liquid level falls upon heating.
- If $\gamma_{\text{real}} = \gamma_{\text{vessel}}$, liquid level remains invariant regardless of temperature.

4. Density Variation with Temperature: Let mass M occupy volume V_0 at T_0 ($\rho_0 = M/V_0$). At temperature $T_0 + \Delta T$:

$$\rho(T) = \frac{M}{V(T)} = \frac{M}{V_0(1 + \gamma\Delta T)} = \rho_0(1 + \gamma\Delta T)^{-1} \approx \rho_{\mathbf{0}}(\mathbf{1} - \gamma\boldsymbol{\Delta}\mathbf{T}) \tag{3.9}$$

Thermal Stress in Rigidly Clamped Rods & Bimetallic Strips

1. Thermal Stress Formulation: When a rod of length L_0, cross-sectional area A, and Young's modulus Y is clamped rigidly between two immovable walls and heated by ΔT:

- Free thermal expansion prevented: $\Delta L = L_0\alpha\Delta T$.
- Compressive thermal strain induced: $\varepsilon_{\text{thermal}} = \frac{\Delta L}{L_0} = \alpha\Delta T$.
- Compressive stress developed:

$$\sigma_{\text{th}} = \frac{\mathbf{F}}{\mathbf{A}} = \mathbf{Y}\varepsilon_{\text{thermal}} = \mathbf{Y}\alpha\boldsymbol{\Delta}\mathbf{T} \tag{3.10}$$

- Thermal force exerted on clamps: $\mathbf{F} = \mathbf{YA}\alpha\boldsymbol{\Delta}\mathbf{T}$. Note that σ_{th} is completely independent of the rod length L_0.

2. Bimetallic Strip Curvature: Two strips of equal thickness d, made of metals with expansion coefficients $\alpha_1 > \alpha_2$, are riveted together. Upon heating,

metal 1 expands more than metal 2, forcing the composite strip into an arc of radius R:

$$R \approx \frac{d}{(\alpha_1 - \alpha_2)\Delta T} \tag{3.11}$$

The strip bends with the higher-α metal on the convex outer side. Upon cooling ($\Delta T < 0$), it bends in the reverse direction (higher-α metal on the concave inner side).

Calorimetry, Specific Heat Capacity & Latent Heat

1. Heat Capacities:

- **Specific Heat Capacity (s or c):** Heat required per unit mass to raise temperature by 1 K:

$$dQ = msdT \implies Q = m\int_{T_1}^{T_2} s(T)dT = ms\Delta T \quad (\text{if } s = \text{const}) \tag{3.12}$$

  For water: $s_w = 1.0\,\text{cal/(g}\cdot^\circ\text{C)} = 4186\,\text{J/(kg}\cdot\text{K)}$. For ice: $s_{\text{ice}} \approx 0.5\,\text{cal/(g}\cdot^\circ\text{C)} = 2100\,\text{J/(kg}\cdot\text{K)}$.
- **Molar Heat Capacity (C):** $C = \frac{1}{n}\frac{dQ}{dT} = M_{\text{molar}}s$.
- **Water Equivalent of Calorimeter (W):** Mass of water that absorbs/releases the same heat as the calorimeter for an identical temperature change:

$$W = m_{\text{cal}}\frac{s_{\text{cal}}}{s_w} \implies Q = (m_w + W)s_w\Delta T \tag{3.13}$$

2. Latent Heat & Phase Changes (Isothermal Heat Addition):

$$Q = mL \tag{3.14}$$

- **Latent Heat of Fusion of Ice (L_f):** $L_f \approx 80\,\text{cal/g} = 3.33 \times 10^5\,\text{J/kg}$ at 0°C.
- **Latent Heat of Vaporization of Water (L_v):** $L_v \approx 540\,\text{cal/g} = 2.26 \times 10^6\,\text{J/kg}$ at 100°C.

3. Principle of Calorimetry (Conservation of Energy): In a thermally isolated calorimeter:

$$\sum \mathbf{Q}_{\text{lost by hot bodies}} = \sum \mathbf{Q}_{\text{gained by cold bodies}} \tag{3.15}$$

Calorimetry Pitfalls: Ice-Water-Steam Equilibrium Traps

Common Error: Blindly applying $m_1 s_1(T_1 - T_f) = m_2 s_2(T_f - T_2)$ without verifying whether sufficient heat exists to melt all ice or condense all steam.

- **Ice + Water Scenario:** Always calculate $Q_{\text{available}}$ from water cooling to 0°C vs. Q_{melt} required to melt all ice at 0°C. If $Q_{\text{available}} < m_{\text{ice}}L_f$, the final

mixture temperature is **strictly** 0°C, and only a fraction $m' = Q_{\text{available}}/L_f$ of ice melts.

- **Steam + Water Scenario:** Always verify if water can absorb all latent heat of steam (540 cal/g) before heating to 100°C. If not, final equilibrium temperature remains **strictly** 100°C.

3.2 Modes of Heat Transfer & Radiation Laws

Thermal Conduction: Fourier's Law & Composite Slabs

1. Fourier's Law of Steady-State Conduction: Thermal energy flowing per unit time through a cross-sectional area A normal to a temperature gradient dT/dx:

$$\mathbf{H = \frac{dQ}{dt} = -kA\frac{dT}{dx}} \tag{3.16}$$

where k is thermal conductivity ($\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$). For a uniform rod of length L with ends maintained at T_H and T_C:

$$H = \frac{kA(T_H - T_C)}{L} = \frac{T_H - T_C}{R_{\text{th}}}, \quad \text{where } \mathbf{R_{\text{th}} = \frac{L}{kA}} \quad [\text{K} \cdot \text{W}^{-1}] \tag{3.17}$$

R_{th} is the **thermal resistance**, directly analogous to electrical resistance $R_{\text{elec}} = \frac{L}{\sigma A}$.

Series & Parallel Thermal Conductors: Derivations

1. Series Combination of Two Conducting Slabs: Consider two slabs of lengths L_1, L_2, areas A, and conductivities k_1, k_2 placed in end-to-end contact between temperatures T_1 and T_2 ($T_1 > T_2$). Let T_0 be the interface temperature. Under steady state, the heat current H is identical through both:

$$H = \frac{k_1 A(T_1 - T_0)}{L_1} = \frac{k_2 A(T_0 - T_2)}{L_2} \implies R_{\text{eq}} = R_1 + R_2 = \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} \tag{3.18}$$

Equating expressions to solve for the interface temperature T_0:

$$\mathbf{T_0 = \frac{\frac{k_1}{L_1}T_1 + \frac{k_2}{L_2}T_2}{\frac{k_1}{L_1} + \frac{k_2}{L_2}} = \frac{k_1 L_2 T_1 + k_2 L_1 T_2}{k_1 L_2 + k_2 L_1}} \tag{3.19}$$

Equivalent thermal conductivity k_{eq} for total length $L_1 + L_2$:

$$\frac{L_1 + L_2}{k_{\text{eq}} A} = \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} \implies \mathbf{k_{\text{eq}} = \frac{L_1 + L_2}{\frac{L_1}{k_1} + \frac{L_2}{k_2}}} \quad \left(\text{If } L_1 = L_2 : \ k_{\text{eq}} = \frac{2k_1 k_2}{k_1 + k_2}\right) \tag{3.20}$$

2. Parallel Combination of Two Conducting Slabs: Plates of areas A_1, A_2 and common length L connected between reservoirs at T_1 and T_2:

$$H_{\text{net}} = H_1 + H_2 \implies \frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} \implies \mathbf{k_{eq}} = \frac{\mathbf{k_1 A_1 + k_2 A_2}}{\mathbf{A_1 + A_2}} \tag{3.21}$$

Thermal Radiation: Stefan-Boltzmann & Wien's Laws

1. Blackbody Radiation & Prevost's Theory: Prevost's theory asserts that all bodies at temperatures $T > 0\,\text{K}$ emit thermal radiation continuously to, and absorb radiation from, their surroundings. Thermal equilibrium represents dynamic parity between emission and absorption.

- **Stefan-Boltzmann Law:** Total radiant energy emitted per unit surface area per second by a blackbody:

$$E_b = \sigma T^4 \tag{3.22}$$

  where $\sigma = 5.670 \times 10^{-8}\,\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$ is the Stefan-Boltzmann constant.
- For a non-blackbody of surface area A and emissivity e $(0 < e < 1)$:

$$P_{\text{emit}} = e\sigma A T^4 \tag{3.23}$$

- **Net Radiative Heat Transfer with Surroundings at T_0:**

$$\mathbf{P_{net}} = \frac{\mathbf{dQ}}{\mathbf{dt}} = \mathbf{e\sigma A(T^4 - T_0^4)} \tag{3.24}$$

- **Wien's Displacement Law:** The wavelength λ_m corresponding to maximum spectral emissive power is inversely proportional to absolute temperature:

$$\lambda_{\mathbf{m}}\mathbf{T} = \mathbf{b} = \mathbf{2.898 \times 10^{-3}}\,\text{m} \cdot \text{K} \tag{3.25}$$

  As temperature increases, peak emission shifts towards shorter wavelengths (higher frequencies).

Newton's Law of Cooling: Derivation & Analytical Form

1. Calculus Derivation from Stefan-Boltzmann Law: Consider a body of mass m, specific heat capacity s, and surface area A at temperature T cooling in surroundings at temperature T_0, where $\Delta T = T - T_0 \ll T_0$.

$$T^4 - T_0^4 = (T_0 + \Delta T)^4 - T_0^4 = T_0^4 \left[\left(1 + \frac{\Delta T}{T_0}\right)^4 - 1 \right] \tag{3.26}$$

Using the binomial expansion $(1 + x)^4 \approx 1 + 4x$ for $x = \Delta T/T_0 \ll 1$:

$$T^4 - T_0^4 \approx T_0^4 \left(1 + 4\frac{\Delta T}{T_0} - 1\right) = 4T_0^3(T - T_0) \tag{3.27}$$

Substituting this into the net radiant heat loss rate:

$$\frac{dQ}{dt} = -ms\frac{dT}{dt} = e\sigma A \cdot 4T_0^3(T - T_0) \implies \boxed{\frac{\mathbf{dT}}{\mathbf{dt}} = -\mathbf{k}(\mathbf{T} - \mathbf{T_0})} \tag{3.28}$$

where cooling constant $k = \frac{4e\sigma AT_0^3}{ms}$. **2. Exact Exponential Solution:**

$$\int_{T_i}^{T(t)} \frac{dT}{T - T_0} = -k\int_0^t dt \implies \boxed{\mathbf{T(t) - T_0 = (T_i - T_0)e^{-kt}}} \tag{3.29}$$

3. Practical Average Approximation Formula for Problems: If a body cools from T_1 to T_2 in time t:

$$\boxed{\frac{\mathbf{T_1 - T_2}}{\mathbf{t}} = \mathbf{K}\left(\frac{\mathbf{T_1 + T_2}}{\mathbf{2}} - \mathbf{T_0}\right)} \tag{3.30}$$

NCERT Extraction: Anomalous Expansion of Water

Water exhibits anomalous expansion between 0°C and 4°C. When water is heated from 0°C, its volume **decreases** and its density **increases**, reaching a maximum density of $\rho_{\max} \approx 1.000\,\mathrm{g/cm^3}$ at 3.98°C ≈ 4°C. Above 4°C, water expands normally with temperature.

- *Ecological Consequence:* In freezing winters, surface water cools to 4°C and sinks. Once the entire body reaches 4°C, further cooling drops surface water to 0°C where it freezes into ice ($\rho_{\text{ice}} < \rho_{\text{water}}$). The ice layer floats on top, acting as a thermal insulator ($k_{\text{ice}} \approx 2.2\,\mathrm{W/m \cdot K}$), preserving liquid water at 4°C beneath and sustaining aquatic life.

3.3 Thermodynamics: First Law, Processes & Mayer's Relation

Thermodynamic Systems, State Functions & First Law

1. Thermodynamic Equilibrium & Zeroth Law: Two systems in mutual thermal equilibrium with a third system are in thermal equilibrium with each other. This establishes temperature as the unique scalar state property governing thermal equilibrium. **2. Internal Energy (U):** Internal energy is a unique thermodynamic **state function** reflecting the sum of molecular kinetic and intermolecular potential energies. For an ideal gas (zero intermolecular potential):

$$U = U(T) = nC_vT \implies \boxed{\mathbf{dU = nC_vdT}} \tag{3.31}$$

Across any cyclic process, $\Delta U_{\text{cycle}} = \oint dU = 0$. **3. First Law of Thermodynamics (Energy Conservation):**

$$\boxed{\mathbf{\Delta Q = \Delta U + W}} \iff \mathbf{dQ = dU + dW} = nC_vdT + PdV \tag{3.32}$$

Physics Sign Conventions:

- $Q > 0$ when heat is absorbed by the system; $Q < 0$ when heat is released.
- $W = \int PdV > 0$ when work is done **by** the system (expansion, $dV > 0$); $W < 0$ when work is done **on** the system (compression, $dV < 0$).

Work Done in Master Thermodynamic Processes: Step-by-Step Derivations

1. Isochoric Process (Constant Volume, $V = \text{const}$):

$$dV = 0 \implies \mathbf{W = 0}, \quad \mathbf{\Delta Q = \Delta U = nC_v \Delta T} \tag{3.33}$$

2. Isobaric Process (Constant Pressure, $P = \text{const}$):

$$\mathbf{W = P\Delta V = P(V_2 - V_1) = nR(T_2 - T_1)} \tag{3.34}$$

$$\Delta Q = nC_p\Delta T, \quad \Delta U = nC_v\Delta T \tag{3.35}$$

$$\implies \mathbf{\Delta U : W : \Delta Q = C_v : R : C_p = 1 : (\gamma - 1) : \gamma} \tag{3.36}$$

3. Isothermal Process (Constant Temperature, $T = \text{const}$): Since $T = \text{const}$, $\Delta U = 0 \implies \Delta Q = W$. Using $P = \frac{nRT}{V}$:

$$W = \int_{V_1}^{V_2} P\,dV = nRT \int_{V_1}^{V_2} \frac{dV}{V} = nRT \ln\left(\frac{V_2}{V_1}\right) \tag{3.37}$$

$$= nRT \ln\left(\frac{P_1}{P_2}\right) = 2.303\, nRT \log_{10}\left(\frac{V_2}{V_1}\right) \tag{3.38}$$

4. Adiabatic Process (Zero Heat Exchange, $\Delta Q = 0$): Governing equations: $PV^\gamma = \text{const}$, $TV^{\gamma-1} = \text{const}$, $P^{1-\gamma}T^\gamma = \text{const}$. From First Law: $dQ = 0 \implies dW = -dU = -nC_v dT$.

$$W = -\int_{T_1}^{T_2} nC_v dT = -nC_v(T_2 - T_1) = n\left(\frac{R}{\gamma - 1}\right)(T_1 - T_2) \tag{3.39}$$

Using $nRT_1 = P_1V_1$ and $nRT_2 = P_2V_2$:

$$\mathbf{W = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1} = -\Delta U} \tag{3.40}$$

5. General Polytropic Process ($PV^m = \text{const}$):

$$W = \frac{P_1V_1 - P_2V_2}{m - 1}, \qquad \mathbf{C = C_v + \frac{R}{1 - m} = \frac{R}{\gamma - 1} + \frac{R}{1 - m}} \tag{3.41}$$

Mayer's Relation Derivation: $C_p - C_v = R$

Consider 1 mole of an ideal gas ($n = 1$).

1. Heat added at constant volume: $dQ_v = C_v dT$. Since $dV = 0$, $dW = 0$, so by First Law:
$$dU = dQ_v = C_v dT \tag{3.42}$$
Because internal energy of an ideal gas depends solely on temperature, $dU = C_v dT$ holds across **any** process.
2. Heat added at constant pressure: $dQ_p = C_p dT$. By First Law:
$$dQ_p = dU + dW = C_v dT + P dV \tag{3.43}$$
3. Differentiating the ideal gas equation of state $PV = RT$ at constant pressure P:
$$P dV = R dT \tag{3.44}$$
4. Substituting $PdV = RdT$ into the First Law expression:
$$C_p dT = C_v dT + R dT \implies (C_p - C_v)dT = R dT \implies \mathbf{C_p - C_v = R} \tag{3.45}$$

In terms of specific heats (per unit mass): $c_p - c_v = \frac{R}{M_{\text{molar}}}$.

3.3.1 $P-V$ Indicator Diagrams: Isothermal vs. Adiabatic Slopes & Expansion Work

An indicator diagram represents thermodynamic transformations on a coordinate plane of state variables, most commonly pressure P versus volume V. Because work done during any quasi-static process is given by the integral $W = \int P\, dV$, the area beneath the trajectory curve on a $P-V$ diagram measures the mechanical work exchanged with the surroundings.

Mathematical Derivation of Trajectory Slopes: The slope of a process at any state (P, V) on the $P-V$ plane is given by the derivative $\frac{dP}{dV}$:

- **Isothermal Trajectory (T = constant):** For an ideal gas obeying Boyle's law $PV =$ constant:
$$P\, dV + V\, dP = 0 \implies \left(\frac{\mathbf{dP}}{\mathbf{dV}}\right)_{\text{isothermal}} = -\frac{\mathbf{P}}{\mathbf{V}} \tag{3.46}$$
- **Adiabatic Trajectory ($dQ = 0$):** For a reversible adiabatic process obeying Poisson's relation $PV^\gamma =$ constant:
$$V^\gamma\, dP + \gamma P V^{\gamma-1}\, dV = 0 \implies \left(\frac{\mathbf{dP}}{\mathbf{dV}}\right)_{\text{adiabatic}} = -\gamma\frac{\mathbf{P}}{\mathbf{V}} \tag{3.47}$$

The Fundamental Slope Theorem: Dividing the adiabatic slope by the isothermal slope yields the cornerstone theorem of thermodynamic indicator curves:

$$\left(\frac{\mathbf{dP}}{\mathbf{dV}}\right)_{\text{adiabatic}} = \gamma \left(\frac{\mathbf{dP}}{\mathbf{dV}}\right)_{\text{isothermal}} \tag{3.48}$$

Since the adiabatic index $\gamma = \frac{C_p}{C_v} > 1$ for all gases ($\gamma \approx 1.67$ for monoatomic, 1.40 for diatomic), the adiabatic curve is **strictly γ times steeper** than the isothermal curve at every point of intersection.

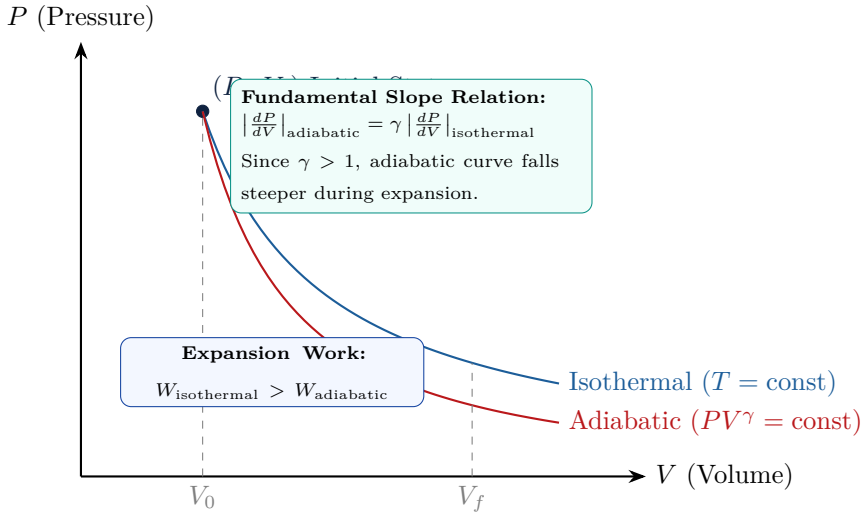


Figure 3.1: $P - V$ Indicator Diagram: Comparison of Isothermal vs. Adiabatic Expansion from Identical Initial State.

Work Comparison & Cyclic Process Conventions

1. Expansion from Common Initial State (P_0, V_0) to Final Volume $V_f > V_0$: Because the isobaric curve remains at P_0, the isothermal curve falls as V^{-1}, and the adiabatic curve plunges as $V^{-\gamma}$:

$$P_{\text{isobaric}} > P_{\text{isothermal}} > P_{\text{adiabatic}} \implies \mathbf{W}_{\text{isobaric}} > \mathbf{W}_{\text{isothermal}} > \mathbf{W}_{\text{adiabatic}} \tag{3.49}$$

2. Compression from Common Initial State (P_0, V_0) to Final Volume $V_f < V_0$: Work done **on** the gas is largest for adiabatic compression:

$$|W_{\text{adiabatic}}| > |W_{\text{isothermal}}| > |W_{\text{isobaric}}| \tag{3.50}$$

3. Net Work in Closed Cycles: The net work executed over any closed cycle on a $P - V$ plane equals the area enclosed by the cyclic trajectory:

$$W_{\text{net}} = \oint P\,dV = \begin{cases} > 0 & \text{Clockwise: net work by gas} \\ < 0 & \text{Counter-clockwise: work on system} \end{cases} \tag{3.51}$$

3.4 Second Law of Thermodynamics & Syllabus Boundaries

Second Law of Thermodynamics: Clausius & Kelvin-Planck Formulations

The First Law dictates energy conservation but fails to predict the spontaneous direction of natural processes. The Second Law establishes irreversibility:

- **Kelvin-Planck Statement:** It is impossible to construct a heat engine operating in a thermodynamic cycle that absorbs thermal energy from a single thermal reservoir and converts 100% of this energy into mechanical work without rejecting any heat to a colder reservoir. (No engine can operate at 100% thermal efficiency; perpetual motion machines of the second kind are impossible).
- **Clausius Statement:** It is impossible to construct a cyclic device that transfers heat spontaneously from a cooler body to a hotter body without requiring external mechanical work input.
- **Equivalence:** A violation of the Kelvin-Planck statement directly entails a violation of the Clausius statement, and vice versa.

Syllabus Boundary Alert: Carnot Engine & Refrigerator Deletion

NMC REVISION WARNING FOR NEET UG: Under the updated NEET UG syllabus issued by the National Medical Commission (NMC):

- **DELETED TOPICS:** The theoretical formulation of the **Carnot Engine**, Carnot cycle efficiency ($\eta = 1 - T_C/T_H$), Carnot theorem proofs, and **Refrigerators / Heat Pumps** (coefficient of performance $\beta = Q_C/W = T_C/(T_H - T_C)$) have been **formally removed**.
- **EXAM ORIENTATION:** Questions will **not** test Carnot cycle calculations. Candidates must focus strictly on General Thermodynamic Cycles (triangular, rectangular $P - V$ loops), First Law applications ($\Delta Q = \Delta U + W$), work in isothermal/adiabatic/isobaric expansions, and the conceptual statements of the Second Law.

3.5 Kinetic Theory of Gases (KTG): Molecular Pressure & Equipartition

Postulates of the Kinetic Theory of an Ideal Gas

An ideal gas is modeled micro-mechanically under the following foundational postulates:

1. The gas comprises an enormous number of identical, tiny, hard spherical molecules in ceaseless, chaotic random motion across all directions with all

possible speeds.

2. The physical volume occupied by the gas molecules themselves is negligibly small compared to the total macroscopic volume of the container ($V_{\text{molecules}} \ll V_{\text{container}}$).
3. No intermolecular forces (attractive or repulsive) act between molecules except during collisions. Hence, potential energy is strictly zero; all internal energy is purely translational/rotational kinetic energy.
4. All intermolecular collisions and collisions with container walls are perfectly elastic ($\Delta K = 0$). Collision durations are negligible compared to transit times between collisions.
5. Gravity has negligible effect on molecular trajectory between collisions due to high speeds ($\sim 10^2 - 10^3$ m/s).

First-Principles Derivation of Ideal Gas Pressure: $P = \frac{1}{3}\rho v_{\text{rms}}^2$

Consider N identical gas molecules, each of mass m, enclosed in a cubical box of side L and volume $V = L^3$.

1. Let a molecule possess velocity vector $\mathbf{v} = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} + v_z\hat{\mathbf{k}}$ with speed squared $v^2 = v_x^2 + v_y^2 + v_z^2$.
2. When this molecule collides elastically with the wall perpendicular to the x-axis at $x = L$:

$$p_{x,\text{initial}} = mv_x, \qquad p_{x,\text{final}} = -mv_x \implies \Delta p_x = -2mv_x \tag{3.52}$$

   Momentum imparted to the wall per collision is $+2mv_x$.
3. Time elapsed between consecutive collisions of this molecule with the same wall:

$$\Delta t = \frac{2L}{v_x} \tag{3.53}$$

4. Average force exerted by this single molecule on the wall:

$$F_1 = \frac{\Delta p_{\text{wall}}}{\Delta t} = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L} \tag{3.54}$$

5. Total force exerted by all N molecules on the wall of area $A = L^2$:

$$F_{\text{total}} = \sum_{i=1}^{N} \frac{mv_{xi}^2}{L} = \frac{m}{L} N\overline{v_x^2} \tag{3.55}$$

6. Since molecular motion is isotropic, $\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} = \frac{1}{3}\overline{v^2} = \frac{1}{3}v_{\text{rms}}^2$.
7. Pressure $P = \frac{F_{\text{total}}}{A} = \frac{mN\overline{v_x^2}}{L \cdot L^2} = \frac{1}{3}\frac{Nm}{V}v_{\text{rms}}^2$:

$$\mathbf{P = \frac{1}{3}\rho v_{\text{rms}}^2 = \frac{1}{3}\frac{M_{\text{total}}}{V}v_{\text{rms}}^2} \tag{3.56}$$

Kinetic Interpretation of Temperature & Molecular Speed Spectrum

1. Average Kinetic Energy per Molecule: Rewriting the pressure equation as $PV = \frac{2}{3}\left(N \cdot \frac{1}{2}mv_{\text{rms}}^2\right) = \frac{2}{3}E_{\text{trans}}$. Equating to the ideal gas law $PV = Nk_BT$ (where $k_B = R/N_A = 1.381 \times 10^{-23}\,\text{J/K}$):

$$\frac{2}{3}E_{\text{trans}} = Nk_BT \implies \overline{\mathbf{K}}_{\text{trans}} = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{m}\mathbf{v}_{\text{rms}}^{\mathbf{2}} = \frac{\mathbf{3}}{\mathbf{2}}\mathbf{k_B T} \tag{3.57}$$

Average translational kinetic energy depends **solely on absolute temperature** T, invariant of molecular mass or gas species! **2. Molecular Speed Hierarchy:**

- **Root-Mean-Square Speed (v_{rms}):**

$$\mathbf{v}_{\text{rms}} = \sqrt{\frac{\mathbf{3k_B T}}{\mathbf{m}}} = \sqrt{\frac{\mathbf{3RT}}{\mathbf{M}_{\text{molar}}}} = \sqrt{\frac{\mathbf{3P}}{\rho}} \tag{3.58}$$

- **Average Speed (v_{avg}):**

$$\mathbf{v}_{\text{avg}} = \sqrt{\frac{\mathbf{8k_B T}}{\pi\mathbf{m}}} = \sqrt{\frac{\mathbf{8RT}}{\pi\mathbf{M}_{\text{molar}}}} \approx 0.921 v_{\text{rms}} \tag{3.59}$$

- **Most Probable Speed (v_{mp}):**

$$\mathbf{v}_{\text{mp}} = \sqrt{\frac{\mathbf{2k_B T}}{\mathbf{m}}} = \sqrt{\frac{\mathbf{2RT}}{\mathbf{M}_{\text{molar}}}} \approx 0.816 v_{\text{rms}} \tag{3.60}$$

- **Golden Speed Ratio:**

$$\mathbf{v}_{\text{mp}} : \mathbf{v}_{\text{avg}} : \mathbf{v}_{\text{rms}} = \sqrt{\mathbf{2}} : \sqrt{\frac{\mathbf{8}}{\pi}} : \sqrt{\mathbf{3}} \approx \mathbf{1 : 1.128 : 1.224} \tag{3.61}$$

Law of Equipartition of Energy, Degrees of Freedom & Specific Heat Ratios

1. Equipartition Principle: In thermal equilibrium at temperature T, the total internal energy is partitioned equally among all quadratic independent degrees of freedom (f), each contributing an average energy of $\frac{1}{2}\mathbf{k_B T}$ per molecule:

$$\overline{E}_{\text{molecule}} = \frac{f}{2}k_BT, \qquad U_{\text{molar}} = \frac{f}{2}RT \tag{3.62}$$

$$\implies \mathbf{C_v} = \frac{\mathbf{f}}{\mathbf{2}}\mathbf{R}, \qquad \mathbf{C_p} = \left(\frac{\mathbf{f}}{\mathbf{2}} + \mathbf{1}\right)\mathbf{R}, \qquad \gamma = \mathbf{1} + \frac{\mathbf{2}}{\mathbf{f}} \tag{3.63}$$

2. Comprehensive Gas Classification Table:

Atomicity	DoF (f)	C_v	C_p	γ	Examples
Monoatomic	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} \approx 1.67$	He, Ne, Ar
Diatomic (Rigid)	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.40$	H_2, N_2, O_2
Diatomic (Vibrating)	7	$\frac{7}{2}R$	$\frac{9}{2}R$	$\frac{9}{7} \approx 1.29$	Cl_2, Br_2
Polyatomic (Non-linear)	6	$3R$	$4R$	$\frac{4}{3} \approx 1.33$	H_2O, NH_3

3. Non-Reacting Mixtures of Gases: For a mixture of n_1 moles of gas 1 (C_{v1}, γ_1) and n_2 moles of gas 2 (C_{v2}, γ_2):

$$\mathbf{C_{v,mix}} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}, \qquad \mathbf{C_{p,mix}} = \frac{n_1 C_{p1} + n_2 C_{p2}}{n_1 + n_2} \tag{3.64}$$

$$\gamma_{\text{mix}} = \frac{C_{p,\text{mix}}}{C_{v,\text{mix}}} = \frac{n_1 C_{p1} + n_2 C_{p2}}{n_1 C_{v1} + n_2 C_{v2}} \tag{3.65}$$

4. Mean Free Path (λ): Average distance traversed by a gas molecule between successive collisions:

$$\lambda = \frac{\mathbf{1}}{\sqrt{\mathbf{2}}\pi \mathbf{n_v d^2}} = \frac{\mathbf{k_B T}}{\sqrt{\mathbf{2}}\pi \mathbf{d^2 P}} \tag{3.66}$$

where $n_v = N/V$ is number density, d is molecular diameter. Notice that $\lambda \propto T$ at constant pressure, and $\lambda \propto \frac{1}{P}$ at constant temperature.

3.6 Simple Harmonic Motion (SHM): Kinematics, Energy & Oscillators

Mathematical Foundations of Simple Harmonic Motion

SHM is bounded periodic motion wherein the restoring force/torque is directly proportional to displacement from the stable equilibrium position and directed towards that equilibrium:

$$F(x) = -kx \implies m\frac{d^2x}{dt^2} + kx = 0 \implies \mathbf{\frac{d^2x}{dt^2} + \omega^2 x = 0} \quad \left(\text{where } \omega = \sqrt{\frac{k}{m}}\right) \tag{3.67}$$

General Solution & Kinematic Equations:

- **Displacement:** $x(t) = A\sin(\omega t + \phi_0)$, where A is amplitude, $\omega = 2\pi/T = 2\pi\nu$ is angular frequency, and ϕ_0 is initial epoch.
- **Velocity:**

$$v(t) = \frac{dx}{dt} = A\omega\cos(\omega t + \phi_0) = \pm\omega\sqrt{\mathbf{A^2 - x^2}} \tag{3.68}$$

  $v_{\max} = \omega A$ occurs at mean position ($x = 0$); $v = 0$ at extreme positions ($x = \pm A$).

- **Acceleration:**

$$a(t) = \frac{dv}{dt} = -\omega^2 A \sin(\omega t + \phi_0) = -\boldsymbol{\omega^2 \mathbf{x}} \tag{3.69}$$

$a_{\max} = \omega^2 A$ occurs at extremes ($x = \pm A$); $a = 0$ at mean position ($x = 0$).
- **Phase Relationships:** Velocity leads displacement by $\pi/2$ radians (90°); acceleration leads displacement by π radians (180°).

Energy Conservation in SHM: Space & Time Averages

1. **Instantaneous Energies as Functions of Position (x):**

- **Potential Energy (U):** $U(x) = \int_0^x kx dx = \boldsymbol{\frac{1}{2}\mathbf{k}\mathbf{x}^2} = \boldsymbol{\frac{1}{2}\mathbf{m}\omega^2\mathbf{x}^2}$ (assuming $U(0) = 0$).
- **Kinetic Energy (K):** $K(x) = \frac{1}{2}mv^2 = \boldsymbol{\frac{1}{2}\mathbf{m}\omega^2(\mathbf{A}^2 - \mathbf{x}^2)}$.
- **Total Mechanical Energy (E):**

$$\mathbf{E} = \mathbf{K}(\mathbf{x}) + \mathbf{U}(\mathbf{x}) = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{m}\omega^2\mathbf{A}^2 = \mathbf{2}\pi^2\mathbf{m}\nu^2\mathbf{A}^2 = \text{constant} \tag{3.70}$$

Total energy is completely independent of time t and position x.

2. **Position where $K = U$:**

$$\frac{1}{2}m\omega^2(A^2 - x^2) = \frac{1}{2}m\omega^2 x^2 \implies 2x^2 = A^2 \implies \mathbf{x} = \pm\frac{\mathbf{A}}{\sqrt{\mathbf{2}}} \tag{3.71}$$

3. **Space Averages vs. Time Averages over One Cycle:**

- **Time Averages ($\langle \dots \rangle_t = \frac{1}{T}\int_0^T \dots dt$):**

$$\langle \mathbf{K} \rangle_\mathbf{t} = \frac{\mathbf{1}}{\mathbf{4}}\mathbf{m}\omega^2\mathbf{A}^2 = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{E}, \qquad \langle \mathbf{U} \rangle_\mathbf{t} = \frac{\mathbf{1}}{\mathbf{4}}\mathbf{m}\omega^2\mathbf{A}^2 = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{E} \tag{3.72}$$

- **Space Averages ($\langle \dots \rangle_x = \frac{1}{2A}\int_{-A}^{A} \dots dx$):**

$$\langle \mathbf{K} \rangle_\mathbf{x} = \frac{\mathbf{2}}{\mathbf{3}}\mathbf{E}, \qquad \langle \mathbf{U} \rangle_\mathbf{x} = \frac{\mathbf{1}}{\mathbf{3}}\mathbf{E} \tag{3.73}$$

- **Frequency of Oscillations:** If displacement oscillates with frequency ν, both kinetic energy $K(t)$ and potential energy $U(t)$ oscillate with **double frequency** 2ν.

Mechanical Oscillators: Spring-Mass Systems & Simple Pendulums

1. **Spring-Block Oscillators:**

$$T = 2\pi\sqrt{\frac{m}{k}} \tag{3.74}$$

- **Series Combination:** $\frac{1}{k_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2} \implies k_{\text{eq}} = \frac{k_1 k_2}{k_1+k_2} \implies T = 2\pi\sqrt{\frac{m(k_1+k_2)}{k_1 k_2}}$.
- **Parallel Combination:** $k_{\text{eq}} = k_1 + k_2 \implies T = 2\pi\sqrt{\frac{m}{k_1+k_2}}$.
- **Spring Cutting:** Spring constant is inversely proportional to length: $k \cdot L = \text{const}$. If cut into length ratio $m : n$, spring constants become $k_1 = \frac{m+n}{m}k$ and $k_2 = \frac{m+n}{n}k$.
- **Two-Body Oscillator Connected by Spring:** Reduced mass $\mu = \frac{m_1 m_2}{m_1+m_2} \implies \mathbf{T = 2\pi\sqrt{\frac{\mu}{k}}}$.

2. Simple Pendulum Derivation: A bob of mass m suspended by a light string of length L. For angular displacement θ:

$$\tau_{\text{restoring}} = -mgL\sin\theta \approx -mgL\theta \quad \text{(for paraxial small angles } \theta \ll 1) \tag{3.75}$$

Using $\tau = I\alpha = (mL^2)\frac{d^2\theta}{dt^2}$:

$$mL^2\frac{d^2\theta}{dt^2} + mgL\theta = 0 \implies \frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0 \implies \omega = \sqrt{\frac{\mathbf{g}}{\mathbf{L}}}, \quad \mathbf{T = 2\pi\sqrt{\frac{L}{g}}} \tag{3.76}$$

Accelerating Frames & Effective Gravity:

- Elevator accelerating upwards with a: $g_{\text{eff}} = g + a \implies T = 2\pi\sqrt{\frac{L}{g+a}}$ (period decreases).
- Elevator accelerating downwards with a: $g_{\text{eff}} = g - a \implies T = 2\pi\sqrt{\frac{L}{g-a}}$ (period increases; free fall $a = g \implies T \to \infty$).
- Horizontally accelerating vehicle: $g_{\text{eff}} = \sqrt{g^2+a^2} \implies T = 2\pi\sqrt{\frac{L}{\sqrt{g^2+a^2}}}$.
- Pendulum in liquid of density σ (bob density ρ): $g_{\text{eff}} = g\left(1 - \frac{\sigma}{\rho}\right) \implies T = 2\pi\sqrt{\frac{L}{g(1-\sigma/\rho)}}$.

3. Seconds Pendulum: A pendulum whose time period is exactly $T = 2.0\,\text{s}$ (1 s tick each half-swing):

$$L = \frac{gT^2}{4\pi^2} = \frac{9.8 \times 4}{4\pi^2} \approx \mathbf{0.993}\,\text{m} \approx \mathbf{1.0}\,\text{m} \tag{3.77}$$

Phase Circle (Phasor) Technique for Instantaneous Transit Times

To compute the transit time between any two positions x_1 and x_2 in SHM, project the motion onto a reference circle of radius A rotating counterclockwise at angular velocity ω:

$$x = A\sin\theta, \quad \theta = \omega t \implies t = \frac{\Delta\theta}{\omega} = \frac{\Delta\theta}{2\pi}T \tag{3.78}$$

High-Yield Benchmarks (Starting from Mean Position $x = 0$):

- From $x = 0$ to $x = \frac{A}{2}$: $\theta_1 = 0 \to \theta_2 = 30^\circ = \frac{\pi}{6} \implies \mathbf{t = \frac{T}{12}}$.
- From $x = \frac{A}{2}$ to $x = A$: $\Delta\theta = 90^\circ - 30^\circ = 60^\circ = \frac{\pi}{3} \implies \mathbf{t = \frac{T}{6}}$.
- Notice: Traveling the outer half of amplitude takes **twice as long** as the inner half ($T/6$ vs $T/12$) due to deceleration.
- From $x = 0$ to $x = \frac{A}{\sqrt{2}}$: $\theta = 45^\circ = \frac{\pi}{4} \implies \mathbf{t = \frac{T}{8}}$.

3.7 Wave Motion, Stretched Strings & Organ Pipes

Kinematics of 1D Harmonic Progressive Waves

A 1D transverse or longitudinal sinusoidal traveling wave propagating along the $+x$ direction:

$$\mathbf{y(x,t) = A}\sin(\mathbf{kx - \omega t} + \phi) = \mathbf{A}\sin\left[\mathbf{2\pi}\left(\frac{\mathbf{x}}{\lambda} - \nu\mathbf{t}\right) + \phi\right] \tag{3.79}$$

where $k = \frac{2\pi}{\lambda}$ is the angular wave number, $\omega = 2\pi\nu$ is angular frequency.

- **Wave Speed (v):** Speed of propagation of constant phase profile:

$$kx - \omega t = \text{const} \implies k\frac{dx}{dt} - \omega = 0 \implies \mathbf{v} = \frac{\omega}{\mathbf{k}} = \nu\lambda \tag{3.80}$$

- **Particle Velocity (v_p) vs. Wave Velocity (v):**

$$v_p = \frac{\partial y}{\partial t} = -\omega A\cos(kx - \omega t), \qquad \frac{\partial y}{\partial x} = kA\cos(kx - \omega t) \tag{3.81}$$

$$\mathbf{v_p} = -\mathbf{v}\left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}}\right) = -\mathbf{v} \times (\text{Wave profile slope}) \tag{3.82}$$

  If wave moves right ($v > 0$) and slope is positive ($\partial y/\partial x > 0$), particle moves downwards ($v_p < 0$).

- **Particle Acceleration:** $a_p = \frac{\partial^2 y}{\partial t^2} = -\omega^2 y$.

Speeds of Transverse Waves on Strings & Sound in Gases

1. Transverse Wave on Stretched String:

$$\mathbf{v} = \sqrt{\frac{\mathbf{T}}{\mu}} \tag{3.83}$$

where T is tension (N), and $\mu = \frac{M}{L} = \rho A$ is linear mass density (kg/m). *Derivation via Centripetal Equilibrium of Arc Element:* Consider a small curved element of string of length $\Delta l = R\Delta\theta$ moving with wave speed v along a circular

crest of radius R. Mass of element $\Delta m = \mu \Delta l = \mu R \Delta\theta$. The inward restoring component of tension:

$$F_{\text{inward}} = 2T \sin\left(\frac{\Delta\theta}{2}\right) \approx T\Delta\theta \tag{3.84}$$

Equating to centripetal force:

$$T\Delta\theta = \frac{\Delta m \cdot v^2}{R} = \frac{(\mu R \Delta\theta)v^2}{R} = \mu v^2 \Delta\theta \implies \mathbf{v = \sqrt{\frac{T}{\mu}}} \tag{3.85}$$

2. Speed of Longitudinal Waves (Sound) in Fluids & Gases:

- General relation: $v = \sqrt{\frac{B}{\rho}}$, where B is bulk modulus of elasticity.
- **Newton's Isothermal Formula (Erroneous):** Newton assumed sound compressions/rarefactions occur isothermally ($B = P$).

$$v_{\text{Newton}} = \sqrt{\frac{P}{\rho}} \approx \sqrt{\frac{1.013 \times 10^5}{1.293}} \approx 280\,\text{m/s} \tag{3.86}$$
$$\approx 15\% \text{ below experimental value } (332\,\text{m/s})$$

- **Laplace's Adiabatic Correction (Accurate):** Compressions and rarefactions occur so rapidly that heat exchange is zero ($dQ = 0 \implies B_{\text{adiabatic}} = \gamma P$):

$$\mathbf{v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M_{\text{molar}}}}} \tag{3.87}$$

  For air ($\gamma \approx 1.4, M \approx 28.8 \times 10^{-3}\,\text{kg/mol}$ at $273\,\text{K}$): $v = 332\,\text{m/s}$ (perfect agreement).
- **Dependence on Environmental Factors:**
  - *Pressure:* Invariant of pressure at constant temperature (since $P/\rho = RT/M = \text{const}$).
  - *Temperature:* $v \propto \sqrt{T} \implies v_t \approx v_0 + 0.61t$ (for small temperatures $t°\text{C}$).
  - *Humidity:* Moist air is less dense than dry air ($\rho_{\text{moist}} < \rho_{\text{dry}}$), so speed of sound **increases** with humidity.

Standing Waves: Strings & Organ Pipes with End Corrections

1. Superposition & Standing Wave Equation on String Fixed at Both Ends: Two identical waves traveling in opposite directions: $y_1 = A\sin(kx - \omega t)$ and $y_2 = A\sin(kx + \omega t)$.

$$y_{\text{net}} = y_1 + y_2 = [\mathbf{2A}\sin(\mathbf{kx})]\cos(\omega \mathbf{t}) \tag{3.88}$$

- **Nodes ($y = 0$):** $\sin(kx) = 0 \implies kx = n\pi \implies \mathbf{x = \frac{n\lambda}{2}}$ $(n = 0, 1, 2, \dots)$.

- **Antinodes ($|y| = 2A$):** $|\sin(kx)| = 1 \implies \mathbf{x = (2n-1)\frac{\lambda}{4}}$.
- Boundary condition: nodes at $x = 0$ and $x = L \implies L = n\frac{\lambda}{2} \implies \nu_{\mathbf{n}} = \mathbf{n}\left(\frac{\mathbf{v}}{\mathbf{2L}}\right) = \mathbf{n}\nu_{\mathbf{1}}$ (**All harmonics present**: $1:2:3:\dots$).

2. Closed Organ Pipe (One End Closed, One End Open): Boundary condition: Node at closed end, Antinode at open end. Length $L = (2n-1)\frac{\lambda}{4}$:

$$\nu_{\mathbf{n}} = (\mathbf{2n-1})\frac{\mathbf{v}}{\mathbf{4L}} = (\mathbf{2n-1})\nu_{\mathbf{1}} \qquad \text{where fundamental } \nu_1 = \frac{v}{4L} \tag{3.89}$$

Only odd harmonics exist in a closed organ pipe ($\nu_1 : \nu_3 : \nu_5 = 1:3:5:\dots$). No even harmonics. **3. Open Organ Pipe (Both Ends Open):** Boundary condition: Antinodes at both open ends. Length $L = n\frac{\lambda}{2}$:

$$\nu_{\mathbf{n}} = \mathbf{n}\frac{\mathbf{v}}{\mathbf{2L}} = \mathbf{n}\nu_{\mathbf{1}} \qquad \text{where fundamental } \nu_1 = \frac{v}{2L} \tag{3.90}$$

All harmonics exist ($\nu_1 : \nu_2 : \nu_3 = 1:2:3:\dots$). Tone is richer/sweeter than closed pipe. **4. End Correction (e):** Antinode forms slightly outside the physical open tube boundary due to finite acoustic inertia of moving air:

$$e \approx 0.6r \quad \text{(where } r \text{ is tube inner radius)} \tag{3.91}$$

- For Closed Pipe: $L_{\text{eff}} = L + e = L + 0.6r$.
- For Open Pipe: $L_{\text{eff}} = L + 2e = L + 1.2r$.
- **Resonance Tube Method to eliminate e:** Using 1st resonance length l_1 and 2nd resonance length l_2 for a tuning fork of frequency ν:

$$l_1 + e = \frac{\lambda}{4}, \qquad l_2 + e = \frac{3\lambda}{4} \implies l_2 - l_1 = \frac{\lambda}{2} \implies \lambda = \mathbf{2(l_2 - l_1)} \tag{3.92}$$

$$\mathbf{v} = \mathbf{2}\nu(\mathbf{l_2 - l_1}), \qquad \mathbf{e} = \frac{\mathbf{l_2 - 3l_1}}{\mathbf{2}} \tag{3.93}$$

Acoustic Beats: Analytical Derivation

Consider superposition of two harmonic sound waves of equal amplitude A and slightly differing frequencies ν_1, ν_2 ($\omega_1 \approx \omega_2$):

$$y_1 = A\sin(\omega_1 t), \qquad y_2 = A\sin(\omega_2 t) \tag{3.94}$$

$$y_{\text{net}} = y_1 + y_2 = 2A\cos\left(\frac{\omega_1 - \omega_2}{2}t\right)\sin\left(\frac{\omega_1 + \omega_2}{2}t\right) = A_{\text{mod}}(t)\sin(\omega_{\text{avg}}t) \tag{3.95}$$

The modulated amplitude is $A_{\text{mod}}(t) = 2A\cos(\pi(\nu_1 - \nu_2)t)$. Intensity $I \propto A_{\text{mod}}^2 \propto \cos^2(\pi(\nu_1 - \nu_2)t) = \frac{1+\cos(2\pi(\nu_1-\nu_2)t)}{2}$. Maximum intensity occurs whenever $2\pi(\nu_1 - \nu_2)t = n\pi \implies t = \frac{n}{|\nu_1 - \nu_2|}$. Time interval between consecutive waxing/waning: $\Delta t = \frac{1}{|\nu_1 - \nu_2|}$.

$$\nu_{\text{beat}} = |\nu_{\mathbf{1}} - \nu_{\mathbf{2}}| \tag{3.96}$$

Tuning Fork Rules:

- Loading fork with wax increases inertia $\implies \nu$ decreases.
- Filing fork prongs decreases mass $\implies \nu$ increases.

Syllabus Boundary Alert: Complete Deletion of Doppler Effect

NMC REVISION WARNING FOR NEET UG: Under the revised NMC curriculum for NEET UG 2027:

- **STRICTLY DELETED:** The **Doppler Effect in Sound Waves** (apparent frequency shifts $f' = f\frac{v \pm v_0}{v \mp v_s}$, moving observers, moving sources, echo reflection from cliffs, acoustic shockwaves) has been **completely eliminated** from the syllabus.
- **EXAM ORIENTATION:** Candidates must not solve Doppler frequency formulas. Instead, master Standing Waves, Harmonics in Organ Pipes, Resonance Tube end corrections, and Beats.

3.8 Master High-Yield Problem Archetypes & Worked Solutions

Archetype 1: Ice-Water-Steam Calorimetry Mixing Equilibrium

Problem: 10 g of ice at $-10°$C is mixed with 10 g of steam at 100°C in an insulated calorimeter of negligible heat capacity. Determine the final equilibrium temperature and the constitution of the mixture. **Step-by-Step Solution:**

1. Heat required to bring ice from $-10°$C to liquid water at 100°C:
   - Warm ice to 0°C: $Q_1 = m_{\text{ice}} s_{\text{ice}} \Delta T = 10 \times 0.5 \times 10 = 50$ cal.
   - Melt ice at 0°C: $Q_2 = m_{\text{ice}} L_f = 10 \times 80 = 800$ cal.
   - Warm water to 100°C: $Q_3 = m_{\text{ice}} s_w \Delta T = 10 \times 1.0 \times 100 = 1000$ cal.
   - Total heat absorbed to reach 100°C water: $Q_{\text{absorb}} = 50 + 800 + 1000 = \mathbf{1850}$ cal.
2. Total latent heat available from steam condensing at 100°C:

$$Q_{\text{condense, total}} = m_{\text{steam}} L_v = 10 \times 540 = \mathbf{5400}\,\text{cal} \tag{3.97}$$

3. Notice that $Q_{\text{condense, total}} = 5400$ cal $> Q_{\text{absorb}} = 1850$ cal. Therefore, not all steam condenses! The mixture reaches dynamic equilibrium at $T_{\mathbf{eq}} = 100°$C.
4. Mass of steam condensed to supply 1850 cal:

$$m_{\text{condensed}} = \frac{1850\,\text{cal}}{540\,\text{cal/g}} \approx 3.43\,\text{g} \tag{3.98}$$

5. Surviving steam: $m_{\text{steam, remaining}} = 10 - 3.43 = \mathbf{6.57}$ g.
6. Total water at 100°C: $m_{\text{water}} = 10 + 3.43 = \mathbf{13.43}$ g.

Archetype 2: Mixture of Distinct Polyatomic & Monoatomic Gases

Problem: 2 moles of Helium (monoatomic, $\gamma_1 = 5/3$) are mixed with 3 moles of Oxygen (diatomic, rigid, $\gamma_2 = 7/5$). Calculate $C_{v,\text{mix}}$, $C_{p,\text{mix}}$, and the adiabatic exponent γ_{mix}. **Step-by-Step Solution:**

1. Molar heat capacities of individual gases:
   - Helium (monoatomic): $C_{v1} = \frac{3}{2}R$, $C_{p1} = \frac{5}{2}R$.
   - Oxygen (diatomic): $C_{v2} = \frac{5}{2}R$, $C_{p2} = \frac{7}{2}R$.
2. Total moles: $n_{\text{total}} = 2 + 3 = 5\,\text{mol}$.
3. Equivalent $C_{v,\text{mix}}$:

$$C_{v,\text{mix}} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2} = \frac{2\left(\frac{3}{2}R\right) + 3\left(\frac{5}{2}R\right)}{5} \tag{3.99}$$

$$= \frac{10.5R}{5} = \mathbf{2.1R} = \frac{21}{10}R \tag{3.100}$$

4. Equivalent $C_{p,\text{mix}}$:

$$C_{p,\text{mix}} = C_{v,\text{mix}} + R = 2.1R + R = \mathbf{3.1R} = \frac{31}{10}R \tag{3.101}$$

5. Equivalent adiabatic index γ_{mix}:

$$\gamma_{\text{mix}} = \frac{C_{p,\text{mix}}}{C_{v,\text{mix}}} = \frac{3.1R}{2.1R} = \frac{31}{21} \approx \mathbf{1.476} \tag{3.102}$$

Archetype 3: Resonance Tube End Correction & Speed of Sound

Problem: A tuning fork of frequency $\nu = 512\,\text{Hz}$ excites the first two resonances in a resonance water tube at lengths $l_1 = 15.2\,\text{cm}$ and $l_2 = 48.0\,\text{cm}$. Compute the speed of sound in air and the inner pipe radius r. **Step-by-Step Solution:**

1. The fundamental difference between resonances isolates the half-wavelength:

$$\frac{\lambda}{2} = l_2 - l_1 = 48.0\,\text{cm} - 15.2\,\text{cm} = 32.8\,\text{cm} \implies \lambda = 65.6\,\text{cm} = 0.656\,\text{m} \tag{3.103}$$

2. Speed of sound in air:

$$v = \nu\lambda = 512\,\text{s}^{-1} \times 0.656\,\text{m} = \mathbf{335.87}\,\text{m/s} \tag{3.104}$$

3. End correction e:

$$e = \frac{l_2 - 3l_1}{2} = \frac{48.0 - 3(15.2)}{2} = \frac{48.0 - 45.6}{2} = \frac{2.4}{2} = \mathbf{1.2}\,\text{cm} \tag{3.105}$$

4. Inner radius r of the tube using $e \approx 0.6r$:

$$r = \frac{e}{0.6} = \frac{1.2\,\text{cm}}{0.6} = \mathbf{2.0}\,\text{cm} \tag{3.106}$$

PART IV

Electrodynamics & Magnetism

CHAPTER 4

Electrostatics, Capacitance, Current Electricity, Magnetism & Alternating Currents

Electrodynamics represents over 25% of the NEET Physics paper. In this chapter, we build from Coulomb's law and continuous charge distributions to Gauss's flux theorem, dielectric physics and capacitance networks, Ohm's drift dynamics and DC circuits, the magnetic effects of moving charges via Biot-Savart and Ampère laws, electromagnetic induction, LCR resonance in AC circuits, and Maxwell's electromagnetic wave spectrum.

A Personal Note from Satya: The Art of Knowing When to Stop (The 45-Second Rule)

Before we enter the world of circuits, flux, and resonance, let me share a crucial truth about exam psychology.
NEET UG is not an IQ contest; it is a rapid-fire elimination game played under intense time pressure (180 questions in 180 minutes). The single most common reason why dedicated students score 520 instead of 650+ is not that they didn't know the theory—it's that *they didn't know when to put their pen down.*
Here is the cognitive trap: You encounter a tricky question—say, an asymmetrical resistor cube or a magnetic force on an oblique helical trajectory. You spend 60 seconds setting up the equations. The algebra gets tangled. You feel stuck. But your ego whispers: *"I've already spent two minutes on this! If I skip now, those two minutes are wasted!"* So you stay for another two minutes. You end up calculating in panic, and you mark an option.
What just happened?

- You probably marked a wrong option under panic and got -1 instead of $+4$ (a devastating 5-mark swing).
- Even worse: you consumed 4 full minutes on one question. In that exact time, you could have read and solved **six** direct, guaranteed NCERT questions in Biology or Inorganic Chemistry!

The 45-Second Golden Rule: Read the problem. If you cannot visualize the complete, crystal-clear solution path within 45 seconds, **circle the question number on your test booklet, put your pen down, and move on immediately.** Do not hesitate. Do not feel guilty.
NEET top ranks belong to students who sweep through the paper collecting all the easy 4-pointers first in Round 1, and return to battle the complex derivations in Round 2. Knowing when to stop is not surrender; it is tactical mastery.

4.1 Electrostatics, Coulomb's Law & Continuous Charge Distributions

Foundations of Electric Charge & Coulomb's Law in Vector Form

1. Fundamental Charge Properties:

- **Quantization of Charge:** Charge exists strictly as integer multiples of the elementary charge $e \approx 1.602 \times 10^{-19}\,\text{C}$:

$$q = \pm ne \quad (n \in \mathbb{N}) \tag{4.1}$$

- **Conservation of Charge:** The algebraic sum of fundamental charges within an isolated physical system remains invariant over time.
- **Invariance of Charge:** The magnitude of electric charge is independent of the reference frame speed ($q_{\text{rest}} = q_{\text{motion}}$), unlike relativistic mass.

2. Coulomb's Inverse-Square Law in Vector Form: The electrostatic force exerted by point charge q_1 at position vector $\mathbf{r}_1$ on charge q_2 at $\mathbf{r}_2$:

$$\mathbf{F}_{12} = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{q_1 q_2}{|\mathbf{r}_2 - \mathbf{r}_1|^3}(\mathbf{r}_2 - \mathbf{r}_1) = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{q_1 q_2}{r_{12}^2}\hat{\mathbf{r}}_{12} \tag{4.2}$$

where $\varepsilon_0 = 8.854 \times 10^{-12}\,\text{C}^2 \cdot \text{N}^{-1} \cdot \text{m}^{-2}$ is the permittivity of free space, and $\frac{1}{4\pi\varepsilon_0} \approx 8.988 \times 10^9\,\text{N} \cdot \text{m}^2 \cdot \text{C}^{-2}$. In a dielectric medium of relative permittivity (dielectric constant) $\kappa = \varepsilon_r > 1$:

$$F_{\text{medium}} = \frac{F_{\text{vacuum}}}{\kappa} \tag{4.3}$$

3. Superposition Principle: The resultant force acting on any test charge q_0 due to a collection of discrete charges $q_1, q_2, \ldots, q_n$ is the vector sum:

$$\mathbf{F}_{\text{net}} = \sum_{i=1}^{n} \mathbf{F}_{i0} = \frac{q_0}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{|\mathbf{r}_0 - \mathbf{r}_i|^3}(\mathbf{r}_0 - \mathbf{r}_i) \tag{4.4}$$

Electric Field & Potential of Continuous Charge Geometries: Ring & Disk

1. Electric Field Definition: $\mathbf{E} = \lim_{q_0 \to 0}\frac{\mathbf{F}}{q_0} = -\boldsymbol{\nabla} V$. **2. Uniformly Charged Thin Circular Ring on its Symmetry Axis:** Consider a ring of total charge Q and radius R positioned in the $y-z$ plane with symmetry axis along x.

- By radial symmetry, perpendicular field components cancel: $dE_\perp = 0$.
- Axial field element: $dE_x = dE\cos\theta = \frac{1}{4\pi\varepsilon_0}\frac{dq}{x^2+R^2}\frac{x}{\sqrt{x^2+R^2}}$.
- Total integrated axial electric field:

$$\mathbf{E}(\mathbf{x}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}}\frac{\mathbf{Qx}}{(\mathbf{x^2 + R^2})^{\mathbf{3/2}}}\hat{\mathbf{i}} \tag{4.5}$$

- **At Ring Center ($x = 0$):** $E(0) = 0$.
- **Far-Field Regime ($x \gg R$):** $E(x) \approx \frac{1}{4\pi\varepsilon_0}\frac{Q}{x^2}$ (acts as point charge).
- **Location of Maximum Electric Field:**

$$\frac{dE}{dx} = \frac{Q}{4\pi\varepsilon_0}\frac{(x^2+R^2)^{3/2} - 3x^2(x^2+R^2)^{1/2}}{(x^2+R^2)^3} = 0$$
$$\implies x^2 + R^2 - 3x^2 = 0 \implies \mathbf{x = \pm\frac{R}{\sqrt{2}}} \tag{4.6}$$

$$\mathbf{E_{max} = \frac{1}{4\pi\varepsilon_0}\frac{2Q}{3\sqrt{3}R^2}} \tag{4.7}$$

- **Electrostatic Potential on Ring Axis:**

$$\mathbf{V(x) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{x^2+R^2}}} \implies V_{\text{center}} = \frac{Q}{4\pi\varepsilon_0 R} \tag{4.8}$$

3. Uniformly Charged Flat Circular Disk on its Axis: Integrating concentric rings of radius r and charge $dq = \sigma(2\pi r dr)$ from $r = 0$ to R:

$$\mathbf{E(x) = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{\sqrt{x^2+R^2}}\right)\hat{i}}, \qquad \mathbf{V(x) = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{x^2+R^2} - x\right)} \tag{4.9}$$

For $R \to \infty$ (infinite sheet), $E = \frac{\sigma}{2\varepsilon_0}$ (constant, independent of x).

Electric Dipole Dynamics: Field, Torque & Potential Energy

An electric dipole consists of two equal and opposite point charges $+q$ and $-q$ separated by a displacement vector $2\mathbf{a}$ pointing from $-q$ to $+q$:

$$\mathbf{p} = q(2\mathbf{a}) \quad [\text{C} \cdot \text{m}] \tag{4.10}$$

1. Electric Field of Dipole in Coordinate Space ($r \gg a$):

- **Axial Point (End-on Position, $\theta = 0°$):**

$$\mathbf{E_{axial} = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}} \quad \text{(Collinear and parallel to } \mathbf{p}) \tag{4.11}$$

- **Equatorial Point (Broadside-on Position, $\theta = 90°$):**

$$\mathbf{E_{eq} = -\frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}} \quad \text{(Antiparallel to } \mathbf{p}) \tag{4.12}$$

- **General Coordinate Point (r, θ):**

$$|\mathbf{E}| = \frac{\mathbf{1}}{4\pi\varepsilon_0}\frac{\mathbf{p}}{\mathbf{r^3}}\sqrt{\mathbf{1} + \mathbf{3}\cos^2\theta}, \qquad \tan\alpha = \frac{1}{2}\tan\theta \tag{4.13}$$

where α is the angle between $\mathbf{E}$ and position vector $\mathbf{r}$.

- **Electrostatic Potential at (r, θ):**

$$\mathbf{V}(\mathbf{r}, \theta) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{\mathbf{r^2}} = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{p} \cos\theta}{\mathbf{r^2}} \tag{4.14}$$

  Notice: $V_{\text{eq}} = V(r, 90^\circ) = 0$ everywhere along the equatorial bisector plane.

2. External Uniform Electric Field Interactions:

- Net Force: $\mathbf{F}_{\text{net}} = \mathbf{qE} + (-\mathbf{q})\mathbf{E} = \mathbf{0}$ (zero translational acceleration).
- Torque:

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E} \implies |\boldsymbol{\tau}| = pE \sin\theta \tag{4.15}$$

  Maximum torque occurs at $\theta = 90^\circ$ ($\tau_{\max} = pE$).
- Potential Energy of Dipole in External Field:

$$\mathbf{U} = -\mathbf{p} \cdot \mathbf{E} = -\mathbf{pE} \cos\theta \tag{4.16}$$

  - **Stable Equilibrium ($\theta = 0^\circ$):** $\mathbf{p} \parallel \mathbf{E} \implies \boldsymbol{\tau} = 0$, $U_{\min} = -pE$.
  - **Unstable Equilibrium ($\theta = 180^\circ$):** $\mathbf{p} \parallel -\mathbf{E} \implies \boldsymbol{\tau} = 0$, $U_{\max} = +pE$.
- Work done by external agent in rotating dipole from θ_1 to θ_2:

$$\mathbf{W}_{\text{ext}} = \boldsymbol{\Delta}\mathbf{U} = \mathbf{pE}(\cos\boldsymbol{\theta_1} - \cos\boldsymbol{\theta_2}) \tag{4.17}$$

4.2 Gauss's Law, Electric Flux & Boundary Value Applications

Gauss's Law Formulation & Electric Flux

1. Electric Flux (Φ_E): Surface integral of the electric field vector through an oriented surface element $d\mathbf{A} = \hat{\mathbf{n}} dA$:

$$\Phi_E = \iint_S \mathbf{E} \cdot d\mathbf{A} \quad [\text{N} \cdot \text{m}^2 \cdot \text{C}^{-1} \text{ or V} \cdot \text{m}] \tag{4.18}$$

2. Gauss's Law (Maxwell's First Equation): The net outward electric flux through any closed Gaussian surface enclosing net charge q_{enclosed}:

$$\oint_{\partial \mathbf{V}} \mathbf{E} \cdot \mathbf{dA} = \frac{\mathbf{q}_{\text{enclosed}}}{\boldsymbol{\varepsilon_0}} \tag{4.19}$$

Key Physical Insights:

- Net flux is completely independent of the shape, size, or geometry of the Gaussian boundary.
- Charges residing outside the closed surface contribute zero net flux through the surface (entering flux equals exiting flux).

- Electric field $\mathbf{E}$ on the left-hand side is the **net resultant field** produced by all charges (both inside and outside).

Master Applications of Gauss's Law: Exact Field Derivations

1. Infinitely Long Line Charge of Uniform Density λ (C/m): Using a coaxial Gaussian cylinder of radius r and length L:

$$\oint \mathbf{E} \cdot d\mathbf{A} = E(2\pi rL) = \frac{\lambda L}{\varepsilon_0} \implies \mathbf{E}(\mathbf{r}) = \frac{\lambda}{\mathbf{2\pi\varepsilon_0 r}}\hat{\mathbf{r}} \tag{4.20}$$

2. Uniformly Charged Infinite Thin Flat Sheet of Density σ (C/m^2): Using a Gaussian pillbox of cross-sectional area A piercing normal to the sheet:

$$\oint \mathbf{E} \cdot d\mathbf{A} = 2EA = \frac{\sigma A}{\varepsilon_0} \implies \mathbf{E} = \frac{\sigma}{\mathbf{2\varepsilon_0}}\hat{\mathbf{n}} \quad \text{(Independent of distance)} \tag{4.21}$$

3. Surface of a Charged Conductor: Inside the conductor, $E = 0$. Flux exits only from the outer face:

$$EA = \frac{\sigma A}{\varepsilon_0} \implies \mathbf{E}_{\text{conductor}} = \frac{\sigma}{\boldsymbol{\varepsilon_0}}\hat{\mathbf{n}} \quad \text{(Twice that of a non-conducting sheet)} \tag{4.22}$$

4. Thin Spherical Shell of Radius R and Total Charge Q:

- **Interior Region ($r < R$):** Gaussian surface encloses $q_{\text{encl}} = 0 \implies \mathbf{E}_{\text{in}} = \mathbf{0}$.
- Potential is constant throughout interior: $\mathbf{V}_{\text{in}} = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}}\frac{\mathbf{Q}}{\mathbf{R}}$.
- **Exterior Region ($r \geq R$):** Gaussian sphere encloses Q:

$$\mathbf{E}_{\text{out}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}}\frac{\mathbf{Q}}{\mathbf{r^2}}\hat{\mathbf{r}}, \qquad \mathbf{V}_{\text{out}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}}\frac{\mathbf{Q}}{\mathbf{r}}$$

5. Uniformly Charged Non-Conducting Solid Sphere of Radius R and Total Charge Q:

- Volume charge density: $\rho = \frac{Q}{\frac{4}{3}\pi R^3}$.
- **Interior Region ($r \leq R$):** $q_{\text{encl}} = \rho\left(\frac{4}{3}\pi r^3\right) = Q\frac{r^3}{R^3}$:

$$E(r) \cdot 4\pi r^2 = \frac{Qr^3}{\varepsilon_0 R^3} \implies \mathbf{E}_{\text{in}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}}\frac{\mathbf{Qr}}{\mathbf{R^3}}\hat{\mathbf{r}} = \frac{\rho\mathbf{r}}{\mathbf{3\varepsilon_0}}\hat{\mathbf{r}} \tag{4.23}$$

  Electric field increases linearly with radial distance r from center.
- **Interior Potential:** $V_{\text{in}}(r) = -\int_{\infty}^{R} E_{\text{out}}dr - \int_{R}^{r} E_{\text{in}}dr$:

$$\mathbf{V}_{\text{in}}(\mathbf{r}) = \frac{\mathbf{Q}}{\mathbf{4\pi\varepsilon_0}}\left[\frac{\mathbf{3R^2 - r^2}}{\mathbf{2R^3}}\right] \implies \mathbf{V}_{\text{center}} = \frac{\mathbf{3}}{\mathbf{2}}\mathbf{V}_{\text{surface}} = \mathbf{1.5V}_{\text{surface}} \tag{4.24}$$

- **Exterior Region ($r \geq R$):** Identical to point charge:

$$\mathbf{E}_{\text{out}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{Q}}{\mathbf{r^2}} \hat{\mathbf{r}}, \qquad \mathbf{V}_{\text{out}}(\mathbf{r}) = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}} \frac{\mathbf{Q}}{\mathbf{r}} \tag{4.25}$$

NCERT Extraction: Electrostatic Shielding & Cavity Boundary Properties

Inside any hollow conducting cavity of arbitrary shape, the electric field is **identically zero** ($\mathbf{E} = 0$), provided no charge is enclosed within the cavity. The potential throughout the cavity remains constant and strictly equals the potential on the conducting surface ($V_{\text{cavity}} = V_{\text{surface}}$).

- *Faraday Cage Principle:* Any sensitive electrical apparatus enclosed in a hollow metallic enclosure is completely shielded from external electrical disturbances.
- *Practical Safety:* During lightning storms, it is significantly safer to remain inside a closed metallic car or bus than under a tree or in the open, as lightning charges flow along the outer conducting body directly to the ground without penetrating the passenger interior.

Syllabus Boundary Alert: Van de Graaff Generator Deletion

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG syllabus issued by the National Medical Commission:

- **STRICTLY DELETED:** The **Van de Graaff Generator** (high voltage production via belt charge transport, corona discharge, insulating column, spherical dome collection) has been **completely eliminated**.
- **EXAM ORIENTATION:** Candidates must master Gauss's law boundary applications (spherical shells, solid spheres, line charges, and conductor electrostatic shielding) without studying the obsolete Van de Graaff machine.

4.3 Capacitance, Dielectric Physics & Energy Storage

Dielectric Polarization & Capacitance Principles

1. Capacitance Definition: The capacity of a conductor to store electric charge per unit potential rise:

$$C = \frac{Q}{V} \quad [\text{Farad}, \text{F} = \text{C} \cdot \text{V}^{-1}] \tag{4.26}$$

For an isolated conducting sphere of radius R: $C = 4\pi\varepsilon_0 R$. **2. Parallel Plate Capacitor in Vacuum:** Plates of area A separated by vacuum gap d with uniform field $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$ and $V = Ed$:

$$\mathbf{C_0} = \frac{\varepsilon_0 \mathbf{A}}{\mathbf{d}} \tag{4.27}$$

3. Dielectric Material Polarization: When an insulating dielectric is inserted into an external electric field $\mathbf{E}_0$, bound molecular dipoles align, generating a macroscopic polarization vector $\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$. This induces surface bound charge density $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ producing an internal opposing depolarizing field $\mathbf{E}_{\text{induced}} = \frac{\sigma_b}{\varepsilon_0}$.

$$\mathbf{E}_{\text{net}} = \mathbf{E_0} - \mathbf{E}_{\text{induced}} = \frac{\mathbf{E_0}}{\kappa} \quad \text{(Field is attenuated by factor } \kappa) \tag{4.28}$$

where $\kappa = 1 + \chi_e = \varepsilon_r$ is the dielectric constant. **4. Bound Induced Surface Charge:**

$$E_{\text{net}} = \frac{\sigma - \sigma_b}{\varepsilon_0} = \frac{\sigma}{\kappa \varepsilon_0} \implies \sigma_b = \sigma \left(1 - \frac{1}{\kappa}\right) \implies \mathbf{q_b} = \mathbf{q}\left(\mathbf{1} - \frac{\mathbf{1}}{\kappa}\right) \tag{4.29}$$

Capacitor with Partial Dielectric Slab & Network Combinations

1. Partial Dielectric Slab Insertion (Thickness $t < d$): Across the vacuum gap $(d - t)$, electric field is $E_0 = \frac{Q}{\varepsilon_0 A}$. Inside the slab of thickness t, field is $E = \frac{E_0}{\kappa}$. Total potential difference between plates:

$$V = E_0(d - t) + Et = E_0 \left[(d - t) + \frac{t}{\kappa}\right] = \frac{Q}{\varepsilon_0 A}\left[d - t\left(1 - \frac{1}{\kappa}\right)\right] \tag{4.30}$$

$$\mathbf{C} = \frac{\mathbf{Q}}{\mathbf{V}} = \frac{\varepsilon_0 \mathbf{A}}{\mathbf{d} - \mathbf{t}\left(\mathbf{1} - \frac{\mathbf{1}}{\kappa}\right)} \tag{4.31}$$

Special Cases:

- Complete dielectric filling ($t = d$): $C = \kappa \frac{\varepsilon_0 A}{d} = \kappa C_0$.
- Conducting slab inserted ($t < d$, $\kappa \to \infty$): $C = \frac{\varepsilon_0 A}{d - t}$.

2. Grouping of Capacitors:

- **Series Combination (Charge Q is identical):**

$$V = V_1 + V_2 + \cdots \implies \frac{\mathbf{1}}{\mathbf{C}_{\text{eq}}} = \sum_{\mathbf{i=1}}^{\mathbf{n}} \frac{\mathbf{1}}{\mathbf{C_i}} \quad \left(\text{For two: } C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}\right) \tag{4.32}$$

- **Parallel Combination (Potential difference V is identical):**

$$Q = Q_1 + Q_2 + \cdots \implies \mathbf{C}_{\text{eq}} = \sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{C_i} \tag{4.33}$$

3. Electrostatic Energy Stored & Field Energy Density: Work required to assemble charge Q on plates against potential V:

$$\mathbf{U} = \int_{\mathbf{0}}^{\mathbf{Q}} \mathbf{V(q)dq} = \int_{\mathbf{0}}^{\mathbf{Q}} \frac{\mathbf{q}}{\mathbf{C}} \mathbf{dq} = \frac{\mathbf{Q^2}}{\mathbf{2C}} = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{CV^2} = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{QV} \tag{4.34}$$

4. Energy Density in Electric Field: Dividing energy U by the volume of the dielectric space $(A \cdot d)$:

$$u_E = \frac{U}{Ad} = \frac{\frac{1}{2}(\kappa\varepsilon_0 A/d)(Ed)^2}{Ad} \implies \mathbf{u_E = \frac{1}{2}\kappa\varepsilon_0 E^2} \tag{4.35}$$

5. Attractive Mechanical Force Between Plates:

$$\mathbf{F = -\frac{dU}{dx} = \frac{Q^2}{2\varepsilon_0 A} = \frac{1}{2}QE_0 = \frac{1}{2}\varepsilon_0 A E_0^2} \tag{4.36}$$

Dielectric Insertion: Battery Connected vs. Battery Disconnected

Physical Parameter	**Case A: Battery Disconnected (Q = const)**	**Case B: Battery Remains Connected (V = const)**
Charge (Q)	Unchanged: $Q = Q_0$	Increases: $Q = \kappa Q_0$
Capacitance (C)	Increases: $C = \kappa C_0$	Increases: $C = \kappa C_0$
Potential Difference (V)	Decreases: $V = V_0/\kappa$	Unchanged: $V = V_0$
Electric Field (E)	Decreases: $E = E_0/\kappa$	Unchanged: $E = E_0$
Stored Energy (U)	Decreases: $U = U_0/\kappa$	Increases: $U = \kappa U_0$
Battery Work Done	Zero (battery isolated)	$W_{\text{battery}} = \Delta Q \cdot V_0 = (\kappa - 1)C_0 V_0^2$

4.4 Current Electricity, Drift Dynamics & DC Resistive Networks

Drift Velocity, Microscopic Ohm's Law & Mobility

1. Microscopic Mechanism of Conduction in Metals: Under zero external field, conduction electrons execute chaotic thermal motion at speeds $v_{\text{th}} \sim 10^5\,\text{m/s}$ with random directions ($\langle \mathbf{v}_{\text{th}} \rangle = 0$). When an external electric field $\mathbf{E}$ is applied, each electron experiences an electrostatic force $\mathbf{F} = -e\mathbf{E}$, imparting constant acceleration $\mathbf{a} = -\frac{e\mathbf{E}}{m_e}$. Electrons repeatedly scatter elastically from vibrating crystal lattice ions. The average time elapsed between consecutive scattering collisions is the **relaxation time** $\tau \sim 10^{-14}\,\text{s}$. **2. Drift Velocity ($\mathbf{v}_d$) Derivation:**

$$\mathbf{v}_d = \langle \mathbf{v}_{\text{initial}} + \mathbf{a}t \rangle = 0 + \mathbf{a}\tau \implies \mathbf{v_d = -\frac{eE\tau}{m_e}} \tag{4.37}$$

Typically, drift speed is remarkably slow: $v_d \sim 10^{-4}\,\text{m/s} = 0.1\,\text{mm/s}$. **3. Current Density & Microscopic Ohm's Law:** Consider a conductor of cross-sectional area A with free electron density n. In time Δt, charge passing through is $\Delta Q = neAv_d\Delta t$.

$$I = \frac{\Delta Q}{\Delta t} = \mathbf{neAv_d} \implies \mathbf{J} = \frac{\mathbf{I}}{\mathbf{A}} = \mathbf{nev_d} = \left(\frac{\mathbf{ne^2}\tau}{\mathbf{m_e}}\right)\mathbf{E} \tag{4.38}$$

$$\mathbf{J} = \sigma\mathbf{E} = \frac{\mathbf{1}}{\rho}\mathbf{E} \quad \text{(Vector / Microscopic Ohm's Law)} \tag{4.39}$$

where electrical conductivity σ and resistivity ρ:

$$\sigma = \frac{\mathbf{ne^2}\tau}{\mathbf{m_e}}, \qquad \rho = \frac{\mathbf{m_e}}{\mathbf{ne^2}\tau} \tag{4.40}$$

4. Electron Mobility (μ): Drift speed per unit electric field:

$$\mu = \frac{\mathbf{v_d}}{\mathbf{E}} = \frac{\mathbf{e}\tau}{\mathbf{m_e}} \quad [\text{m}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}] \tag{4.41}$$

Temperature Dependence, Cell Grouping & Power Transfer Theorem

1. Temperature Dependence of Resistance: For metallic conductors, increasing temperature intensifies lattice vibrations, reducing relaxation time τ:

$$R(T) = R_0[1 + \alpha(T - T_0)] \implies \alpha = \frac{\mathbf{R_T} - \mathbf{R_0}}{\mathbf{R_0\Delta T}} \quad [^\circ\text{C}^{-1} \text{ or } \text{K}^{-1}] \tag{4.42}$$

For metals, $\alpha > 0$ (positive). For semiconductors and electrolytes, charge carrier density surges exponentially with temperature ($n(T) \propto e^{-E_g/2k_BT}$), overriding τ drops, giving $\alpha < \mathbf{0}$ (negative). **2. Terminal Potential Difference of a Real Cell ($\mathcal{E}, r$):**

- **Discharging (Supplying current I to load R):** $\mathbf{V = \mathcal{E} - Ir} = \frac{\mathcal{E}R}{R+r}$.
- **Charging (External charger driving current into cell):** $\mathbf{V = \mathcal{E} + Ir}$.
- **Open Circuit ($I = 0$):** $V = \mathcal{E}$. Short circuit ($R = 0$): $I_{\text{sc}} = \frac{\mathcal{E}}{r}, V = 0$.

3. Mixed Grouping of Cells (m rows of n series cells each): Total cells $N = n \times m$. Total EMF $= n\mathcal{E}$; total internal resistance $= \frac{nr}{m}$. Current through load R:

$$I = \frac{n\mathcal{E}}{R + \frac{nr}{m}} = \frac{mn\mathcal{E}}{mR + nr} \tag{4.43}$$

To maximize current, minimize the denominator $(mR+nr) = (\sqrt{mR} - \sqrt{nr})^2 + 2\sqrt{mnRr}$. The minimum occurs when $\sqrt{mR} = \sqrt{nr}$:

$$\mathbf{R} = \frac{\mathbf{nr}}{\mathbf{m}} \quad \text{(Load resistance equals total internal resistance)} \tag{4.44}$$

4. Maximum Power Transfer Theorem: Power delivered to load: $P(R) = I^2R = \frac{\mathcal{E}^2 R}{(R+r)^2}$. Setting $\frac{dP}{dR} = 0$:

$$\frac{dP}{dR} = \mathcal{E}^2 \frac{(R+r)^2 - 2R(R+r)}{(R+r)^4} = 0 \implies R + r - 2R = 0 \implies \mathbf{R = r} \quad (4.45)$$

$$\mathbf{P_{max} = \frac{\mathcal{E}^2}{4r}} \quad (4.46)$$

Kirchhoff's Laws & Measurement Bridges

1. Kirchhoff's Laws:

- **Junction Rule (KCL):** At any circuit node: $\sum I_{\text{in}} = \sum I_{\text{out}}$ (Conservation of Charge).
- **Loop Rule (KVL):** Around any closed loop: $\sum \Delta V = \sum \mathcal{E} - \sum IR = 0$ (Conservation of Energy).

2. Wheatstone Bridge Balance Derivation: A quadrilateral network of four resistors P, Q, R, S with a galvanometer connected between junctions B and D. At balance, no current traverses the galvanometer ($I_g = 0 \implies V_B = V_D$). Applying KVL to loop $ABDA$: $I_1P - I_2R = 0 \implies I_1P = I_2R$. Applying KVL to loop $BCDB$: $I_1Q - I_2S = 0 \implies I_1Q = I_2S$. Dividing the two equations yields the classical balance condition:

$$\mathbf{\frac{P}{Q} = \frac{R}{S}} \quad (4.47)$$

3. Metre Bridge (Slide-Wire Bridge): A uniform wire of length 100 cm acts as ratio arms $P = \rho_w l$ and $Q = \rho_w(100 - l)$.

$$\frac{R}{X} = \frac{l}{100 - l} \implies \mathbf{X = R\left(\frac{100 - l}{l}\right)} \quad (4.48)$$

Maximum experimental precision occurs at the central null point ($l \approx 50$ cm).

Syllabus Boundary Alert: Potentiometer & Carbon Resistor Color Codes Deletion

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG syllabus issued by the National Medical Commission:

- **STRICTLY DELETED:** The **Potentiometer** (principle of fall of potential $V \propto l$, comparison of EMFs of two cells $\mathcal{E}_1/\mathcal{E}_2 = l_1/l_2$, and determination of cell internal resistance $r = R(l_1/l_2 - 1)$) has been **completely deleted**.
- **STRICTLY DELETED: Color Coding of Carbon Resistors** (BBROYGBVGW mnemonic bands, tolerance multipliers) has been **completely deleted**.
- **RETAINED:** The **Metre Bridge** and **Wheatstone Bridge** are fully re-

tained and form the exclusive focus for bridge measurements.

4.5 Magnetic Effects of Electric Current & Moving Charges

Biot-Savart Law & Magnetic Lorentz Force

1. Biot-Savart Law: The infinitesimal magnetic field $d\mathbf{B}$ generated by an elementary current length vector $Id\mathbf{l}$ at displacement $\mathbf{r}$:

$$\mathbf{dB} = \frac{\mu_0}{4\pi}\frac{\mathbf{I(dl \times \hat{r})}}{\mathbf{r^2}} = \frac{\mu_0}{4\pi}\frac{\mathbf{I(dl \times r)}}{\mathbf{r^3}} \tag{4.49}$$

where $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T \cdot m \cdot A^{-1}}$ is vacuum permeability. **2. Magnetic Lorentz Force on a Moving Charge:**

$$\mathbf{F = q(E + v \times B)} \tag{4.50}$$

- Pure magnetic force: $\mathbf{F}_B = q(\mathbf{v} \times \mathbf{B})$. Since $\mathbf{F}_B \perp \mathbf{v}$, the instantaneous power is $P = \mathbf{F}_B \cdot \mathbf{v} = 0$.
- **Work Done by Magnetic Force is Identically Zero:** Magnetic forces change the direction of motion, never particle kinetic energy or speed.
- **Helical Trajectory in Uniform Field B:** When particle velocity $\mathbf{v}$ makes angle θ with $\mathbf{B}$ ($v_\perp = v\sin\theta, v_\parallel = v\cos\theta$):

$$\mathbf{r} = \frac{\mathbf{mv}_\perp}{\mathbf{qB}} = \frac{\mathbf{mv}\sin\theta}{\mathbf{qB}}, \qquad \mathbf{T} = \frac{\mathbf{2\pi m}}{\mathbf{qB}}, \qquad \text{Pitch } \mathbf{p} = \mathbf{v}_\parallel \mathbf{T} = \frac{\mathbf{2\pi mv}\cos\theta}{\mathbf{qB}} \tag{4.51}$$

Master Magnetic Field Formulas: Straight Wires, Loops, Solenoids & Toroids

1. Finite Straight Wire Subtending Angles θ_1, θ_2 at Distance d:

$$\mathbf{B} = \frac{\mu_0 \mathbf{I}}{4\pi \mathbf{d}}(\sin\theta_1 + \sin\theta_2) \tag{4.52}$$

- **Infinitely Long Straight Wire ($\theta_1 = \theta_2 = 90°$):** $\mathbf{B} = \frac{\mu_0 \mathbf{I}}{2\pi \mathbf{d}}$.
- **Semi-Infinite Wire End ($\theta_1 = 90°, \theta_2 = 0°$):** $\mathbf{B} = \frac{\mu_0 \mathbf{I}}{4\pi \mathbf{d}}$.

2. Circular Current Loop of Radius R on Symmetry Axis (x):

$$\mathbf{B(x)} = \frac{\mu_0 \mathbf{IR^2}}{\mathbf{2(R^2 + x^2)^{3/2}}}\hat{\mathbf{i}} \implies \mathbf{B}_{\text{center}} = \frac{\mu_0 \mathbf{I}}{\mathbf{2R}} \tag{4.53}$$

3. Ampere's Circuital Law:

$$\oint \mathbf{B} \cdot \mathbf{dl} = \mu_0 \mathbf{I}_{\text{enclosed}} \tag{4.54}$$

- **Ideal Long Solenoid (n turns/meter):** Interior field is axial and uniform: $\mathbf{B} = \mu_0 \mathbf{nI}$. Near ends: $B_{\text{end}} = \frac{1}{2}\mu_0 nI$.
- **Toroid of Mean Radius R and N Total Turns:** $n = \frac{N}{2\pi R} \implies \mathbf{B} = \mu_0 \mathbf{nI} = \frac{\mu_0 \mathbf{NI}}{\mathbf{2\pi R}}$.

4. Force Between Parallel Currents (Definition of Ampere):

$$\frac{\mathbf{F}}{\mathbf{L}} = \frac{\mu_0 \mathbf{I_1 I_2}}{\mathbf{2\pi d}} \tag{4.55}$$

Parallel currents **attract**; antiparallel currents **repel**.

Moving Coil Galvanometer Conversion & Magnetic Materials

1. Moving Coil Galvanometer (MCG): Deflecting torque balances radial restoring suspension torque: $\tau = NIAB = C\theta \implies \theta = \left(\frac{NAB}{C}\right) I$.

- **Conversion to Ammeter (Extending range to $I > I_g$):** Connect small shunt resistance S in parallel:

$$(I - I_g)S = I_g R_g \implies \mathbf{S} = \frac{\mathbf{I_g R_g}}{\mathbf{I - I_g}} \tag{4.56}$$

- **Conversion to Voltmeter (Extending range to V):** Connect high multiplier resistance R in series:

$$V = I_g(R_g + R) \implies \mathbf{R} = \frac{\mathbf{V}}{\mathbf{I_g}} - \mathbf{R_g} \tag{4.57}$$

2. Classification of Magnetic Materials:

- **Diamagnetic:** $\chi_m < 0$ (small, negative, independent of temperature). $\mu_r < 1$. Repelled by magnetic fields. Examples: Bi, Cu, H_2O, NaCl.
- **Paramagnetic:** $\chi_m > 0$ (small, positive). Follows **Curie's Law**: $\chi_m = \frac{C}{T}$. $\mu_r > 1$. Weakly attracted. Examples: Al, Na, O_2.
- **Ferromagnetic:** $\chi_m \gg 1$ (enormous, positive). Follows **Curie-Weiss Law** above Curie temperature T_c: $\chi_m = \frac{C}{T - T_c}$. Exhibits domains and hysteresis. Examples: Fe, Co, Ni.

Syllabus Boundary Alert: Cyclotron & Earth's Magnetism Deletion

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG syllabus issued by the National Medical Commission:

- **STRICTLY DELETED:** The **Cyclotron** (resonating RF frequency $\nu_c = \frac{qB}{2\pi m}$, kinetic energy $K_{\max} = \frac{q^2B^2R^2}{2m}$, relativistic limits) has been **completely removed**.
- **STRICTLY DELETED: Earth's Magnetic Elements** (Magnetic Declination θ, Angle of Dip / Inclination δ, Horizontal Component B_H, relation $B = \sqrt{B_H^2 + B_V^2}$) have been **completely removed**.

4.6 Electromagnetic Induction (EMI) & Inductive Circuits

Faraday's Law, Lenz's Law & Motional EMF

1. Magnetic Flux (Φ_B):

$$\Phi_B = \iint \mathbf{B} \cdot d\mathbf{A} = BA\cos\theta \quad [\text{Weber}, \text{Wb} = \text{T} \cdot \text{m}^2] \tag{4.58}$$

2. Faraday's Law of Electromagnetic Induction: Induced EMF in any closed circuit equals the negative time rate of change of magnetic flux:

$$\mathcal{E} = -\frac{\mathbf{d\Phi_B}}{\mathbf{dt}} = -\mathbf{N}\frac{\mathbf{d\Phi_B}}{\mathbf{dt}} \tag{4.59}$$

3. Lenz's Law (Conservation of Energy): The polarity of the induced EMF is such that it produces a current whose magnetic field opposes the original flux change that created it. **4. Motional EMF Derivations:**

- **Straight Conducting Rod of Length L Moving with Velocity $\mathbf{v} \perp \mathbf{B}$:** Lorentz force on free electrons $\mathbf{F} = -e(\mathbf{v} \times \mathbf{B})$ drives charge accumulation until internal electrostatic field cancels magnetic force: $eE = evB \implies \mathcal{E} = \mathbf{BvL}$.
- **Conducting Rod of Length L Rotating at Angular Velocity ω in Uniform B:** Integrating elemental motional EMF $d\mathcal{E} = Bv(r)dr = B(\omega r)dr$:

$$\mathcal{E} = \int_{\mathbf{0}}^{\mathbf{L}} \mathbf{B}\omega\mathbf{r}\mathbf{dr} = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{B}\omega\mathbf{L^2} \tag{4.60}$$

Self-Inductance, Mutual Inductance & Magnetic Energy Density

1. Self-Inductance (L): Total flux linkage is proportional to circulating current: $N\Phi_B = LI$.

$$\mathcal{E}_\mathbf{L} = -\mathbf{L}\frac{\mathbf{dI}}{\mathbf{dt}} \quad [\text{Henry}, \text{H} = \text{V} \cdot \text{s} \cdot \text{A}^{-1}] \tag{4.61}$$

For a long air-core solenoid of length l, area A, and turn density n:

$$N\Phi = (nl)(\mu_0 nIA) = \mu_0 n^2 AlI$$
$$\implies \mathbf{L = \mu_0 n^2 Al = \mu_0 \frac{N^2}{l} A} \tag{4.62}$$

2. Mutual Inductance (M): $N_2\Phi_2 = M_{21}I_1 \implies \mathcal{E}_2 = -M\frac{dI_1}{dt}$. For coaxial solenoids:

$$\mathbf{M = \mu_0 n_1 n_2 Al = \frac{\mu_0 N_1 N_2 A}{l}} \tag{4.63}$$

Coupling coefficient: $\mathbf{k = \frac{M}{\sqrt{L_1 L_2}}}$ $(0 \le k \le 1)$. **3. Energy Stored in an Inductor:**

$$\mathbf{U = \int_0^I Lidi = \frac{1}{2} LI^2} \tag{4.64}$$

4. Magnetic Energy Density in a Magnetic Field:

$$\mathbf{u_B = \frac{U}{\text{Volume}} = \frac{B^2}{2\mu_0}} \tag{4.65}$$

NCERT Extraction: Eddy Currents & Practical Technological Applications

When bulk conductors are subjected to changing magnetic flux, circulating closed induced current loops (**Eddy Currents**) are established in planes perpendicular to the magnetic lines of force.

- **Electromagnetic Damping:** The metallic non-magnetic frame of a moving coil galvanometer develops opposing eddy currents as it deflects, rapidly bringing the pointer to rest at its steady reading without oscillatory hunting (dead-beat galvanometer).
- **Induction Furnaces:** High-frequency alternating magnetic fields generate massive eddy currents within metallic charges, producing Joule heat sufficient to melt metals in vacuum.
- **Magnetic Braking in Trains:** Electromagnets activated above rails induce opposing eddy currents in the rotating metallic wheels/rails, producing smooth, wear-free deceleration.
- **Mitigation in Transformers:** To minimize eddy current heating losses in transformers and motors, solid iron cores are replaced with thin, individually varnished/insulated metallic laminations stacked parallel to magnetic flux lines.

4.7 Alternating Currents (AC) & Resonant Circuits

Alternating Current Fundamentals: Phasors & RMS Values

A sinusoidal alternating EMF and current:

$$V(t) = V_0 \sin(\omega t), \qquad I(t) = I_0 \sin(\omega t - \phi) \tag{4.66}$$

1. Mean (Average) Value over Half-Cycle:

$$I_{\text{avg}} = \frac{1}{\pi/\omega} \int_0^{\pi/\omega} I_0 \sin(\omega t) dt = \mathbf{\frac{2}{\pi} I_0 \approx 0.637 I_0} \tag{4.67}$$

Full cycle average is strictly zero: $\langle I \rangle_{\text{full}} = 0$. **2. Root-Mean-Square (RMS / Virtual) Value:** Equating Joule heating over a full cycle $H = \int_0^T I^2 R dt = I_{\text{rms}}^2 RT$:

$$I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T I_0^2 \sin^2(\omega t) dt} = \mathbf{\frac{I_0}{\sqrt{2}} \approx 0.707 I_0}, \qquad \mathbf{V_{\text{rms}} = \frac{V_0}{\sqrt{2}}} \tag{4.68}$$

AC meters (hot-wire ammeters and voltmeters) always read **RMS values**.

Series LCR Circuit: Impedance, Resonance & Quality Factor

In a series LCR circuit with supply voltage $V(t) = V_0 \sin(\omega t)$:

$$V_R = IR, \qquad V_L = IX_L = I(\omega L), \qquad V_C = IX_C = I\left(\frac{1}{\omega C}\right) \tag{4.69}$$

Since V_L leads current by $\pi/2$ and V_C lags current by $\pi/2$:

$$V_0 = \sqrt{V_R^2 + (V_L - V_C)^2} = I_0 \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \tag{4.70}$$

$$\mathbf{Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}, \qquad \tan\phi = \frac{\mathbf{X_L - X_C}}{\mathbf{R}} \tag{4.71}$$

1. Electrical Resonance Condition: When inductive reactance equals capacitive reactance:

$$X_L = X_C \implies \omega_0 L = \frac{1}{\omega_0 C} \implies \mathbf{\omega_0 = \frac{1}{\sqrt{LC}}}, \quad \mathbf{\nu_0 = \frac{1}{2\pi\sqrt{LC}}} \tag{4.72}$$

At resonance:

- Impedance is minimum and purely resistive: $\mathbf{Z}_{\min} = \mathbf{R}$.
- Current amplitude is maximum and in phase with voltage: $\mathbf{I_{0,\max}} = \frac{\mathbf{V_0}}{\mathbf{R}}$, $\phi = 0$.
- Power factor is unity: $\cos\phi = 1$.

2. Sharpness of Resonance & Quality Factor (Q):

$$\mathbf{Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{\omega_0}{2\Delta\omega} = \frac{\text{Resonant Frequency}}{\text{Bandwidth}}} \tag{4.73}$$

High Q-factor indicates sharp selectivity and low damping. **3. Average Power in AC Circuit & Power Factor:**

$$\mathbf{P_{avg} = V_{rms} I_{rms} \cos\phi} \tag{4.74}$$

where $\cos\phi = \frac{R}{Z}$ is the **power factor**.

- Pure L or pure C: $\phi = \pm\pi/2 \implies \cos\phi = 0 \implies P_{avg} = 0$ (**Wattless Current** $I_w = I_{rms}\sin\phi$).
- Pure R or LCR at resonance: $\phi = 0 \implies \cos\phi = 1 \implies P_{avg} = V_{rms} I_{rms}$.

4. Transformers:

$$\mathbf{\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = K} \quad \text{(Transformation Ratio)} \tag{4.75}$$

Efficiency: $\eta = \frac{P_{output}}{P_{input}} = \frac{V_s I_s \cos\phi_s}{V_p I_p \cos\phi_p}$.

4.8 Electromagnetic Waves & Maxwell's Equations

Maxwell's Displacement Current & Unified Field Equations

1. Maxwell's Displacement Current (I_d): Ampere's Circuital Law ($\oint \mathbf{B}\cdot d\mathbf{l} = \mu_0 I$) failed when applied to a charging capacitor boundary. Maxwell resolved this inconsistency by postulating that a time-varying electric field produces an equivalent displacement current:

$$\Phi_E = EA = \left(\frac{Q}{\varepsilon_0 A}\right)A = \frac{Q}{\varepsilon_0} \implies I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \frac{dQ}{dt} = I_c \tag{4.76}$$

$$\mathbf{\oint B\cdot dl = \mu_0(I_c + I_d) = \mu_0 I_c + \mu_0\varepsilon_0\frac{d\Phi_E}{dt}} \quad \text{(Ampere-Maxwell Law)} \tag{4.77}$$

2. Transverse Nature & Speed of EM Waves: Electric and magnetic vectors oscillate perpendicular to each other and perpendicular to the wave propagation direction:

$$\mathbf{c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = \frac{E_0}{B_0} \approx 3.00\times 10^8\ m/s} \tag{4.78}$$

3. Poynting Vector & Radiation Pressure: Energy flux vector: $\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E}\times\mathbf{B})$. Average intensity: $I = \frac{1}{2}\varepsilon_0 c E_0^2 = \frac{E_0 B_0}{2\mu_0}$.

$$\text{Radiation Pressure: } P_{rad} = \begin{cases} \frac{I}{c} & \text{Perfectly Absorbing Surface} \\ \frac{2I}{c} & \text{Perfectly Reflecting Surface} \end{cases} \tag{4.79}$$

The Electromagnetic Spectrum: Master Classification

Wave	λ	ν	Source & Application
Radio	$> 0.1\,\text{m}$	$< 3 \times 10^9\,\text{Hz}$	LC antennas; Radio/TV communication
Micro	1 mm–0.1 m	10^9–$10^{11}\,\text{Hz}$	Klystrons; Radar, Microwave ovens
Infrared	700 nm–1 mm	10^{11}–$4 \times 10^{14}\,\text{Hz}$	Molecular vibration; Remote controls
Visible	400–700 nm	4–$7.5 \times 10^{14}\,\text{Hz}$	Outer shell transitions; Vision
UV	1–400 nm	10^{15}–$10^{17}\,\text{Hz}$	Inner shell transitions; Sterilization
X-Rays	0.01–10 nm	10^{16}–$10^{20}\,\text{Hz}$	Decelerated electrons; Radiography
γ-Rays	$< 10^{-3}\,\text{nm}$	$> 10^{19}\,\text{Hz}$	Nuclear transitions; Radiotherapy

4.9 Master High-Yield Problem Archetypes & Worked Solutions

Archetype 1: Cube Resistor Network Equivalent Resistances

Problem: Twelve identical resistors, each of resistance R, form the edges of a skeleton cube. Compute equivalent resistance between: (i) diagonally opposite body corners, (ii) opposite face corners, and (iii) adjacent edge corners. **Step-by-Step Solution:**

1. **Body Diagonal (Points A and G):** Inject current I at A and extract at G. By threefold rotational symmetry around the body diagonal:
   - 3 edges leave A, each carrying $I/3$.
   - Each splits into 2 edges, each carrying $I/6$ across the middle 6 edges.
   - These recombine into 3 edges entering G, each carrying $I/3$.

   Total potential drop: $V_{AG} = \left(\frac{I}{3}\right) R + \left(\frac{I}{6}\right) R + \left(\frac{I}{3}\right) R = IR\left(\frac{1}{3} + \frac{1}{6} + \frac{1}{3}\right) = \frac{5}{6} IR$.

$$\mathbf{R}_{\text{body}} = \frac{\mathbf{5}}{\mathbf{6}}\mathbf{R} \tag{4.80}$$

2. **Face Diagonal (Points A and C):** By mirror symmetry across the plane bisecting the face:

$$\mathbf{R}_{\text{face}} = \frac{\mathbf{3}}{\mathbf{4}}\mathbf{R} \tag{4.81}$$

3. **Adjacent Edge (Points A and B):**

$$\mathbf{R}_{\text{edge}} = \frac{\mathbf{7}}{\mathbf{12}}\mathbf{R} \tag{4.82}$$

Archetype 2: Series LCR Circuit Resonance & Power Factor Calculation

Problem: A series LCR circuit has $R = 10\,\Omega$, $L = 0.1\,\text{H}$, and $C = 100\,\mu\text{F}$ connected to an AC source $V(t) = 200\sin(100t)\,\text{V}$. Calculate: (i) Inductive

reactance X_L, (ii) Capacitive reactance X_C, (iii) Total impedance Z, (iv) Peak current I_0, (v) Resonant frequency ω_0, and (vi) Quality factor Q. **Step-by-Step Solution:**

1. $\omega = 100\,\text{rad/s}$, $V_0 = 200\,\text{V}$.
2. $X_L = \omega L = 100 \times 0.1 = \mathbf{10\,\Omega}$.
3. $X_C = \frac{1}{\omega C} = \frac{1}{100\times 100\times 10^{-6}} = \frac{1}{10^{-2}} = \mathbf{100\,\Omega}$.
4. Total impedance Z:

$$\begin{aligned} Z &= \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{10^2 + (10-100)^2} \\ &= \sqrt{100 + 8100} = \sqrt{8200} \approx \mathbf{90.55\,\Omega} \end{aligned} \tag{4.83}$$

   Circuit is strongly capacitive ($X_C > X_L$), current leads voltage.
5. Peak current: $I_0 = \frac{V_0}{Z} = \frac{200}{90.55} \approx \mathbf{2.21}\,\text{A}$.
6. Resonant angular frequency ω_0:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.1 \times 10^{-4}}} = \frac{1}{\sqrt{10^{-5}}} = \frac{1000}{\sqrt{10}} \approx \mathbf{316.2}\,\text{rad/s} \tag{4.84}$$

7. Quality factor Q:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{10}\sqrt{\frac{0.1}{10^{-4}}} = \frac{1}{10}\sqrt{1000} = \frac{31.62}{10} \approx \mathbf{3.16} \tag{4.85}$$

PART V

Optics & Modern Physics

CHAPTER 5

Ray Optics, Wave Optics, Modern Physics & Semiconductor Electronics

This chapter covers geometric and wave optics alongside modern quantum and solid-state physics. We investigate Snell's law, total internal reflection, the lens-maker's equation, optical instruments, Huygens' wavelets, Young's double-slit interference, single-slit diffraction, Einstein's photoelectric effect, de Broglie matter waves, Bohr's atomic postulates and spectral emissions, nuclear binding energy, and p-n junction diode electronics with digital logic gates.

A Personal Note from Satya: The 100-Mistakes Principle (Why Trying Again Actually Rewires Your Brain)

You are more than halfway through this book. Let's talk frankly about the days when self-doubt creeps in.

Every single medical student currently walking the corridors of AIIMS, MAMC, or KGMU had days where they scored 480 in a Sunday grand test, stared at their marks sheet in despair, and wondered if they simply weren't smart enough.

Here is what neuroscience tells us about high-stakes learning: *your brain does not grow when you get a question right.* Getting a question right simply exercises an existing, pre-built neural circuit. But when you tackle a difficult problem, get it wrong, feel that sting of frustration, and then sit down to trace *exactly* where your reasoning derailed—that is the precise biological moment where your brain releases acetylcholine and dopamine, signaling your synapses to rebuild and strengthen.

How to Turn Mistakes into a 700+ Score:

1. **Maintain a Ruthless Error Logbook:** Never just say "Oh, that was a silly calculation mistake." Categorize every single error into one of three buckets:
   - *Conceptual Blind Spot:* You didn't know the law or rule (e.g., forgetting that de Broglie wavelength for a charged particle depends on qV).
   - *Question Trap Slip:* You knew the concept, but NTA phrased it with "incorrect statement" or switched units from μm to nm.
   - *Execution Panic:* You rushed through basic arithmetic under clock pressure.
2. **Re-solve Every Mistake 48 Hours Later:** An error is only truly eliminated when you can re-solve the question on a blank sheet of paper two days later without looking at the answer key.

Every single mistake you catch and correct in this book is one fewer trap waiting for you on the actual NEET exam day. Treat every wrong answer not as an indictment of your talent, but as a solved mystery. Keep your head high, turn the page, and let's master Modern Physics.

5.1 Ray Optics, Reflection, Refraction & Total Internal Reflection

Foundations of Geometric Optics & Sign Conventions

1. Cartesian Sign Convention (Strict Standard):

- All optical distances are measured along the principal axis starting from the optical pole (P) or optical centre (O).
- Distances measured in the direction of incident light are **positive** ($+$); distances opposite to incident light are **negative** ($-$).
- Heights measured perpendicularly upwards above the principal axis are **positive** ($+$); heights downwards are **negative** ($-$).

2. Reflection at Spherical Mirrors:

$$\frac{\mathbf{1}}{\mathbf{v}} + \frac{\mathbf{1}}{\mathbf{u}} = \frac{\mathbf{1}}{\mathbf{f}} = \frac{\mathbf{2}}{\mathbf{R}} \tag{5.1}$$

- **Concave Mirror:** $f < 0, R < 0$ (converging). Produces real inverted images for $u > |f|$, and virtual erect magnified image for $u < |f|$.
- **Convex Mirror:** $f > 0, R > 0$ (diverging). Produces virtual, erect, diminished images for all real object locations.
- **Transverse (Linear) Magnification:**

$$\mathbf{m} = \frac{\mathbf{h_i}}{\mathbf{h_o}} = -\frac{\mathbf{v}}{\mathbf{u}} = \frac{\mathbf{f}}{\mathbf{f - u}} = \frac{\mathbf{f - v}}{\mathbf{f}} \tag{5.2}$$

  $m < 0 \implies$ real, inverted; $m > 0 \implies$ virtual, erect.
- **Longitudinal Magnification for Small Objects (du):** $m_L = \frac{dv}{du} = -m^2 = -\left(\frac{v}{u}\right)^2$.

Refraction at Plane Interfaces, Apparent Depth & Total Internal Reflection

1. Snell's Law of Refraction:

$$\mathbf{n_1} \sin \mathbf{i} = \mathbf{n_2} \sin \mathbf{r} \implies \frac{\sin \mathbf{i}}{\sin \mathbf{r}} = \frac{\mathbf{n_2}}{\mathbf{n_1}} = {}^{\mathbf{1}}\mathbf{n_2} = \frac{\mathbf{v_1}}{\mathbf{v_2}} = \frac{\boldsymbol{\lambda_1}}{\boldsymbol{\lambda_2}} \tag{5.3}$$

Frequency ν remains invariant across any refractive boundary: $\nu_1 = \nu_2 = \nu$. **2. Apparent Depth & Normal Shift in Dense Media:** An object placed at real depth d in a medium of refractive index μ viewed normally from air:

$$\mathbf{d}_{\text{apparent}} = \frac{\mathbf{d}}{\mu}, \qquad \text{Normal Shift } \boldsymbol{\Delta t} = \mathbf{d} - \mathbf{d}_{\text{app}} = \mathbf{d}\left(\mathbf{1} - \frac{\mathbf{1}}{\mu}\right) \tag{5.4}$$

For composite multiple horizontal layers: $d_{\text{apparent, total}} = \sum_{i=1}^{n} \frac{d_i}{\mu_i}$. **3. Total Internal Reflection (TIR):** When light propagates from an optically denser medium (n_1) toward a rarer medium ($n_2 < n_1$):

- Critical angle i_c: angle of incidence where angle of refraction $r = 90^\circ$ (for air, $n_2 = 1$):

$$n_1 \sin i_c = n_2 \sin 90^\circ \implies \sin \mathbf{i_c} = \frac{\mathbf{n_2}}{\mathbf{n_1}} = \frac{\mathbf{1}}{\mu} \tag{5.5}$$

- For $i > i_c$, 100% of incident light energy reflects back into the denser medium with zero transmission loss.
- **Optical Fibres:** Comprise a central cylindrical core (μ_1) surrounded by a cladding ($\mu_2 < \mu_1$). Light admitted within the cone of acceptance undergoes repeated total internal reflections.

$$\text{Numerical Aperture (NA)} = \sin \theta_{\mathbf{a}} = \sqrt{\mu_{\mathbf{1}}^{\mathbf{2}} - \mu_{\mathbf{2}}^{\mathbf{2}}} \tag{5.6}$$

Refraction at a Single Spherical Surface: First-Principles Derivation

Consider a convex spherical refracting interface of radius of curvature R separating medium 1 (μ_1) from medium 2 (μ_2). A point object O lies on the principal axis in medium 1 at distance $-u$. Light refracts to form an image I at $+v$ in medium 2.

1. Let ray from O strike surface at M near pole P. Let angles with principal axis be α (at O), β (at I), and θ (at center of curvature C).
2. In triangle OMC, exterior angle is incidence angle: $i = \alpha + \theta$.
3. In triangle IMC, exterior angle is θ: $\theta = r + \beta \implies r = \theta - \beta$.
4. Under paraxial approximations ($\tan\phi \approx \phi \approx \sin\phi$ and $M \approx P$):

$$\alpha \approx \frac{PM}{u} = -\frac{h}{u}, \qquad \beta \approx \frac{PM}{v} = \frac{h}{v}, \qquad \theta \approx \frac{PM}{R} = \frac{h}{R} \tag{5.7}$$

5. By Snell's Law for small angles: $\mu_1 i = \mu_2 r$.

$$\mu_1(\alpha + \theta) = \mu_2(\theta - \beta) \implies \mu_1\alpha + \mu_2\beta = (\mu_2 - \mu_1)\theta \tag{5.8}$$

6. Substituting the paraxial arc angles:

$$\mu_1 \left(-\frac{h}{u}\right) + \mu_2 \left(\frac{h}{v}\right) = (\mu_2 - \mu_1)\frac{h}{R} \tag{5.9}$$

Cancelling h:

$$\frac{\mu_{\mathbf{2}}}{\mathbf{v}} - \frac{\mu_{\mathbf{1}}}{\mathbf{u}} = \frac{\mu_{\mathbf{2}} - \mu_{\mathbf{1}}}{\mathbf{R}} \tag{5.10}$$

This master governing formula holds universally for both convex and concave surfaces, whether light travels from rarer to denser or denser to rarer, provided proper Cartesian signs are used for u, v, R.

5.2 Thin Lenses, Lens-Maker's Formula, Prisms & Optical Instruments

Lens-Maker's Formula & Thin Lens Equation: Step-by-Step Proof

Consider a thin lens of refractive index μ_2 bounded by spherical surfaces of radii R_1 and R_2 immersed in a surrounding medium of refractive index μ_1.

1. **Refraction at First Surface (R_1):** Object at u, forms intermediate image at v':
$$\frac{\mu_2}{v'} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1} \qquad \text{(Eq. 1)}$$
2. **Refraction at Second Surface (R_2):** The intermediate image v' serves as a virtual object for the second interface, forming final image at v:
$$\frac{\mu_1}{v} - \frac{\mu_2}{v'} = \frac{\mu_1 - \mu_2}{R_2} = -\frac{\mu_2 - \mu_1}{R_2} \qquad \text{(Eq. 2)}$$
3. **Adding Equations (1) and (2):** The intermediate term $\frac{\mu_2}{v'}$ cancels completely:
$$\mu_1\left(\frac{1}{v} - \frac{1}{u}\right) = (\mu_2 - \mu_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad (5.11)$$
Dividing through by μ_1:
$$\frac{1}{v} - \frac{1}{u} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad (5.12)$$
4. Setting $u \to -\infty \implies v = f$:
$$\mathbf{\frac{1}{f} = \left(\frac{\mu_{lens}}{\mu_{medium}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)} \quad \text{(Lens-Maker's Formula)} \qquad (5.13)$$
$$\mathbf{\frac{1}{v} - \frac{1}{u} = \frac{1}{f}} \quad \text{(Gaussian Thin Lens Formula)} \qquad (5.14)$$

3. Optical Power (P):

$$\mathbf{P = \frac{1}{f\,[m]}} \quad [\text{Dioptres, D} = \text{m}^{-1}] \qquad (5.15)$$

4. Combination of Thin Lenses:

- **Lenses in Contact:**
$$P_{eq} = P_1 + P_2 \implies \frac{1}{F_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}, \qquad m = m_1 \times m_2$$
- **Separated by Distance d:**
$$P_{eq} = P_1 + P_2 - dP_1P_2 \implies \frac{1}{F_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

5.2.1 Refraction Through Triangular Prisms, Angular Deviation & Dispersion

A prism is a transparent optical medium bounded by two non-parallel refracting planes inclined at an angle A, termed the angle of the prism or refracting angle.

Ray Geometry and Angular Deviation: Consider a monochromatic ray of light incident at angle i on the first refracting face AB of a triangular glass prism (Figure 5.1). The ray bends towards the normal at angle r_1, traverses through the prism, and meets the second face AC at angle of incidence r_2. It emerges into air at angle of emergence e, bending away from the normal.

From the geometry of the quadrilateral formed by the prism apex A and the normals at the two refracting surfaces, the opposite angles sum to 180°. Therefore, within the interior ray triangle:

$$r_1 + r_2 = A \tag{5.16}$$

The total angular deviation δ suffered by the ray is the sum of the deviations at both refracting interfaces ($\delta = (i - r_1) + (e - r_2)$):

$$\delta = \mathbf{i} + \mathbf{e} - \mathbf{A} \tag{5.17}$$

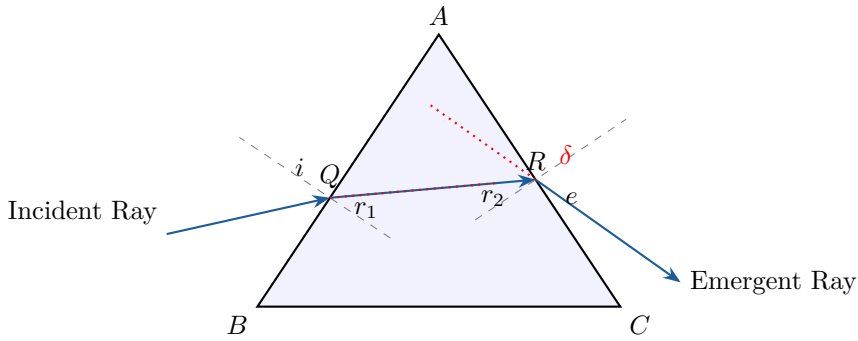


Figure 5.1: Refraction of a Monochromatic Light Ray Through a Triangular Glass Prism.

Condition for Minimum Deviation & The Prism Formula

1. Minimum Deviation ($\delta = \delta_m$): Experimentally, as the angle of incidence i increases, the deviation δ first decreases to a unique minimum value δ_m, and then increases. By the principle of optical reversibility, at minimum deviation the light path is strictly symmetric:

$$i = e \quad \text{and} \quad r_1 = r_2 = \frac{A}{2} \tag{5.18}$$

The refracted ray QR inside the prism travels **strictly parallel to the base** BC for an equilateral or isosceles prism. Substituting into the deviation relation:

$$\delta_m = 2i - A \implies i = \frac{A + \delta_m}{2} \tag{5.19}$$

Applying Snell's law at the first refracting surface ($\sin i = \mu \sin r_1$) yields the celebrated **Prism Formula**:

$$\mu = \frac{\sin\left(\frac{\mathbf{A}+\delta_\mathbf{m}}{\mathbf{2}}\right)}{\sin\left(\frac{\mathbf{A}}{\mathbf{2}}\right)} \tag{5.20}$$

2. Thin Prism Approximation ($A \leq 10^\circ$): For small refracting angles, the small-angle approximation $\sin\theta \approx \theta$ applies:

$$\frac{A+\delta}{2} \approx \mu\frac{A}{2} \implies \delta = (\mu - \mathbf{1})\mathbf{A} \tag{5.21}$$

3. Angular Dispersion & Dispersive Power (ω): Because refractive index depends inversely on wavelength ($\mu_V > \mu_R$ by Cauchy's dispersion formula $\mu \approx a + \frac{b}{\lambda^2}$), violet light deviates most and red least:

- Angular Dispersion: $\theta = \delta_\mathbf{V} - \delta_\mathbf{R} = (\mu_\mathbf{V} - \mu_\mathbf{R})\mathbf{A}$
- Dispersive Power: $\omega = \frac{\delta_\mathbf{V}-\delta_\mathbf{R}}{\delta_\mathbf{Y}} = \frac{\mu_\mathbf{V}-\mu_\mathbf{R}}{\mu_\mathbf{Y}-\mathbf{1}}$ (Completely independent of prism angle A).

Magnifying Power of Optical Instruments: Microscopes & Telescopes

1. Simple Microscope (Magnifying Glass):

- Image formed at Near Point ($D = 25\,\text{cm}$): $\mathbf{m = 1 + \frac{D}{f}}$ (Maximum magnification, strained eye).
- Image formed at Infinity (Normal Adjustment): $\mathbf{m = \frac{D}{f}}$ (Relaxed eye).

2. Compound Microscope: Objective lens (f_o, small aperture) + Eyepiece (f_e, larger aperture).

- Total Magnifying Power: $m = m_o \times m_e = \left(-\frac{v_o}{u_o}\right) m_e$.
- **Near Point Adjustment (D):** $\mathbf{m = -\frac{v_o}{u_o}\left(1 + \frac{D}{f_e}\right)}$, Tube length $L = v_o + |u_e|$.
- **Normal Adjustment (Infinity):** $\mathbf{m \approx -\frac{L}{f_o}\frac{D}{f_e}}$, Tube length $L = v_o + f_e$.

3. Astronomical Refracting Telescope: Objective (f_o, large aperture) + Eyepiece (f_e, small aperture).

- **Normal Adjustment (Image at Infinity):** $\mathbf{m = -\frac{f_o}{f_e}}$, Tube length $\mathbf{L = f_o + f_e}$.

- **Near Point Adjustment (D):** $\mathbf{m = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)}$, Tube length $L = f_o + \frac{f_e D}{f_e + D}$.

Syllabus Boundary Alert: Deletion of Resolving Power of Optical Instruments

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG curriculum:

- **STRICTLY DELETED:** The theoretical formulas for **Resolving Power of a Microscope** (RP $= \frac{2\mu \sin\theta}{1.22\lambda}$) and **Resolving Power of an Astronomical Telescope** (RP $= \frac{D}{1.22\lambda}$), along with Rayleigh's criterion angular limit $\theta = \frac{1.22\lambda}{D}$, have been **completely deleted**.
- **EXAM ORIENTATION:** Candidates must focus on Magnifying Power derivations, tube lengths, and ray diagrams of microscopes/telescopes without computing resolving powers.

5.3 Wave Optics: Huygens' Principle, Interference & Diffraction

Wavefronts, Huygens' Secondary Wavelets & Interference Physics

1. Wavefront Classification: A wavefront is the locus of all adjacent points in a medium vibrating with identical instantaneous phase:

- **Spherical Wavefront:** Emitted by an isotropic point source; intensity falls as $I \propto \frac{1}{r^2}$, amplitude $A \propto \frac{1}{r}$.
- **Cylindrical Wavefront:** Emitted by a linear slit source; intensity $I \propto \frac{1}{r}$, amplitude $A \propto \frac{1}{\sqrt{r}}$.
- **Plane Wavefront:** Formed at infinite distance from any source; intensity and amplitude remain constant.
- Light rays are everywhere perpendicular (normal) to the advancing wavefront.

2. Huygens' Principle:

1. Every point on a primary wavefront acts as a source of secondary spherical wavelets spreading in all forward directions at the speed of the wave.
2. The forward envelope (tangential surface) touching all these secondary wavelets at any subsequent instant defines the new position of the wavefront.

3. Superposition of Two Coherent Waves: Let two waves of identical frequency have amplitudes a_1, a_2 and relative phase difference ϕ:

$$y_1 = a_1 \sin(\omega t), \qquad y_2 = a_2 \sin(\omega t + \phi) \tag{5.22}$$

Resultant amplitude A_R and resultant intensity I_R:

$$\mathbf{A_R = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\phi}}, \qquad \mathbf{I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi} \tag{5.23}$$

If two sources are identical ($I_1 = I_2 = I_0$):

$$\mathbf{I_R = 2I_0(1 + \cos\phi) = 4I_0\cos^2\left(\frac{\phi}{2}\right)} \tag{5.24}$$

- **Constructive Interference (Bright Fringes):** $\phi = 2n\pi \iff \Delta x = n\lambda \implies \mathbf{I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2 = 4I_0}$.
- **Destructive Interference (Dark Fringes):** $\phi = (2n-1)\pi \iff \Delta x = (2n-1)\frac{\lambda}{2} \implies \mathbf{I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2 = 0}$.
- **Fringe Contrast / Visibility:** $V = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$.

Young's Double Slit Experiment (YDSE): Path Difference & Fringe Width

Two narrow slits S_1, S_2 separated by distance d illuminate a screen at distance $D \gg d$. Let P be a point at height y from the central line.

1. Path difference $\Delta x = S_2P - S_1P$:

$$S_2P^2 - S_1P^2 = \left[D^2 + \left(y + \frac{d}{2}\right)^2\right] - \left[D^2 + \left(y - \frac{d}{2}\right)^2\right] = 2yd \tag{5.25}$$

$$(S_2P - S_1P)(S_2P + S_1P) = 2yd \implies \Delta x \approx \frac{2yd}{2D} \implies \mathbf{\Delta x = \frac{yd}{D} = d\sin\theta} \tag{5.26}$$

2. **Bright Fringe Positions (Maxima):**

$$\Delta x = n\lambda \implies \frac{y_n d}{D} = n\lambda \implies \mathbf{y_n = \frac{n\lambda D}{d}} \quad (n = 0, \pm1, \pm2, \dots) \tag{5.27}$$

3. **Dark Fringe Positions (Minima):**

$$\Delta x = (2n-1)\frac{\lambda}{2} \implies \mathbf{y_n' = (2n-1)\frac{\lambda D}{2d}} \quad (n = 1, 2, 3, \dots) \tag{5.28}$$

4. **Fringe Width (β):** Distance separating any two consecutive bright or dark fringes:

$$\beta = \mathbf{y_{n+1} - y_n = \frac{\lambda D}{d}} \tag{5.29}$$

   β is completely uniform across all fringe orders on the screen.

5. **Angular Fringe Width (θ_0):**

$$\theta_0 = \frac{\beta}{D} = \frac{\lambda}{d} \quad \text{(Independent of screen distance } D\text{)} \tag{5.30}$$

6. **Fringe Shift due to Transparent Thin Sheet of Thickness t and Index μ:** Introducing the sheet in front of one slit introduces additional optical path $(\mu - 1)t$:

$$\boldsymbol{\Delta y_0 = \frac{(\mu - 1)tD}{d} = \frac{(\mu - 1)t}{\lambda}\beta} \tag{5.31}$$

The entire fringe pattern shifts toward the covered slit without changing fringe width β.

Single-Slit Fraunhofer Diffraction: Minima & Central Maximum Width

Light of wavelength λ illuminates a single rectangular slit of width a.

1. **Diffraction Minima Condition:** The slit aperture is divided into $2n$ equal zones that cancel pairwise:

$$\boldsymbol{a \sin\theta = n\lambda} \quad (n = \pm 1, \pm 2, \pm 3, \dots) \tag{5.32}$$

For small angles: $\theta_n \approx \frac{n\lambda}{a}$.

2. **Secondary Maxima Condition:**

$$\boldsymbol{a \sin\theta \approx (2n + 1)\frac{\lambda}{2}} \quad (n = 1, 2, 3, \dots) \tag{5.33}$$

Peak intensities drop precipitously: $I_0 : \frac{I_0}{22} : \frac{I_0}{61} : \dots$

3. **Angular Width of Central Maximum ($2\theta_0$):** Bounded by the first diffraction minima on either side ($n = \pm 1$):

$$\boldsymbol{2\theta_0 = \frac{2\lambda}{a}} \tag{5.34}$$

4. **Linear Width of Central Diffraction Maximum (W):**

$$\boldsymbol{W = 2y_1 = 2\left(\frac{\lambda D}{a}\right) = \frac{2\lambda D}{a}} \tag{5.35}$$

Notice: The central diffraction peak is **twice as wide** as secondary maxima ($W = 2\beta_{\text{diff}}$).

Syllabus Boundary Alert: Complete Deletion of Light Polarisation

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG syllabus issued by the National Medical Commission:

- **STRICTLY DELETED:** The entire topic of **Polarisation of Light** has been **completely eliminated**.
- Deleted concepts include: Brewster's Law of polarisation by reflection ($\tan i_p = \mu$), Brewster's angle, Malus' Law of transmitted intensity ($I = I_0 \cos^2\theta$), Po-

laroid sheets, polarisation by scattering, and optical activity.
- **EXAM ORIENTATION:** Candidates must focus solely on Wavefronts, Huygens' Principle, YDSE Interference, and Single-Slit Fraunhofer Diffraction.

5.4 Dual Nature of Matter & Radiation: Photoelectric Effect & de Broglie Waves

Photoelectric Effect: Experimental Laws & Einstein's Quantum Derivation

1. Key Experimental Observations (Hertz, Hallwachs, Lenard):

- **Instantaneous Emission:** Electron emission occurs without any measurable time lag ($< 10^{-9}\,\text{s}$), contradicting classical wave continuous energy buildup.
- **Threshold Frequency (ν_0):** For each photosensitive metal, there exists a minimum characteristic threshold frequency ν_0 below which **no photoelectrons are emitted**, regardless of incident light intensity or duration.
- **Kinetic Energy vs. Frequency:** Maximum kinetic energy $K_{\max}$ is strictly linear with incident frequency ν and completely independent of light intensity.
- **Saturation Current vs. Intensity:** Photoelectric saturation current is strictly proportional to incident light intensity (photon flux).

2. Einstein's Photoelectric Equation Derivation: Light consists of discrete energy quanta called **photons**, each carrying energy $E = h\nu$. When a photon of energy $h\nu$ collides with a conduction electron, its entire energy is transferred to that single electron. A portion equal to the metal's **work function** ($\Phi_0 = h\nu_0$) is consumed to overcome surface binding; the remainder appears as maximum electron kinetic energy:

$$h\nu = \Phi_0 + K_{\max} \implies \mathbf{K_{\max} = h\nu - \Phi_0 = h(\nu - \nu_0) = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}} \tag{5.36}$$

where $h = 6.626 \times 10^{-34}\,\text{J}\cdot\text{s}$ is Planck's constant ($hc \approx 12400\,\text{eV}\cdot\text{Å} \approx 1240\,\text{eV}\cdot\text{nm}$). **3. Stopping Potential (V_0):** The negative retarding collector potential required to halt the most energetic photoelectrons:

$$K_{\max} = eV_0 \implies \mathbf{V_0 = \left(\frac{h}{e}\right)\nu - \frac{\Phi_0}{e} = \left(\frac{h}{e}\right)(\nu - \nu_0)} \tag{5.37}$$

Universal Graph Property: In the V_0 vs. ν straight-line graph, the slope is a universal constant:

$$\text{Slope} = \frac{\mathbf{h}}{\mathbf{e}} \approx \mathbf{4.14 \times 10^{-15}}\,\text{V}\cdot\text{s} \tag{5.38}$$

The horizontal axis intercept is threshold frequency ν_0; vertical axis intercept is $-\Phi_0/e$.

de Broglie Matter Waves: Particle-Wave Duality Expressions

Louis de Broglie postulated that any moving material particle of mass m and velocity v possesses an associated matter wave:

$$\lambda = \frac{\mathbf{h}}{\mathbf{p}} = \frac{\mathbf{h}}{\mathbf{mv}} = \frac{\mathbf{h}}{\sqrt{\mathbf{2mK}}} \tag{5.39}$$

where K is kinetic energy. **1. De Broglie Wavelength of Charged Particles Accelerated through Potential V:** $K = qV \implies \lambda = \frac{h}{\sqrt{2mqV}}$.

- **Electron** ($m = 9.109 \times 10^{-31}\,\text{kg}, q = e$)**:**

$$\lambda_{\mathbf{e}} = \frac{\mathbf{1.227}}{\sqrt{\mathbf{V}}}\,\text{nm} = \frac{\mathbf{12.27}}{\sqrt{\mathbf{V}}}\,\text{Å} = \sqrt{\frac{\mathbf{150}}{\mathbf{V}}}\,\text{Å} \tag{5.40}$$

- **Proton** ($m_p = 1.673 \times 10^{-27}\,\text{kg}, q = e$)**:**

$$\lambda_{\mathbf{p}} = \frac{\mathbf{0.286}}{\sqrt{\mathbf{V}}}\,\text{Å} \tag{5.41}$$

- **Deuteron** ($m_d = 2m_p, q = e$)**:**

$$\lambda_{\mathbf{d}} = \frac{\mathbf{0.202}}{\sqrt{\mathbf{V}}}\,\text{Å} \tag{5.42}$$

- **Alpha Particle** ($m_\alpha = 4m_p, q = 2e$)**:**

$$\lambda_\alpha = \frac{\mathbf{0.101}}{\sqrt{\mathbf{V}}}\,\text{Å} \tag{5.43}$$

2. Thermal Neutrons in Equilibrium at Temperature T: Mean translational kinetic energy $K = \frac{3}{2}k_BT$:

$$\lambda_{\text{thermal}} = \frac{\mathbf{h}}{\sqrt{\mathbf{3mk_BT}}} \tag{5.44}$$

Syllabus Boundary Alert: Deletion of Davisson-Germer Experiment

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG curriculum:

- **STRICTLY DELETED:** The **Davisson-Germer Experiment** (nickel crystal electron diffraction, Bragg peak at 54 V and $\theta = 50°$, accelerating electron gun details) has been **formally removed**.

- **EXAM ORIENTATION:** Candidates must be proficient in Einstein's photoelectric equation, photon momentum calculations, and de Broglie wavelength formulas for accelerated particles without studying the experimental apparatus of Davisson and Germer.

5.5 Atomic Physics: Bohr's Postulates, Spectral Series & Energy Levels

Rutherford Scattering & Bohr's Atomic Postulates

1. Rutherford Scattering (Geiger-Marsden Experiment): Distance of closest approach r_0 for an α-particle ($q_\alpha = 2e$) directed head-on with kinetic energy K toward a nucleus of atomic number Z:

$$K = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r_0} \implies \mathbf{r_0 = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K}} \tag{5.45}$$

Impact parameter: $b = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2\cot(\theta/2)}{K}$. **2. Bohr's Foundational Postulates for Hydrogen-Like Species:**

1. **Stable Stationary Orbits:** An orbital electron circles the positive nucleus solely under the Coulomb attraction without radiating electromagnetic energy, despite continuous centripetal acceleration.
2. **Quantization of Orbital Angular Momentum:** Stable orbits are restricted strictly to those where orbital angular momentum is an integer multiple of $\hbar = h/2\pi$:

$$\mathbf{L = m_e vr = \frac{nh}{2\pi}} \quad (n = 1, 2, 3, \dots) \tag{5.46}$$

3. **Frequency Condition (Radiative Transitions):** Photons are emitted or absorbed exclusively when an electron undergoes a discrete transition between stationary energy levels E_i and E_f:

$$\mathbf{\Delta E = E_i - E_f = h\nu = \frac{hc}{\lambda}} \tag{5.47}$$

Bohr Atom Formulas: Rigorous Analytical Derivations for Hydrogen-Like Species

For a hydrogen-like atom of nuclear charge $+Ze$ and electron mass m:

1. **Electrostatic Centripetal Equilibrium:**

$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2} \implies mv^2 r = \frac{Ze^2}{4\pi\varepsilon_0} \tag{Eq. 1}$$

2. **Angular Momentum Quantization:**

$$mvr = \frac{nh}{2\pi} \implies v = \frac{nh}{2\pi mr} \qquad \text{(Eq. 2)}$$

3. **Orbital Radius Derivation:** Substituting Eq. 2 into Eq. 1:

$$m\left(\frac{n^2h^2}{4\pi^2m^2r^2}\right)r = \frac{Ze^2}{4\pi\varepsilon_0} \implies \mathbf{r_n = \frac{\varepsilon_0 n^2 h^2}{\pi m Z e^2} = r_0\frac{n^2}{Z} = 0.529\frac{n^2}{Z}}\ \text{Å} \quad (5.48)$$

where $r_0 = 0.529177\,\text{Å}$ is the first Bohr radius of hydrogen ($n = 1, Z = 1$).

4. **Orbital Velocity Derivation:**

$$\mathbf{v_n = \frac{Ze^2}{2\varepsilon_0 nh} = v_0\frac{Z}{n} = (2.18\times10^6\,\text{m/s})\frac{Z}{n} = c\alpha\frac{Z}{n}} \qquad (5.49)$$

where $\alpha = \frac{e^2}{2\varepsilon_0 hc} \approx \frac{1}{137}$ is the fine-structure constant.

5. **Energies in the n-th Stationary Orbit:**
   - **Kinetic Energy:** $\mathbf{K_n = \frac{1}{2}mv_n^2 = \frac{mZ^2e^4}{8\varepsilon_0^2 n^2h^2} = +13.6\frac{Z^2}{n^2}}$ eV.
   - **Potential Energy:** $\mathbf{U_n = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n} = -\frac{mZ^2e^4}{4\varepsilon_0^2 n^2h^2} = -27.2\frac{Z^2}{n^2}}$ eV.
   - **Total Mechanical Energy:**

$$\mathbf{E_n = K_n + U_n = -\frac{mZ^2e^4}{8\varepsilon_0^2 n^2h^2} = -13.6\frac{Z^2}{n^2}}\ \text{eV} \qquad (5.50)$$

   - **Golden Energy Ratios:**

$$\mathbf{E_n = -K_n = \frac{U_n}{2}} \qquad (5.51)$$

Hydrogen Spectral Emission Series & Rydberg Equation

When an electron drops from an upper stationary orbit n_2 to a lower orbit n_1 ($n_2 > n_1$):

$$\Delta E = E_{n_2} - E_{n_1} = 13.6Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)\ \text{eV} \qquad (5.52)$$

$$\mathbf{\frac{1}{\lambda} = \overline{\nu} = R_\infty Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)} \qquad (5.53)$$

where $R_\infty = \frac{me^4}{8\varepsilon_0^2 ch^3} \approx 1.09737\times10^7\,\text{m}^{-1}$ is the Rydberg constant ($\frac{1}{R_\infty} \approx 912\,\text{Å}$).

Series	n_1	n_2	Domain	Range (λ)
Lyman	1	$2, 3, \ldots$	**UV**	91.2–121.6 nm
Balmer	2	$3, 4, \ldots$	**Visible**	364.6–656.3 nm
Paschen	3	$4, 5, \ldots$	**Near IR**	820.4–1875 nm
Brackett	4	$5, 6, \ldots$	**Mid IR**	1458–4051 nm
Pfund	5	$6, 7, \ldots$	**Far IR**	2279–7458 nm

Total Spectral Lines Emitted from Excited Level n to Ground State:

$$\mathbf{N = \frac{n(n-1)}{2}} \tag{5.54}$$

5.6 Nuclear Physics: Composition, Density, Mass Defect & Binding Energy

Nuclear Dimensions, Universal Nuclear Density & Strong Nuclear Force

1. Nuclear Constituents: Atomic nucleus contains Z protons and $N = A-Z$ neutrons (nucleons). **2. Nuclear Radius Empirical Law:**

$$\mathbf{R = R_0 A^{1/3}} \tag{5.55}$$

where $R_0 \approx 1.2 \times 10^{-15}\,\text{m} = 1.2\,\text{fm}$. **3. Proof of Constant Nuclear Density (ρ_{nuc}):** Nuclear volume: $V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi(R_0 A^{1/3})^3 = \frac{4}{3}\pi R_0^3 A$. Nuclear mass: $M \approx A \cdot m_N$ (where nucleon mass $m_N \approx 1.66 \times 10^{-27}\,\text{kg}$).

$$\rho_{\text{nuc}} = \frac{\mathbf{M}}{\mathbf{V}} = \frac{\mathbf{A m_N}}{\frac{4}{3}\pi \mathbf{R_0^3 A}} = \frac{\mathbf{3 m_N}}{\mathbf{4\pi R_0^3}} \approx \mathbf{2.3 \times 10^{17}}\,\text{kg/m}^{\mathbf{3}} \tag{5.56}$$

Nuclear density is an **absolute universal constant**, completely independent of mass number A. **4. Properties of the Strong Nuclear Force:**

- Strongest fundamental force in nature ($\sim$ 100 times stronger than electrostatic repulsion).
- Strictly short-ranged: active solely across $1-2\,\text{fm}$. For $r < 0.8\,\text{fm}$, it becomes strongly repulsive.
- Charge-independent: $F_{p-p} = F_{p-n} = F_{n-n}$ under identical quantum states.
- Saturated: each nucleon interacts exclusively with its immediate neighbouring nucleons.

Mass Defect, Nuclear Binding Energy Curve & Energy Reactions

1. Mass Defect (Δm): The rest mass of any stable bound nucleus is strictly less than the combined rest masses of its constituent free protons and neutrons:

$$\mathbf{\Delta m = [Z m_p + (A - Z) m_n] - M_{\text{nucleus}}(A, Z)} \tag{5.57}$$

2. Nuclear Binding Energy (E_b): By Einstein's mass-energy equivalence ($E = \Delta m c^2$):

$$\mathbf{E_b = \Delta m c^2 = \Delta m\,[u] \times 931.5\,MeV} \tag{5.58}$$

where $1\,\text{u} \equiv 931.494\,\text{MeV}/c^2$. **3. Binding Energy per Nucleon (E_b/A Curve):** The stability of a nucleus is dictated strictly by binding energy per nucleon (E_b/A):

- Rapid increase from light nuclei (^{2}H ≈ 1.1 MeV/nucleon) with local peaks at tightly bound alpha-multiples (^{4}He,^{12} C,^{16} O).
- Broad flat maximum in intermediate mass range $30 < A < 120$ with peak at $^{\mathbf{56}}$Fe $\approx$ **8.75** MeV/nucleon.
- Gradual drop toward heavy nuclei due to cumulative Coulomb repulsion between protons, dropping to ≈ 7.6 MeV/nucleon for ^{238}U.
- **Nuclear Fission:** Heavy nucleus splitting ($A \sim 240 \rightarrow A_1 + A_2 \sim 120$) gains ≈ 0.9 MeV/nucleon $\implies Q \approx 240 \times 0.9 \approx$ **200** MeV per fission.
- **Nuclear Fusion:** Two very light nuclei fusing into a heavier nucleus (^{2}H+^{3} H $\rightarrow^4$ He $+ n$) releases $\approx$ **17.6** MeV (3.5 MeV/nucleon).

Syllabus Boundary Alert: Complete Deletion of Radioactivity

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG syllabus issued by the National Medical Commission:

- **STRICTLY DELETED:** The entire subtopic of **Radioactivity** has been **completely eliminated**.
- Deleted topics include: α-decay, β-decay (β^- and β^+), γ-decay, Radioactive Decay Law ($N(t) = N_0 e^{-\lambda t}$), Half-life ($T_{1/2} = \frac{0.693}{\lambda}$), Mean Life ($\tau = \frac{1}{\lambda}$), and Activity ($A = \lambda N$).
- **EXAM ORIENTATION:** Candidates must focus on Mass Defect, Binding Energy per Nucleon, Nuclear Density, and Fission/Fusion Q-values without solving radioactive decay kinematics.

5.7 Semiconductor Physics, Diodes & Digital Logic Gates

Energy Bands, Intrinsic & Extrinsic Semiconductors

1. Energy Band Theory of Solids:

- **Conductors:** Conduction band and valence band overlap, or conduction band is partially filled ($E_g = 0$). High electron density $n \sim 10^{28}$ m^{-3}.
- **Insulators:** Large forbidden energy gap: $\mathbf{E_g > 3.0}$ eV (e.g., Diamond $E_g \approx 5.4$ eV).
- **Semiconductors:** Narrow forbidden energy gap: $\mathbf{E_g \leq 3.0}$ eV (Si : $E_g = 1.1$ eV; Ge : $E_g = 0.72$ eV; GaAs : $E_g = 1.42$ eV).

2. Intrinsic Semiconductors (Pure Crystals): Thermal energy excites valence electrons across E_g into the conduction band, leaving equal numbers of vacant states (**holes**) in the valence band:

$$\mathbf{n_e = n_h = n_i} \tag{5.59}$$

Total electrical conductivity: $\sigma = \mathbf{e}(\mathbf{n_e}\mu_\mathbf{e} + \mathbf{n_h}\mu_\mathbf{h})$. **3. Extrinsic Semiconductors (Doped Crystals):**

- **n-Type:** Doped with pentavalent donor impurities (P, As, Sb). Introduces donor discrete energy level E_d just below conduction band (0.05 eV for Si).

$$\mathbf{n_e} \approx \mathbf{N_D} \gg \mathbf{n_h} \quad \text{(Electrons: majority carriers; Holes: minority carriers)} \tag{5.60}$$

- **p-Type:** Doped with trivalent acceptor impurities (B, Al, In, Ga). Introduces acceptor discrete energy level E_a just above valence band (0.05 eV for Si).

$$\mathbf{n_h} \approx \mathbf{N_A} \gg \mathbf{n_e} \quad \text{(Holes: majority carriers; Electrons: minority carriers)} \tag{5.61}$$

- **Law of Mass Action (Thermal Equilibrium):**

$$\mathbf{n_e} \cdot \mathbf{n_h} = \mathbf{n_i^2(T)} \tag{5.62}$$

Both n-type and p-type semiconductors are strictly **electrically neutral** overall.

p-n Junction Diode Physics, Biasing & Rectifiers

1. p-n Junction Formation: Majority holes from p-side diffuse into n-side, and majority electrons from n-side diffuse into p-side. Uncompensated immobile donor ions (+) on n-side and acceptor ions (−) on p-side form a **depletion layer** and built-in **barrier potential** ($V_0 \approx 0.7$ V for Si, 0.3 V for Ge) pointing from n to p. **2. Biasing Modes:**

- **Forward Bias (p to +, n to −):** External field opposes barrier field $\implies$ barrier potential lowers to $(V_0 - V)$, depletion layer narrows, forward current rises exponentially: $I \approx I_0(e^{eV/\eta k_B T} - 1)$.
- **Reverse Bias (p to −, n to +):** External field aids barrier field $\implies$ barrier increases to $(V_0 + V)$, depletion layer widens, current is tiny minority saturation current $I_0 \sim \mu$A (Si: nA).

3. Diode Rectifiers:

- **Half-Wave Rectifier:** One diode conducting during positive AC half-cycles.

$$\eta_{\max} = \frac{\mathbf{0.406 R_L}}{\mathbf{r_f + R_L}} \approx \mathbf{40.6}\%, \qquad \nu_{\text{ripple}} = \nu_{\text{AC}} \tag{5.63}$$

- **Full-Wave Rectifier (Center-Tapped or Bridge):** Two or four diodes conducting alternately.

$$\eta_{\max} = \frac{\mathbf{0.812 R_L}}{\mathbf{r_f + R_L}} \approx \mathbf{81.2}\%, \qquad \nu_{\text{ripple}} = \mathbf{2}\nu_{\text{AC}} \tag{5.64}$$

Special Purpose Diodes: Zener, Photodiode, LED & Solar Cell

1. Zener Diode as a Voltage Regulator: Heavily doped p-n junction with an ultra-thin depletion layer ($\sim 10^{-6}$ m) that undergoes sharp reverse breakdown at a fixed Zener breakdown voltage V_z. Connected in **reverse bias** parallel with fluctuating load R_L:

$$V_{\text{out}} = V_z = \text{constant}, \qquad I_{\text{supply}} = I_z + I_L \tag{5.65}$$

Series current-limiting resistor R_s:

$$\mathbf{R_s = \frac{V_{\text{unregulated}} - V_z}{I_z + I_L}} \tag{5.66}$$

2. Photodiode: Operates in **reverse bias**. Incident photons ($h\nu > E_g$) generate electron-hole pairs in the depletion region. Fractional change in minority reverse current is much larger than in forward current, making reverse bias ideal for optical detection. **3. Light Emitting Diode (LED):** Operates in **forward bias**. Electrons recombine with holes at junction, releasing recombination energy as spontaneous photons across direct bandgap: $\lambda = \frac{\mathbf{hc}}{\mathbf{E_g}}$. **4. Solar Cell:** Unbiased p-n junction operating in the **fourth quadrant** of $I - V$ characteristic, converting optical solar flux into electrical power.

Digital Logic Gates: Truth Tables & De Morgan's Theorems

1. Fundamental Logic Gates:

- **AND Gate:** $Y = A \cdot B$ (Output is 1 strictly when both inputs are 1).
- **OR Gate:** $Y = A + B$ (Output is 1 when at least one input is 1).
- **NOT Gate (Inverter):** $Y = \overline{A}$ ($0 \to 1, 1 \to 0$).

2. Universal Logic Gates (Can Synthesize Any Boolean Function):

- **NAND Gate:** $Y = \overline{A \cdot B}$.
- **NOR Gate:** $Y = \overline{A + B}$.

3. De Morgan's Master Theorems:

$$\mathbf{\overline{A + B} = \overline{A} \cdot \overline{B}}, \qquad \mathbf{\overline{A \cdot B} = \overline{A} + \overline{B}} \tag{5.67}$$

Syllabus Boundary Alert: Complete Deletion of Transistors

NMC REVISION WARNING FOR NEET UG: Under the revised NEET UG syllabus issued by the National Medical Commission:

- **STRICTLY DELETED:** The entire topic of **Transistors** (bipolar junction transistors npn and pnp, emitter, base, collector actions) has been **completely eliminated**.
- Deleted subtopics include: Common Emitter (CE) input and output charac-

teristics, current gain factors ($\alpha = I_c/I_e, \beta = I_c/I_b, \beta = \frac{\alpha}{1-\alpha}$), transistor as a CE amplifier (voltage gain $A_v = -\beta\frac{R_L}{R_i}$), transistor as a digital switch, and transistor as an oscillator.

- **EXAM ORIENTATION:** Candidates must focus exclusively on Semiconductors, p-n junction diodes, rectifiers, Zener diode regulators, optoelectronic diodes (LED, photodiode, solar cell), and Digital Logic Gates.

NCERT High-Yield Extraction: Optoelectronic Materials & Eye Sensitivity

The semiconductor bandgap for visible LEDs must exceed 1.8 eV ($hc/\lambda_{\max} \approx 1240/700\,\text{nm} \approx 1.8\,\text{eV}$).

- **Gallium Arsenide Phosphide ($GaAs_{1-x}P_x$):** Used for red, amber, and yellow LEDs ($E_g \approx 1.9\,\text{eV}$).
- **Gallium Nitride (GaN):** Direct bandgap of 3.4 eV enables bright blue and ultraviolet LEDs.
- **Photodiode Detection Limit:** To detect an optical signal of wavelength λ, the semiconductor bandgap must satisfy $\mathbf{E_g \leq \frac{hc}{\lambda}}$.

5.8 Master High-Yield Problem Archetypes & Worked Solutions

Archetype 1: Silvered Plano-Convex Lens Equivalent Mirror Focal Length

Problem: A plano-convex lens of refractive index $\mu = 1.5$ has a curved surface of radius of curvature $R = 20\,\text{cm}$. Compute the equivalent focal length when: (i) the flat plane surface is silvered, and (ii) the curved surface is silvered. **Step-by-Step Solution:**

1. Power of the plano-convex lens in air:

$$P_L = \frac{1}{f_L} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.5 - 1)\left(\frac{1}{20} - 0\right) = \frac{0.5}{20} = \frac{1}{40}\,\text{cm}^{-1} \tag{5.68}$$

2. Any silvered lens system acts as an equivalent mirror of power:

$$P_{\text{eq}} = 2P_L + P_M \implies \mathbf{F}_{\text{eq}} = -\frac{\mathbf{1}}{\mathbf{P}_{\text{eq}}} \tag{5.69}$$

3. **Case (i): Flat Plane Surface is Silvered:** Plane mirror has $R_M = \infty \implies P_M = -\frac{1}{f_M} = 0$.

$$P_{\text{eq}} = 2P_L + 0 = 2\left(\frac{1}{40}\right) = \frac{1}{20}\,\text{cm}^{-1} \implies \mathbf{F}_{\text{eq}} = -\mathbf{20}\,\text{cm} \tag{5.70}$$

Acts as a **concave mirror of focal length** 20 cm.

4. **Case (ii): Curved Surface is Silvered:** Concave mirror formed by silvered curved face has $R_M = 20\,\text{cm} \implies f_M = -10\,\text{cm} \implies P_M = -\frac{1}{f_M} = +\frac{1}{10}\,\text{cm}^{-1}$.

$$P_{\text{eq}} = 2P_L + P_M = 2\left(\frac{1}{40}\right) + \frac{1}{10} = \frac{3}{20}\,\text{cm}^{-1} \tag{5.71}$$

$$\implies \mathbf{F}_{\text{eq}} = -\frac{\mathbf{20}}{\mathbf{3}}\,\text{cm} \approx -\mathbf{6.67}\,\text{cm} \tag{5.72}$$

Acts as a **concave mirror of focal length** 6.67 cm.

Archetype 2: Multi-Wavelength Bright Fringe Coincidence in YDSE

Problem: In a YDSE setup with slit separation $d = 1.0\,\text{mm}$ and screen distance $D = 1.0\,\text{m}$, light containing two wavelengths $\lambda_1 = 600\,\text{nm}$ and $\lambda_2 = 480\,\text{nm}$ is used. Determine the minimum non-zero distance from the central maximum where bright fringes of both wavelengths coincide. **Step-by-Step Solution:**

1. For bright fringes to coincide at position y:

$$y = \frac{n_1\lambda_1 D}{d} = \frac{n_2\lambda_2 D}{d} \implies n_1\lambda_1 = n_2\lambda_2 \tag{5.73}$$

2. Taking the ratio of integers:

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{480\,\text{nm}}{600\,\text{nm}} = \frac{4}{5} \tag{5.74}$$

3. Smallest non-zero integers satisfying this condition: $n_1 = 4$ and $n_2 = 5$.
4. Minimum distance y_{min}:

$$y_{\text{min}} = \frac{4 \times (600 \times 10^{-9}\,\text{m}) \times 1.0\,\text{m}}{1.0 \times 10^{-3}\,\text{m}} = 2.4 \times 10^{-3}\,\text{m} = \mathbf{2.4}\,\text{mm} \tag{5.75}$$

The 4th bright fringe of λ_1 coincides with the 5th bright fringe of λ_2 at $y = 2.4\,\text{mm}$.

Archetype 3: Zener Diode Voltage Regulator Design Calculations

Problem: A Zener diode with breakdown voltage $V_z = 10\,\text{V}$ and maximum power dissipation $P_{z,\text{max}} = 1.0\,\text{W}$ is used to regulate a DC supply that fluctuates between 15 V and 25 V. The load resistance is $R_L = 1.0\,\text{k}\Omega$. Calculate: (i) the load current I_L, and (ii) the suitable series resistance R_s to ensure the Zener current remains within safe limits. **Step-by-Step Solution:**

1. The regulated voltage across the load is fixed at $V_L = V_z = 10\,\text{V}$.

2. Constant load current:

$$I_L = \frac{V_z}{R_L} = \frac{10\,\text{V}}{1000\,\Omega} = 10\,\text{mA} = 0.01\,\text{A} \tag{5.76}$$

3. Maximum safe Zener diode current:

$$I_{z,\max} = \frac{P_{z,\max}}{V_z} = \frac{1.0\,\text{W}}{10\,\text{V}} = 100\,\text{mA} = 0.10\,\text{A} \tag{5.77}$$

4. Total maximum supply current: $I_{\text{total,max}} = I_{z,\max} + I_L = 100\,\text{mA} + 10\,\text{mA} = 110\,\text{mA}$.
5. At maximum input voltage $V_{\text{in}} = 25\,\text{V}$:

$$R_{s,\min} = \frac{V_{\text{in,max}} - V_z}{I_{\text{total,max}}} = \frac{25 - 10}{0.11} = \frac{15}{0.11} \approx \mathbf{136.4\,\Omega} \tag{5.78}$$

6. At minimum input voltage $V_{\text{in}} = 15\,\text{V}$ with minimum holding current $I_{z,\min} \approx 5\,\text{mA}$:

$$R_{s,\max} = \frac{V_{\text{in,min}} - V_z}{I_{z,\min} + I_L} = \frac{15 - 10}{0.005 + 0.010} = \frac{5}{0.015} \approx \mathbf{333.3\,\Omega} \tag{5.79}$$

A standard resistance of $R_s \approx 200\,\Omega$ provides robust regulation across the entire input range.

PART VI

Experimental Physics & Laboratory Skills

CHAPTER 6

Experimental Physics & Laboratory Skills: Complete NMC Practicals Manual

Under the revised NMC guidelines, experimental physics forms a dedicated unit tested directly across the compulsory 45-question physics paper. This chapter provides a rigorous, step-by-step analytical manual for all 18 prescribed laboratory experiments, focusing on least count derivations, zero errors, reading formulas, experimental precautions, and high-frequency exam traps.

6.1 Fundamental Mechanics & Measurement Lab Protocols

6.1.1 Practical 1: Vernier Calipers — Diameter and Depth Measurement

Vernier Calipers: Governing Principles, Least Count & Zero Error

1. Theoretical Foundation & Least Count (Vernier Constant, VC): The Vernier caliper measures linear dimensions beyond the resolution of a conventional metric scale. When n Vernier Scale Divisions (VSD) coincide with m Main Scale Divisions (MSD):

$$\text{LC} = \mathbf{1}\,\text{MSD} - \mathbf{1}\,\text{VSD} = \left(\mathbf{1} - \frac{\mathbf{m}}{\mathbf{n}}\right)\text{MSD} \tag{6.1}$$

Standard Metric Instrument: $10\,\text{VSD} = 9\,\text{MSD}$ with $1\,\text{MSD} = 1\,\text{mm} = 0.1\,\text{cm}$:

$$\text{LC} = 1\,\text{mm} - \frac{9}{10}\,\text{mm} = \mathbf{0.1}\,\text{mm} = \mathbf{0.01}\,\text{cm} = \mathbf{100}\,\mu\text{m} \tag{6.2}$$

2. True Measurement Formulation:

$$\text{True Reading} = \text{MSR} + (\text{VSR} \times \text{LC}) - (\text{Zero Error}) \tag{6.3}$$

Where MSR is the main scale reading just preceding the vernier zero mark, and VSR is the vernier division index that aligns with any main scale line.

- **Positive Zero Error ($+e$):** The vernier zero mark lies to the **right** of the main scale zero mark when jaws are in contact.

$$e = +(n \times \text{LC}) \implies \text{True Reading} = \text{Observed Reading} - e \tag{6.4}$$

- **Negative Zero Error ($-e$):** The vernier zero mark lies to the **left** of the main scale zero mark when jaws are in contact.

$$e = -(N - n) \times \text{LC} \implies \text{True Reading} = \text{Observed Reading} + |e| \quad (6.5)$$

3. Operational Jaws and Application Protocols:

- **External Jaws:** Measure outer diameter (D) of cylinders, spheres, and rectangular blocks.
- **Internal Jaws:** Measure inner diameter (d) of beakers, hollow pipes, and test tubes.
- **Metallic Depth Strip (Stem):** Slides out from the rear of the beam to measure the internal depth (h) of vessels:

$$V_{\text{internal}} = \frac{\pi d^2}{4} h \quad (6.6)$$

6.1.2 Practical 2: Screw Gauge — Diameter of Thin Wire & Sheet Thickness

Screw Gauge (Micrometer): Pitch, Backlash Error & Measurements

1. Pitch & Least Count Formulations: A screw gauge utilizes the principle of translating circular rotary motion into precision linear axial displacement:

$$\text{Pitch} = \frac{\text{Distance advanced on pitch scale}}{\text{Complete revolutions of circular cap}} \quad (\text{Typ. } 1.0 \text{ or } 0.5\,\text{mm}) \quad (6.7)$$

$$\text{LC} = \frac{\text{Pitch}}{N} = \frac{1\,\text{mm}}{100} = \mathbf{0.01}\,\text{mm} = \mathbf{10}\,\mu\text{m} \quad (6.8)$$

2. True Diameter and Thickness Measurement:

$$\text{True Reading} = \text{PSR} + (\text{CSR} \times \text{LC}) - (\text{Zero Error}) \quad (6.9)$$

- **Positive Zero Error ($+e$):** The zero of the circular scale lies **below the linear reference line** when the studs touch $\implies e = +(n \times \text{LC})$.
- **Negative Zero Error ($-e$):** The zero of the circular scale lies **above the linear reference line** when the studs touch $\implies e = -(N - n) \times \text{LC}$.
- **Backlash Error:** Arises from mechanical wear and tear causing loose fitting between screw threads and internal nut. *Remedy:* Always turn the thimble in **one uniform direction only**.
- **Ratchet Safety Mechanism:** Always advance the screw using the ratchet knob until three audible clicks sound, ensuring uniform compressive pressure without deforming thin specimens.

6.1.3 Practical 3: Simple Pendulum — Energy Dissipation & Acceleration Due to Gravity (g)

Simple Pendulum: Oscillation Dynamics & Exponential Energy Dissipation

1. Acceleration Due to Gravity (g) Formulation: For small angular displacements ($\theta \leq 4°$, ensuring $\sin\theta \approx \theta$):

$$T = 2\pi\sqrt{\frac{L}{g}} \implies T^2 = \left(\frac{4\pi^2}{g}\right) L \tag{6.10}$$

$$\implies \mathbf{g = 4\pi^2\left(\frac{L}{T^2}\right) = \frac{4\pi^2}{\text{Slope of } T^2 \text{ vs } L \text{ graph}}} \tag{6.11}$$

Where effective length $L = l + h + r$ (l = length of suspension thread, h = hook length, r = radius of spherical bob).

- The graph of L versus T^2 is a **straight line passing through the origin**.

2. Energy Dissipation Protocol (Damped Oscillations): Due to viscous air drag, the mechanical energy E of the oscillating pendulum gradually dissipates:

$$E(t) = \frac{1}{2}m\omega^2 A(t)^2 \implies \mathbf{E \propto A^2} \tag{6.12}$$

The instantaneous amplitude $A(t)$ undergoes exponential damping:

$$A(t) = A_0 e^{-\gamma t} \implies A^2(t) = A_0^2 e^{-2\gamma t} \implies \ln(\mathbf{A^2}) = \ln(\mathbf{A_0^2}) - \mathbf{2\gamma t} \tag{6.13}$$

- A plot of A^2 versus time t shows a smooth **exponential decay curve**.
- A plot of $\ln(A^2)$ versus time t yields a **straight line with negative slope** (-2γ), experimentally verifying that energy dissipation is directly proportional to instantaneous kinetic energy.

6.1.4 Practical 4: Metre Scale — Determination of Mass by Principle of Moments

Principle of Moments: Rotational Equilibrium on a Knife-Edge

1. Governing Principle of Moments: A rigid uniform body supported at a fixed knife-edge pivot attains rotational equilibrium when the net external torque about the pivot vanishes:

$$\sum \boldsymbol{\tau}_{\text{pivot}} = 0 \implies \sum \tau_{\text{clockwise}} = \sum \tau_{\text{anticlockwise}} \tag{6.14}$$

2. Experimental Protocol: Suspend a known standard mass m_1 at position x_1 and an unknown mass m_2 at position x_2 on a uniform metre scale balanced at pivot position x_p:

$$m_1 g(x_p - x_1) = Mg(x_p - x_{\text{cg}}) + m_2 g(x_2 - x_p) \tag{6.15}$$

Where M is the intrinsic mass of the metre scale and x_{cg} is its center of gravity.

- **Simplified Direct Method ($x_p = x_{\text{cg}}$):** If the knife-edge is placed exactly at the center of gravity ($x_{\text{cg}} \approx 50.0\,\text{cm}$), the torque exerted by the scale's own mass vanishes ($Mg(x_p - x_{\text{cg}}) = 0$), yielding:

$$m_1(x_p - x_1) = m_2(x_2 - x_p) \implies \mathbf{m_2 = m_1\left(\frac{x_p - x_1}{x_2 - x_p}\right)} \tag{6.16}$$

- **Determination of Scale Mass M:** Balance the scale without m_2 by placing the knife-edge away from x_{cg} and counter-balancing with a single known mass m_1:

$$M = m_1\left(\frac{x_p - x_1}{x_{\text{cg}} - x_p}\right) \tag{6.17}$$

6.1.5 Practical 5: Young's Modulus of Elasticity by Searle's Apparatus

Searle's Apparatus: Stress-Strain Analysis & Elongation Calculus

1. Two-Wire Differential Suspension System: Searle's apparatus employs two identical wires of identical length L, cross-sectional radius r, and composition:

- **Reference Wire:** Suspends a constant dead load to keep the wire taut and straight. It acts as an automatic compensator for thermal expansion of the building ceiling and support girder.
- **Experimental Wire:** Carries the graduated slotted weights (M) that induce tensile longitudinal strain.

2. Governing Elasticity Equation:

$$\mathbf{Y} = \frac{\text{Tensile Stress}}{\text{Longitudinal Strain}} = \frac{\mathbf{Mg/(\pi r^2)}}{\mathbf{l/L}} = \frac{\mathbf{MgL}}{\mathbf{\pi r^2 l}} \implies \mathbf{l} = \left(\frac{\mathbf{gL}}{\mathbf{\pi r^2 Y}}\right)\mathbf{M} \tag{6.18}$$

- Wire elongation l is measured using a sensitive spirit level and spherometer-micrometer drum attached between the two frames.
- The graph of elongation l versus suspended mass M is a **straight line passing through the origin**. Young's modulus is determined from the reciprocal slope:

$$Y = \frac{gL}{\pi r^2(\text{Slope of } l \text{ vs } M \text{ graph})} \tag{6.19}$$

- **Loading and Unloading Precaution:** Readings must be recorded during both progressive loading and unloading cycles; taking the average prevents errors caused by elastic hysteresis or delayed recovery.

6.2 Hydrodynamics, Thermal Physics & Sound Lab Protocols

6.2.1 Practical 6: Surface Tension by Capillary Rise & Detergent Effects

Capillary Rise (Jurin's Law) & Surfactant Influence on Surface Tension

1. Capillary Ascent Formulation: When a narrow glass capillary tube of internal radius r is immersed vertically into water of density ρ:

$$2\pi r T\cos\theta = \pi r^2(h + r/3)\rho g \implies \mathbf{T = \frac{r(h + r/3)\rho g}{2\cos\theta} \approx \frac{rh\rho g}{2}} \tag{6.20}$$

For pure water in contact with clean glass, the contact angle is negligible ($\theta \approx 0^\circ \implies \cos\theta = 1$). The term $r/3$ is the **meniscus volume correction** for a hemispherical liquid cap.

- The capillary bore radius r is accurately determined by measuring the internal diameter at the liquid level using a horizontal travelling microscope.
- **Jurin's Rule of Constant Product:** $h{\cdot}r = \text{constant}$. Narrower capillaries produce proportionally higher ascents.

2. Effect of Adding Detergent (Surfactant Concentration):

- Detergent molecules consist of hydrophobic non-polar hydrocarbon tails and hydrophilic polar heads. They preferentially concentrate at the water-air interface, disrupting cohesive hydrogen bonding networks between adjacent water molecules.
- **Observation:** Adding detergent causes a **marked decrease in surface tension** ($T_{\text{detergent}} < T_{\text{pure}}$), resulting in an immediate and measurable **drop in capillary ascent height** h.

6.2.2 Practical 7: Viscosity of Liquid by Terminal Velocity (Stokes' Law)

Terminal Velocity in Viscous Media & Dynamic Viscosity Determination

1. Hydrodynamic Balance at Steady State: When a smooth spherical ball of radius r and density ρ falls through a wide column of viscous liquid

(such as glycerol or castor oil) of density σ and dynamic viscosity η, it rapidly accelerates until gravitational weight is exactly balanced by the sum of buoyant upward thrust and viscous drag:

$$W = F_B + F_v \implies \frac{4}{3}\pi r^3 \rho g = \frac{4}{3}\pi r^3 \sigma g + 6\pi\eta r v_t \tag{6.21}$$

$$\eta = \frac{\mathbf{2r^2(\rho - \sigma)g}}{\mathbf{9v_t}} \implies \mathbf{v_t} = \left(\frac{\mathbf{2(\rho - \sigma)g}}{\mathbf{9\eta}}\right)\mathbf{r^2} \tag{6.22}$$

2. Graphical Calculus and Experimental Protocol:

- The fall time between two horizontal reference marks separated by vertical distance h is timed after the sphere has traversed an initial lead-in distance to ensure steady terminal velocity $v_t = h/t$.
- A plot of terminal velocity v_t versus the square of radius (r^2) is a **straight line passing through the origin**:

$$\eta = \frac{2(\rho - \sigma)g}{9 \times (\text{Slope of } v_t \text{ vs } r^2 \text{ graph})} \tag{6.23}$$

- **Wall and Bottom Effect Corrections:** The glass cylinder diameter must be large compared to the sphere diameter ($D_{\text{cylinder}} > 10 \cdot r$) to ensure that boundary wall drag does not retard the terminal velocity.

6.2.3 Practical 8: Speed of Sound in Air using Resonance Tube

Resonance Tube: Closed Acoustic Columns & End Correction Calculus

1. Resonance in a Cylindrical Air Column Closed at One End: Water level in the tube provides a rigid boundary (displacement node / pressure antinode). The open mouth provides an approximate displacement antinode. Due to the inertia of moving air spilling slightly outside the open rim, the true antinode forms at an acoustic distance e outside the tube mouth:

$$l_1 + e = \frac{\lambda}{4} \quad \text{(First Resonance / Fundamental Mode)} \tag{6.24}$$

$$l_2 + e = \frac{3\lambda}{4} \quad \text{(Second Resonance / First Overtone)} \tag{6.25}$$

Subtracting the two equations eliminates the unknown end correction e:

$$l_2 - l_1 = \frac{\lambda}{2} \implies \boldsymbol{\lambda = 2(l_2 - l_1)} \tag{6.26}$$

$$\mathbf{v = f\lambda = 2f(l_2 - l_1)} \tag{6.27}$$

Where f is the verified operating frequency of the excited tuning fork. **2. Acoustic End Correction Calculation:**

$$\mathbf{e = \frac{l_2 - 3l_1}{2} \approx 0.6\,r} \qquad (r = \text{internal tube radius}) \tag{6.28}$$

- **Temperature Correction:** The measured sound speed v is normalized to 0°C via:

$$v_0 = v_t - 0.61\,t \quad (t = \text{ambient air temperature in } ^\circ\text{C}) \tag{6.29}$$

6.2.4 Practical 9: Specific Heat Capacity of Solids & Liquids by Method of Mixtures

Calorimetry: Conservation of Thermal Energy & Water Equivalent

1. Principle of Calorimetry: In an isolated thermodynamic calorimeter with negligible heat exchange with the surrounding environment, thermal equilibrium requires that the heat lost by the warmer solid equals the total heat gained by the cooler water and the containing calorimeter vessel:

$$Q_{\text{lost}} = Q_{\text{gained}} \implies m_s s_s(T_s - T) = (m_w s_w + m_c s_c)(T - T_0) \tag{6.30}$$

Where m_s is the mass of the solid specimen heated in a hypsometer to boiling temperature T_s, s_s is its specific heat capacity, m_w is the initial mass of water at ambient temperature T_0, m_c and s_c are the mass and specific heat of the copper calorimeter, and T is the steady final equilibrium temperature.

$$\mathbf{s_s = \frac{(m_w s_w + W)(T - T_0)}{m_s(T_s - T)}} \tag{6.31}$$

[where $W = m_c s_c$ is the Water Equivalent]

2. Determination of Specific Heat of an Unknown Liquid (s_l): Replace water with a known mass of liquid m_l and drop in a calibrated metal cylinder of known specific heat capacity s_s:

$$\mathbf{s_l = \frac{m_s s_s(T_s - T) - W(T - T_0)}{m_l(T - T_0)}} \tag{6.32}$$

6.2.5 Practical 10: Cooling Curve of a Warm Body (Newton's Law of Cooling)

Newton's Law of Cooling: Thermal Radiation & Logarithmic Decay Rate

1. Differential Cooling Formulation: For a body cooling under forced convection and radiation in a uniform enclosure maintained at ambient temperature T_0, when the excess temperature $\Delta T = (T - T_0)$ is modest ($\Delta T \leq 30^\circ$C):

$$\frac{dQ}{dt} = -kA(T - T_0) \implies ms\frac{dT}{dt} = -kA(T - T_0) \implies \mathbf{\frac{dT}{dt} = -K(T - T_0)} \tag{6.33}$$

Where $K = \frac{kA}{ms}$ is the overall cooling constant. **2. Integrated Logarithmic Law & Graphs:** Integrating from initial temperature T_1 at $t = 0$:

$$\int_{T_1}^{T} \frac{dT}{T - T_0} = -K \int_0^t dt \implies \ln(\mathbf{T - T_0}) = -\mathbf{Kt + C} \tag{6.34}$$

- **Direct Cooling Curve (T vs t):** A non-linear exponential decay curve whose instantaneous tangent slope $|dT/dt|$ decreases progressively as the temperature difference diminishes, becoming asymptotic to T_0.
- **Logarithmic Plot ($\ln(T - T_0)$ vs t):** A **straight line with negative slope** $(-K)$, validating the direct first-order proportionality of cooling rate to temperature excess.
- **Precaution:** Water must be continuously and gently stirred with a copper stirrer to ensure homogeneous internal temperature without splashing.

6.3 Electricity & Magnetism Lab Protocols

6.3.1 Practical 11: Verification of Ohm's Law & Determination of Wire Resistivity

Ohm's Law Verification, Linear $V - I$ Characteristic & Specific Resistance

1. Circuit Architecture & Ohm's Formulation: Connect a uniform high-resistance alloy wire in series with an accumulator/power supply, plug key, rheostat, and ammeter. Connect a high-impedance voltmeter across the test wire:

$$V = IR \implies \mathbf{R} = \text{Slope of } \mathbf{V} \text{ vs } \mathbf{I} \text{ characteristic graph} \tag{6.35}$$

2. Resistivity (Specific Resistance) Formulation:

$$R = \rho \frac{L}{A} = \rho \frac{L}{\pi(d/2)^2} = \frac{4\rho L}{\pi d^2} \implies \rho = \frac{\pi \mathbf{d^2 R}}{\mathbf{4L}} = \frac{\pi \mathbf{r^2 R}}{\mathbf{L}} \tag{6.36}$$

Where wire length L is measured with a metre scale and diameter d is determined at multiple orientations using a calibrated screw gauge.

- **Crucial Experimental Precaution:** Close the plug key only during brief moments of observation. Continuous current flow causes Joule heating ($H = I^2Rt$), which increases wire resistance ($\Delta R = R_0 \alpha \Delta T$) and causes the $V - I$ graph to bend upward away from linearity.

6.3.2 Practical 12: Metre Bridge — Unknown Resistance & Specific Resistance

Metre Bridge: Null-Deflection Wheatstone Bridge & End Corrections

1. Wheatstone Bridge Null Condition: A metre bridge consists of a 100 cm uniform constantan/manganin wire stretched along a graduated metre rule over heavy copper end strips. When the sliding jockey achieves zero galvanometer deflection at balance length l_1:

$$\frac{P}{Q} = \frac{R}{S} \implies \frac{\sigma l_1}{\sigma(100 - l_1)} = \frac{R}{X} \implies \mathbf{X = R\left(\frac{100 - l_1}{l_1}\right)} \tag{6.37}$$

Where R is a known standard resistance from a resistance box, and X is the unknown wire resistance.

$$\text{Specific Resistance: } \rho = \frac{\mathbf{X \cdot \pi r^2}}{\mathbf{L}_{\text{wire}}} \tag{6.38}$$

2. Elimination of End Errors (α, β): Non-zero contact resistance and imperfect solder joints at the thick copper terminal strips introduce end errors α (at zero end) and β (at 100 cm end):

$$\frac{R}{X} = \frac{l_1 + \alpha}{(100 - l_1) + \beta} \tag{6.39}$$

- **Standard Remedial Protocol:** Interchange the positions of standard resistance R and unknown resistance X in the left and right gaps. Taking the arithmetic mean of the two balance lengths effectively cancels the end errors.
- **Maximum Bridge Sensitivity:** Occurs when all four bridge arms have comparable resistance, placing the null point near the exact centre of the wire ($\mathbf{40}$ cm $\leq \mathbf{l_1} \leq \mathbf{60}$ cm).

6.3.3 Practical 13: Galvanometer Resistance, Figure of Merit & Meter Conversions

Moving Coil Galvanometer: Half-Deflection Method & Conversion Protocols

1. Galvanometer Resistance (G) by Half-Deflection Method:

- With high resistance R connected in series with cell EMF E, record initial deflection θ:

$$I_1 = \frac{E}{R + G} = k\theta \tag{6.40}$$

- Connect a low shunt resistance S across the galvanometer and adjust S until deflection halves to $\theta/2$:

$$I_2 = \frac{E}{R + \frac{GS}{G+S}} \cdot \frac{S}{G+S} = k\frac{\theta}{2} \implies \mathbf{G = \frac{R \cdot S}{R - S}} \tag{6.41}$$

When $R \gg G$, this simplifies to $G \approx S$.

2. Figure of Merit (k) and Full-Scale Current (I_g):

$$\mathbf{k = \frac{I_1}{\theta} = \frac{E}{(R+G)\theta}} \tag{6.42}$$

$$\implies \mathbf{I_g = n \cdot k} \quad (n = \text{total scale divisions on one side})$$

3. Conversion of Galvanometer into Ammeter and Voltmeter:

- **Conversion to Ammeter (Range $0 \to I$):** Connect a small **Shunt Resistance (S)** in **parallel** with the galvanometer coil:

$$(I - I_g)S = I_g G \implies \mathbf{S = \frac{I_g G}{I - I_g}} \approx \frac{I_g G}{I} \quad (\text{for } I \gg I_g) \tag{6.43}$$

Effective ammeter resistance: $R_A = \frac{GS}{G+S} \approx S \to 0$ (ideal ammeter has zero internal resistance).

- **Conversion to Voltmeter (Range $0 \to V$):** Connect a large **Multiplier Resistance (R_V)** in **series** with the galvanometer coil:

$$V = I_g(G + R_V) \implies \mathbf{R_V = \frac{V}{I_g} - G} \tag{6.44}$$

Effective voltmeter resistance: $R_{\text{eff}} = G + R_V \approx R_V \to \infty$ (ideal voltmeter has infinite internal resistance).

6.4 Optics & Semiconductor Electronics Lab Protocols

6.4.1 Practical 14: Determination of Focal Length of Mirrors and Lenses by Parallax Method

Optical Bench: Parallax Removal & Focal Length Formulations

1. Removal of Parallax Principle: Parallax is the apparent relative displacement between two objects when the observer's line of sight shifts sideways. Two collinear needle tips exhibit **zero parallax** only when they reside at the exact same physical coordinates in space; moving the head from left to right causes both tips to move strictly together without separating. **2. Convex Lens ($u - v$ Method):**

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \implies \mathbf{f = \frac{uv}{u - v}} \tag{6.45}$$

- Graph of $1/v$ versus $1/u$ is a straight line with slope -1 and equal intercepts $1/f$ on both axes.
- At symmetric object distance $u = -2f$, real inverted image forms at $v = +2f$.

3. Concave Mirror ($u - v$ Method):

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \implies \mathbf{f = \frac{uv}{u + v}} \tag{6.46}$$

- When object needle is located at center of curvature ($u = -2f = -R$), real inverted image coincides with the needle tip ($v = -2f$) without parallax.

4. Convex Mirror (Auxiliary Convex Lens Method): Since a convex mirror forms virtual images, a convex lens is placed between the object needle and convex mirror to produce a convergent incident beam. The mirror reflects the rays back along their incident path to form an inverted image coinciding with the object needle when the rays strike the mirror normally:

$$\mathbf{R = d} \implies \mathbf{f = \frac{R}{2} = \frac{d}{2}} \tag{6.47}$$

Where d is the distance between the convex mirror and the position of the real auxiliary image produced by the lens alone. **5. Concave Lens (Auxiliary Convex Lens Method):** A concave lens is introduced between an auxiliary convex lens and its real image point I_1. The diverging concave lens pushes the real image further out to I_2. The point I_1 serves as a **virtual object** for the concave lens ($u = +d_1$, $v = +d_2$):

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{d_2} - \frac{1}{d_1} \implies \mathbf{f = \frac{d_1 d_2}{d_1 - d_2} < 0} \tag{6.48}$$

6.4.2 Practical 15: Angle of Minimum Deviation of a Triangular Prism

Prism Deviation Geometry, $i - \delta$ Characteristic & Refractive Index

1. Total Deviation in a Triangular Glass Prism: For a prism with refracting angle A, an incident ray undergoes two successive refractions:

$$r_1 + r_2 = A, \qquad \delta = (i_1 + i_2) - A \tag{6.49}$$

2. Minimum Deviation Condition ($\delta = \delta_m$): A plot of angle of incidence i versus deviation δ is U-shaped and exhibits a single absolute minimum:

$$\text{At Minimum Deviation: } i_1 = i_2 = i, \qquad r_1 = r_2 = \frac{A}{2} \tag{6.50}$$

$$i = \frac{A + \delta_m}{2} \implies \mu = \frac{\sin \mathbf{i}}{\sin \mathbf{r}} = \frac{\sin\left(\frac{\mathbf{A}+\delta_\mathbf{m}}{\mathbf{2}}\right)}{\sin\left(\frac{\mathbf{A}}{\mathbf{2}}\right)} \tag{6.51}$$

- At minimum deviation, the refracted ray passes through the prism **strictly parallel to the base** (for an equilateral or isosceles prism).
- Ray path is symmetric and satisfies the principle of optical reversibility.

6.4.3 Practical 16: Refractive Index of Glass Slab using Travelling Microscope

Travelling Microscope: Vertical Vernier & Normal Refraction Shift

1. Apparent Depth Formulation: When looking through a transparent parallel glass slab of real thickness t and refractive index μ, refraction bends light rays away from the normal, causing an apparent upward displacement of the object:

$$d_{\text{apparent}} = \frac{d_{\text{real}}}{\mu} \implies \mu = \frac{\text{Real Thickness}}{\text{Apparent Thickness}} = \frac{\mathbf{t}_{\text{real}}}{\mathbf{t}_{\text{apparent}}} \tag{6.52}$$

2. Three-Reading Travelling Microscope Protocol: Focus the vertical vernier microscope successively on three planes:

- **Reading R_1:** Microscope focused on a fine ink cross mark made on white paper placed on the stage.
- **Reading R_2:** Glass slab placed over the ink cross; microscope racked upward to refocus on the apparent image of the cross seen through the slab.
- **Reading R_3:** Lycopodium powder sprinkled on the top surface of the glass slab; microscope racked upward to focus on the powder particles.

$$\text{Real Thickness: } t_{\text{real}} = R_3 - R_1, \qquad \text{Apparent Thickness: } t_{\text{apparent}} = R_3 - R_2 \tag{6.53}$$

$$\mu = \frac{\mathbf{R_3} - \mathbf{R_1}}{\mathbf{R_3} - \mathbf{R_2}} \qquad \text{Normal Shift: } \mathbf{\Delta t} = \mathbf{R_2} - \mathbf{R_1} = \mathbf{t}\left(\mathbf{1} - \frac{\mathbf{1}}{\mu}\right) \tag{6.54}$$

6.4.4 Practical 17: Forward & Reverse Bias of p-n Junction Diode & Zener Breakdown

Semiconductor Diodes: $I - V$ Characteristics & Zener Voltage Regulation

1. Forward Bias Characteristic Protocol: Connect the p-side of a semiconductor diode to the positive supply terminal and n-side to the negative terminal via a potentiometer, series current-limiting resistor, and milliammeter (mA):

- Applied forward voltage opposes the built-in barrier potential V_0, narrowing the depletion layer.
- Negligible current flows until the **Cut-in / Knee Voltage (V_k)** is exceeded ($\mathbf{0.7}\,\mathrm{V}$ for Silicon, $\mathbf{0.3}\,\mathrm{V}$ for Germanium). Beyond V_k, forward current increases exponentially.
- **Dynamic Forward Resistance:** $\mathbf{r_f = \frac{\Delta V_f}{\Delta I_f}}$ (typically $10 - 100\,\Omega$).

2. Reverse Bias Characteristic Protocol: Connect p-side to negative terminal and n-side to positive terminal with a microammeter (μA):

- Applied voltage reinforces the barrier potential, widening the depletion layer.
- A tiny, voltage-independent **Reverse Saturation Current (I_0)** flows due to thermally generated minority carriers ($\sim$ nA for Silicon, $\sim \mu$A for Germanium).
- **Dynamic Reverse Resistance:** $\mathbf{r_r = \frac{\Delta V_r}{\Delta I_r} > 1\,M\Omega}$.

3. Zener Diode Reverse Breakdown Voltage (V_z): A Zener diode is heavily doped (p^{++}, n^{++}), resulting in an extremely narrow depletion layer ($\sim 10\,\mathrm{nm}$) and high internal electric field ($> 10^6\,\mathrm{V\,m^{-1}}$):

- When reverse voltage reaches the critical **Zener Breakdown Voltage (V_z)**, valence electrons are torn directly from covalent bonds via quantum tunneling (*Zener breakdown*), or impact ionization causes an avalanche multiplication.
- **Characteristic Sharp Knee:** Reverse current increases vertically over several orders of magnitude while the terminal voltage remains locked at V_z, making it an ideal voltage regulator.

6.4.5 Practical 18: Multimeter Identification of Diode, LED, Resistor & Capacitor

Multimeter Protocol: Component Discrimination & Unidirectional Conduction

A Digital Multimeter (DMM) serves as a rapid diagnostic instrument to identify and distinguish two-terminal electronic components from a mixed collection using its rotary selector switch:

1. **Resistor Identification:** Set DMM to Resistance (Ω) mode. Connect probes across component terminals and note reading; then reverse probe polarity:
   - **Signature:** Displays **identical non-zero resistance in both directions** (symmetric bidirectional conduction obeying Ohm's law).
2. **p-n Junction Diode Identification:** Set DMM to Diode Check ($\rightarrow |-$) mode. Connect red probe ($+$) and black probe ($-$), then reverse polarity:

- **Signature:** Displays forward voltage drop ($\sim \mathbf{0.6 - 0.7}\,\mathrm{V}$ for Silicon) in forward bias (red on anode, black on cathode); displays **Overload / Open-Loop (OL)** in reverse bias (strict unidirectional conduction).

3. **Light Emitting Diode (LED) Identification:** In Diode Check mode:
   - **Signature:** Emits a distinct **glow of visible light** (red, green, blue) with a displayed forward voltage drop between $1.8\,\mathrm{V}$ and $3.2\,\mathrm{V}$ when forward biased; remains dark and displays OL when reverse biased. Longer lead indicates anode (+).
4. **Capacitor Identification:** Set DMM to high resistance range ($M\Omega$):
   - **Signature:** Meter initially shows a low resistance reading which **steadily increases over time to infinity / OL** as the multimeter's internal DC test voltage charges the dielectric plates. Reversing probes shows temporary negative resistance / discharge followed by recharging.
5. **Continuity Buzzer Testing:** Set DMM to Continuity Mode. Resistance $< 50\,\Omega$ triggers an audible continuous buzzer tone, verifying circuit continuity, unblown fuses, and unbroken connecting wires.

Appendix: High-Yield Physics Formula & Equation Compendium

APPENDIX A

High-Yield Formula Compendium: Mechanics, Gravitation, Thermal Physics & Waves

This chapter brings together every governing equation, parameter definition, validity boundary condition, and NEET trap across Class 11 Physics (Units 2 through 10). Use this compendium for rapid revision before full-length tests and final preparation cycles.

A.1 Kinematics (Motion in a Straight Line & Plane)

A.1.1 Rectilinear Motion under Uniform Acceleration

Master Kinematic Equations (Constant Acceleration $a = \text{const}$)

$$v = u + at \tag{A.1}$$

$$s = ut + \frac{1}{2}at^2 \tag{A.2}$$

$$v^2 = u^2 + 2as \tag{A.3}$$

$$s_n = u + \frac{a}{2}(2n - 1) \quad \text{(Displacement in the } n\text{-th second)} \tag{A.4}$$

$$s = \left(\frac{u + v}{2}\right) t \tag{A.5}$$

Parameter Specification:

- u: Initial velocity vector at $t = 0$ [$\mathrm{m\,s^{-1}}$]
- v: Instantaneous velocity vector at time t [$\mathrm{m\,s^{-1}}$]
- a: Constant acceleration vector [$\mathrm{m\,s^{-2}}$]
- s: Net displacement vector from origin [m]
- s_n: Displacement accumulated strictly between $t = (n - 1)$ and $t = n$ [m]

Applicability Conditions & Shortcuts

- **Criterion of Validity:** Strictly applicable **only** when acceleration a is constant in both magnitude and direction. If $a = f(t)$ or $a = f(x)$, kinematic

equations fail; one must integrate:

$$v = \int a\,dt, \quad s = \int v\,dt, \quad \int v\,dv = \int a\,dx$$

- **Galileo's Law of Odd Numbers:** For a body starting from rest ($u = 0$) under constant acceleration, the distances traversed in equal successive intervals of time Δt follow the ratio:

$$s_1 : s_2 : s_3 : \cdots : s_n = 1 : 3 : 5 : \cdots : (2n-1)$$

- **Stopping Distance & Stopping Time:**

$$d_{\text{stop}} = \frac{u^2}{2a} \implies d_{\text{stop}} \propto u^2, \qquad t_{\text{stop}} = \frac{u}{a} \implies t_{\text{stop}} \propto u$$

NEET Traps: Distance vs. Displacement & Turning Points

- **Trap 1 (Distance vs Displacement):** The formula $s = ut + \frac{1}{2}at^2$ calculates **displacement**, NOT total distance traveled. If velocity changes sign ($v = 0$ at some $t_{\text{turn}} < t_{\text{total}}$), distance must be calculated piecewise:

$$\text{Distance} = |s(0 \to t_{\text{turn}})| + |s(t_{\text{turn}} \to t_{\text{total}})|$$

- **Trap 2 (n-th Second Dimensionality):** In $s_n = u + \frac{a}{2}(2n-1)$, u is multiplied by an implicit (1 s), preserving dimensional homogeneity ($[L] = [LT^{-1} \cdot T] = [L]$). Do not drop signs for u and a.

A.1.2 Projectile Motion in a Uniform Gravitational Field

Oblique Projectile Motion (Launch angle θ with horizontal)

$$T = \frac{2u\sin\theta}{g} = \frac{2u_y}{g} \qquad \text{[Total Time of Flight]} \quad \text{(A.6)}$$

$$H_{\max} = \frac{u^2\sin^2\theta}{2g} = \frac{u_y^2}{2g} \qquad \text{[Maximum Height]} \quad \text{(A.7)}$$

$$R = \frac{u^2\sin 2\theta}{g} = \frac{2u_x u_y}{g} \qquad \text{[Horizontal Range]} \quad \text{(A.8)}$$

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} = x\tan\theta\left(1 - \frac{x}{R}\right) \qquad \text{[Trajectory Equation]} \quad \text{(A.9)}$$

High-Yield Variations & Relations

- **Relation Between $H_{\max}$ and R:**

$$\frac{H_{\max}}{R} = \frac{\tan\theta}{4} \implies R\tan\theta = 4H_{\max}$$

  When launch angle $\theta = 45^\circ$, $R_{\max} = 4H_{\max}$.
- **Complementary Angles of Projection:** For identical launch speed u, angles θ and $(90^\circ - \theta)$ yield identical horizontal ranges ($R_1 = R_2$). The corresponding flight times T_1, T_2 and heights H_1, H_2 satisfy:

$$R = \frac{1}{2}gT_1T_2 = 4\sqrt{H_1H_2}$$

- **Velocity at Highest Point:** $v_{\text{top}} = u_x = u\cos\theta$ (purely horizontal, acceleration g is perpendicular: $\mathbf{a} \perp \mathbf{v}$).
- **Radius of Curvature:** At apex: $\rho_{\text{apex}} = \frac{u^2\cos^2\theta}{g}$; At launch: $\rho_{\text{launch}} = \frac{u^2}{g\cos\theta}$.

Common Projectile Traps

- **Kinetic Energy at Apex:** If initial KE is $K_0 = \frac{1}{2}mu^2$, KE at highest point is **not zero**; it is $K_{\text{apex}} = K_0\cos^2\theta$. For $\theta = 60^\circ$, $K_{\text{apex}} = \frac{1}{4}K_0$.
- **Angle Given with Vertical:** If problem states "projected at angle α to the vertical", replace θ with $(90^\circ - \alpha)$. Always double check the reference axis!

A.2 Laws of Motion & Friction

A.2.1 Newton's Second Law, Momentum & Impulse

Momentum, Force & Impulse Formulations

$$\mathbf{F}_{\text{net}} = \frac{d\mathbf{p}}{dt} = m\frac{d\mathbf{v}}{dt} + \mathbf{v}_{\text{rel}}\frac{dm}{dt} \tag{A.10}$$

$$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F}\, dt = \Delta\mathbf{p} = \mathbf{p}_2 - \mathbf{p}_1 \quad \text{[Impulse-Momentum Theorem]} \tag{A.11}$$

Variable Mass System (Rocket Propulsion):

$$v = u + v_{\text{rel}} \ln\left(\frac{m_0}{m}\right) - gt, \qquad F_{\text{thrust}} = v_{\text{rel}}\left(-\frac{dm}{dt}\right) \tag{A.12}$$

A.2.2 Friction Mechanics

Static, Limiting & Kinetic Friction Laws

$$f_s \le f_{s,\max} = \mu_s N \quad \text{[Static: } 0 \le f_s \le \mu_s N\text{]} \tag{A.13}$$

$$f_k = \mu_k N \quad \text{[Kinetic: } \mu_k \le \mu_s\text{]} \tag{A.14}$$

$$\tan\lambda = \mu_s \quad \text{[Angle of Friction } \lambda\text{]} \tag{A.15}$$

$$\theta_r = \lambda = \tan^{-1}(\mu_s) \quad \text{[Angle of Repose } \theta_r\text{]} \tag{A.16}$$

Motion on an Inclined Plane of Inclination θ

- If $\theta \le \theta_r$: Block remains at rest on incline; static friction $f_s = mg\sin\theta$.
- If $\theta > \theta_r$: Block accelerates downward:

$$a_{\text{down}} = g(\sin\theta - \mu_k\cos\theta)$$

- If projected upward along incline:

$$a_{\text{up}} = g(\sin\theta + \mu_k\cos\theta) \implies a_{\text{up}} > a_{\text{down}} \implies t_{\text{up}} < t_{\text{down}}$$

- **Ratio of Times (n-times longer down a rough plane vs smooth plane):**

$$\mu = \tan\theta\left(1 - \frac{1}{n^2}\right)$$

Friction Calculation Traps

- **Normal Reaction Modification:** $N \neq mg$ in general! On incline: $N = mg\cos\theta$. When pulling at angle α: $N = mg - F\sin\alpha$. When pushing at angle α: $N = mg + F\sin\alpha$. (Hence pulling is always easier than pushing).
- **Static Friction is NOT Always $\mu_s N$:** $\mu_s N$ is the **maximum limit** (impending motion). If applied force $F_{\text{applied}} < \mu_s N$, the friction force is exactly equal to F_{applied}, NOT $\mu_s N$.

A.2.3 Circular Dynamics & Highway Banking

Vehicle Dynamics on Circular Tracks

- **Unbanked Flat Road:** $v_{\max,\text{flat}} = \sqrt{\mu_s g R}$
- **Banked Road (Zero Friction):** $v_{\text{opt}} = \sqrt{gR\tan\theta}$
- **Banked Road (Upper Limit):**

$$v_{\max,\text{banked}} = \sqrt{gR\left(\frac{\mu_s + \tan\theta}{1 - \mu_s\tan\theta}\right)}$$

- **Banked Road (Lower Limit):**

$$v_{\min,\text{banked}} = \sqrt{gR\left(\frac{\tan\theta - \mu_s}{1 + \mu_s\tan\theta}\right)}$$

A.3 Work, Energy, Power & Collisions

A.3.1 Work-Energy Theorem & Potential Energy

Fundamental Work & Potential Relations

$$W_{\text{net}} = \Delta K = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \quad \text{[Work-Energy Theorem]} \tag{A.17}$$

$$W_{\text{cons}} = -\Delta U = -(U_f - U_i) \quad \text{[Conservative Force]} \tag{A.18}$$

$$\mathbf{F} = -\boldsymbol{\nabla} U = -\left(\frac{\partial U}{\partial x}\hat{\mathbf{i}} + \frac{\partial U}{\partial y}\hat{\mathbf{j}} + \frac{\partial U}{\partial z}\hat{\mathbf{k}}\right) \tag{A.19}$$

Spring Elastic Potential Energy:

$$U_{\text{spring}} = \frac{1}{2}kx^2, \qquad W_{\text{spring}} = -\frac{1}{2}k(x_f^2 - x_i^2) \tag{A.20}$$

Equilibrium Criteria from Potential Function $U(x)$

- **Equilibrium Condition:** $\frac{dU}{dx} = 0$ (Force $F = 0$).
- **Stable Equilibrium:** $\frac{d^2U}{dx^2} > 0$ (Potential energy is minimum; restoring force acts on perturbation; SHM possible).
- **Unstable Equilibrium:** $\frac{d^2U}{dx^2} < 0$ (Potential energy is maximum; destabilizing force acts on perturbation).
- **Neutral Equilibrium:** $\frac{d^2U}{dx^2} = 0$ (Potential energy is locally constant).

A.3.2 Vertical Circular Motion (Mass m on Inextensible String Length L)

Critical Velocity & Tension Conditions

$$v_{\text{bottom,min}} = \sqrt{5gL} \quad \text{[Condition to complete full circle]} \tag{A.21}$$

$$v_{\text{top,min}} = \sqrt{gL} \quad \text{[Velocity at top for non-slack string } (T_{\text{top}} \geq 0)\text{]} \tag{A.22}$$

$$v_{\text{horizontal}} = \sqrt{3gL} \quad \text{[Velocity when string is horizontal]} \tag{A.23}$$

$$T_{\text{bottom}} - T_{\text{top}} = 6mg \quad \text{[Tension difference is independent of launch speed!]} \tag{A.24}$$

String vs Light Rigid Rod in Vertical Circle

- **String (can slack):** Minimum velocity at bottom to complete circle is $\sqrt{5gL}$; minimum velocity at apex is $\sqrt{gL}$.
- **Light Rigid Rod (cannot slack):** Velocity at highest point can drop to **zero** ($v_{\text{top}} = 0$). Hence minimum velocity at bottom is:

$$v_{\text{bottom,min}} = \sqrt{4gL} = 2\sqrt{gL}$$

Never confuse the string case with the rigid rod / sphere on track case!

A.3.3 1D & 2D Collisions

Coefficient of Restitution & Collision Velocities

$$e = \frac{v_{\text{sep}}}{v_{\text{app}}} = \frac{v_2 - v_1}{u_1 - u_2} \quad (0 \le e \le 1) \tag{A.25}$$

1D Head-On Collision Velocities (m_1 moving with u_1, m_2 moving with u_2):

$$v_1 = \left(\frac{m_1 - em_2}{m_1 + m_2}\right) u_1 + \frac{(1+e)m_2}{m_1 + m_2} u_2 \tag{A.26}$$

$$v_2 = \frac{(1+e)m_1}{m_1 + m_2} u_1 + \left(\frac{m_2 - em_1}{m_1 + m_2}\right) u_2 \tag{A.27}$$

Loss of Kinetic Energy in Inelastic Collision (ΔK):

$$\Delta K_{\text{loss}} = \frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)(u_1 - u_2)^2(1 - e^2) = \frac{1}{2}\mu v_{\text{rel}}^2(1 - e^2) \tag{A.28}$$

Special Elastic Collision Limits ($e = 1$)

- **Equal Masses ($m_1 = m_2$):** Velocities simply interchange after collision:

$$v_1 = u_2, \qquad v_2 = u_1$$

- **Heavy Target at Rest ($m_2 \gg m_1, u_2 = 0$):** $v_1 \approx -u_1$ (rebounds with same speed), $v_2 \approx 0$.
- **Heavy Projectile against Light Target ($m_1 \gg m_2, u_2 = 0$):** $v_1 \approx u_1$, $v_2 \approx 2u_1$ (target launched at twice projectile speed).
- **Rebound from Floor (Floor $m_2 \to \infty, u_2 = 0$):**

$$v_n = e^n v_0, \qquad h_n = e^{2n} h_0$$

$$t_{\text{total}} = t_0\left(\frac{1+e}{1-e}\right), \qquad H_{\text{total}} = h_0\left(\frac{1+e^2}{1-e^2}\right)$$

A.4 Rotational Dynamics & Centre of Mass (Unit 5)

A.4.1 Centre of Mass (CM) Formulations

Discrete and Continuous Centre of Mass Coordinates

$$\mathbf{r}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{M} = \frac{1}{M}\int \mathbf{r}\, dm \tag{A.29}$$

$$\mathbf{v}_{\text{cm}} = \frac{\sum m_i \mathbf{v}_i}{M}, \qquad \mathbf{a}_{\text{cm}} = \frac{\mathbf{F}_{\text{ext}}}{M}, \qquad \mathbf{r}_{\text{rem}} = \frac{M\mathbf{r}_{\text{orig}} - m\mathbf{r}_{\text{cut}}}{M-m} \tag{A.30}$$

Body	CM (y_{cm})	Body	CM (y_{cm})
Semicircular Ring	$\frac{2R}{\pi}$ (base)	Semicircular Disc	$\frac{4R}{3\pi}$ (base)
Hollow Hemisphere	$\frac{R}{2}$ (center)	Solid Hemisphere	$\frac{3R}{8}$ (center)
Hollow Cone	$\frac{h}{3}$ (base)	Solid Cone	$\frac{h}{4}$ (base)

Table A.1: Exact Centre of Mass Coordinates from Base / Symmetric Center along Axis.

A.4.2 Rotational Kinematics, Dynamics & Angular Momentum

Governing Torque and Angular Momentum Relations

$$\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F} = I\boldsymbol{\alpha} = \frac{d\mathbf{L}}{dt} \qquad \text{[Torque Equation]} \tag{A.31}$$

$$\mathbf{L} = \mathbf{r}\times\mathbf{p} = I\boldsymbol{\omega} \qquad \text{[Angular Momentum]} \tag{A.32}$$

$$W_{\text{rot}} = \int \tau\, d\theta = \frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_i^2 \qquad \text{[Rotational Work-Energy]} \tag{A.33}$$

Conservation of Angular Momentum (When $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$):

$$\mathbf{L} = I_1\boldsymbol{\omega}_1 = I_2\boldsymbol{\omega}_2 = \text{constant} \implies \omega \propto \frac{1}{I} \tag{A.34}$$

A.4.3 Moment of Inertia Theorems & Standard Geometrical Values

Moment of Inertia (MI) Fundamental Theorems

$$I = \sum m_i r_i^2 = \int r^2\, dm = Mk^2 \quad [k = \text{Radius of Gyration}] \tag{A.35}$$

$$I = I_{\text{cm}} + Md^2 \quad [\text{Parallel Axis Theorem}] \tag{A.36}$$

$$I_z = I_x + I_y \quad [\text{Perpendicular Axis Theorem}] \tag{A.37}$$

Body	Axis of Rotation	MI (I)	k^2/R^2
Thin Rod (L)	Perpendicular (center)	$\frac{1}{12}ML^2$	$L^2/(12R^2)$
Thin Rod (L)	Perpendicular (end)	$\frac{1}{3}ML^2$	$L^2/(3R^2)$
Thin Ring	Central $\perp$ axis	MR^2	1
Thin Ring	Across diameter	$\frac{1}{2}MR^2$	1/2
Circular Disc	Central $\perp$ axis	$\frac{1}{2}MR^2$	1/2
Circular Disc	Across diameter	$\frac{1}{4}MR^2$	1/4
Solid Cylinder	Longitudinal axis	$\frac{1}{2}MR^2$	1/2
Hollow Cylinder	Longitudinal axis	MR^2	1
Solid Sphere	Any diameter	$\frac{2}{5}MR^2$	0.40
Hollow Sphere	Any diameter	$\frac{2}{3}MR^2$	≈ 0.67

Table A.2: Standard Moments of Inertia and Dimensionless Gyration Factor k^2/R^2.

A.4.4 Pure Rolling Dynamics (Rolling Without Slipping)

Rolling Motion Governing Equations

$$v_{\text{cm}} = R\omega, \qquad a_{\text{cm}} = R\alpha \quad [\text{No-slip Kinematic Constraint}] \tag{A.38}$$

$$K_{\text{total}} = K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2 = \frac{1}{2}Mv_{\text{cm}}^2\left(1 + \frac{k^2}{R^2}\right) \tag{A.39}$$

$$\frac{K_{\text{trans}}}{K_{\text{total}}} = \frac{1}{1 + k^2/R^2}, \qquad \frac{K_{\text{rot}}}{K_{\text{total}}} = \frac{k^2/R^2}{1 + k^2/R^2} \tag{A.40}$$

Rolling Down an Incline of Angle θ and Height h:

$$a = \frac{g\sin\theta}{1 + \frac{k^2}{R^2}}, \qquad v_{\text{bottom}} = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}} \tag{A.41}$$

$$\mu_{\min} = \frac{\tan\theta}{1 + \frac{R^2}{k^2}} \tag{A.42}$$

Race Down an Incline (Identical M, R, θ, h)

- Ordering of k^2/R^2: Solid Sphere (0.40) $<$ Disc/Cylinder (0.50) $<$ Hollow Sphere (0.67) $<$ Ring/Hollow Cylinder (1.00).
- Acceleration Order: $a_{\text{solid sphere}} > a_{\text{disc}} > a_{\text{hollow sphere}} > a_{\text{ring}}$.
- Time of Descent ($t = \sqrt{2s/a}$): $t_{\text{solid sphere}} < t_{\text{disc}} < t_{\text{hollow sphere}} < t_{\text{ring}}$. Solid sphere reaches first!
- Velocity at Bottom: $v_{\text{solid sphere}} > v_{\text{disc}} > v_{\text{hollow sphere}} > v_{\text{ring}}$.

Rotational Traps: Friction & Perpendicular Axis Theorem

- **Friction in Pure Rolling:** Static friction acts at the instantaneous point of contact. Because the point of contact is instantaneously at rest ($v_{\text{contact}} = 0$), **work done by friction in pure rolling is identically zero** ($W_f = 0$). Mechanical energy is conserved.
- **Perpendicular Axis Theorem Trap:** Cannot be applied to 3D bodies like spheres, solid cylinders, or cones. It is strictly valid for 2D plane laminar sheets ($I_z = I_x + I_y$).

A.5 Gravitation (Unit 6)

A.5.1 Universal Gravitation, Acceleration (g) & Field Variations

Newton's Law & Surface Acceleration g

$$F = \frac{GM_1M_2}{r^2} \qquad [G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}] \tag{A.43}$$

$$g = \frac{GM}{R^2} = \frac{4}{3}\pi G\rho R \qquad [\text{Earth: } g \approx 9.8\,\mathrm{m\,s^{-2}}] \tag{A.44}$$

Variations of g (Altitude, Depth & Latitude):

$$g_h = g\left(1 + \frac{h}{R}\right)^{-2} \approx g\left(1 - \frac{2h}{R}\right) \qquad [h \ll R] \tag{A.45}$$

$$g_d = g\left(1 - \frac{d}{R}\right) \qquad [0 \le d \le R] \tag{A.46}$$

$$g' = g - \omega^2 R\cos^2\lambda \qquad [\lambda = \text{latitude}] \tag{A.47}$$

Exact Comparison & Extreme Earth Rotation Cases

- **Equal Reduction at Altitude and Depth:** For small $h \ll R$, $g_h = g_d \implies d = 2h$. A height of 1km causes the same reduction in g as a depth of 2km.
- **Weightlessness at Equator:** For bodies at equator to become weightless ($g' = 0$):

  $$\omega = \sqrt{\frac{g}{R}} \approx \frac{1}{800}\mathrm{rad\,s^{-1}}, \qquad T = 2\pi\sqrt{\frac{R}{g}} \approx 84.6 \text{ minutes} \approx 1.41 \text{ hours}$$

  This requires the Earth to rotate roughly **17 times faster** than its current angular velocity.
- **Earth Stops Rotating:** If $\omega \to 0$, g increases by $\omega^2 R$ at the equator, but **remains unchanged at the poles**.

A.5.2 Gravitational Potential, Escape & Orbital Velocities

Potential, Escape Speed, Orbital Speed & Satellite Energies

$$V(r) = -\frac{GM}{r}, \qquad U(r) = -\frac{GMm}{r} \tag{A.48}$$

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2\,\mathrm{km\,s^{-1}} \qquad \text{[Escape Speed]} \tag{A.49}$$

$$v_o = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR^2}{R+h}} \qquad \text{[Orbital Speed]} \tag{A.50}$$

$$v_e = \sqrt{2}v_o \approx 1.414 v_o \qquad \text{[Near-Earth orbit]} \tag{A.51}$$

$$T = 2\pi\sqrt{\frac{r^3}{GM}} \implies T^2 \propto r^3 \qquad \text{[Kepler's Third Law]} \tag{A.52}$$

Satellite Energy Spectrum (Orbit radius $r = R + h$):

$$K = \frac{GMm}{2r}, \qquad U = -\frac{GMm}{r} \tag{A.53}$$

$$E_{\text{total}} = -\frac{GMm}{2r} = -K = \frac{U}{2}, \qquad \text{B.E.} = \frac{GMm}{2r} \tag{A.54}$$

Gravitation Traps: Launch Angle & Solid Sphere Potential

- **Escape Velocity Independence:** Escape velocity $v_e = \sqrt{2gR}$ is **completely independent** of the mass of the projectile m and the angle of launch θ. A body launched at $30°, 60°$ or $90°$ requires the identical escape speed $11.2\,\mathrm{km\,s^{-1}}$.
- **Gravitational Potential Inside Uniform Solid Sphere (Earth):**

$$V_{\text{in}}(r) = -\frac{GM}{2R^3}\left(3R^2 - r^2\right) \implies V_{\text{center}} = -\frac{3}{2}\frac{GM}{R} = 1.5\,V_{\text{surface}}$$

Potential is NOT zero at the centre; its magnitude is 1.5 times the surface value.

A.6 Properties of Solids & Fluids (Unit 7)

A.6.1 Elasticity & Moduli of Elasticity

Hooke's Law, Elastic Moduli & Energy Density

$$Y = \frac{FL}{A\Delta L} \qquad \text{[Young's Modulus]} \tag{A.55}$$

$$B = -V\frac{\Delta P}{\Delta V}, \qquad K = \frac{1}{B} \qquad \text{[Bulk Modulus \& Compressibility]} \tag{A.56}$$

$$\eta = \frac{F/A}{\theta} \qquad \text{[Modulus of Rigidity]} \tag{A.57}$$

$$\sigma = -\frac{\Delta d/d}{\Delta L/L} \qquad \text{[Poisson's Ratio; } 0 < \sigma < 0.5\text{]} \tag{A.58}$$

Elastic Potential Energy Stored in Stretched Wire:

$$U = \frac{1}{2}F\Delta L = \frac{1}{2}k(\Delta L)^2 \tag{A.59}$$

$$u = \frac{U}{V} = \frac{1}{2} \times \text{stress} \times \text{strain} = \frac{(\text{stress})^2}{2Y} \tag{A.60}$$

Thermal Stress Developed in Clamped Rod:

$$\text{Stress} = Y\alpha\Delta T, \qquad F_{\text{tension}} = YA\alpha\Delta T \tag{A.61}$$

A.6.2 Fluid Statics, Pascal's Principle & Buoyancy

Pressure, Pascal's Law & Archimedes' Principle

$$P = P_0 + \rho g h \qquad [P_{\text{gauge}} = \rho g h] \tag{A.62}$$

$$F_2 = F_1\left(\frac{A_2}{A_1}\right) \qquad \text{[Hydraulic Lift]} \tag{A.63}$$

$$F_B = V_{\text{sub}}\,\rho_{\text{liq}}\,g \qquad \text{[Buoyant Force]} \tag{A.64}$$

$$\frac{V_{\text{sub}}}{V} = \frac{\rho_{\text{body}}}{\rho_{\text{liq}}}, \qquad W_{\text{app}} = W\left(1 - \frac{\rho_{\text{liq}}}{\rho_{\text{body}}}\right) \tag{A.65}$$

A.6.3 Fluid Dynamics, Viscosity & Surface Tension

Continuity, Bernoulli, Viscosity & Capillarity

$$A_1 v_1 = A_2 v_2 = \text{constant} \qquad \text{[Continuity Equation]} \tag{A.66}$$

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant} \qquad \text{[Bernoulli's Theorem]} \tag{A.67}$$

$$v_{\text{efflux}} = \sqrt{2gh}, \quad R_{\text{horiz}} = 2\sqrt{h(H-h)} \qquad \text{[Torricelli's Law]} \tag{A.68}$$

$$F_v = 6\pi\eta r v \qquad \text{[Stokes' Viscous Drag]} \tag{A.69}$$

$$v_t = \frac{2r^2(\rho-\sigma)g}{9\eta} \implies v_t \propto r^2 \qquad \text{[Terminal Velocity]} \tag{A.70}$$

$$R_e = \frac{\rho v d}{\eta} \qquad \text{[Reynolds Number]} \tag{A.71}$$

Surface Tension, Excess Pressure & Capillary Rise:

$$T = \frac{F}{L} = \frac{\Delta W}{\Delta A}, \quad W_{\text{split}} = 4\pi R^2 T(n^{1/3} - 1) \tag{A.72}$$

$$\Delta P_{\text{drop}} = \frac{2T}{R}, \qquad \Delta P_{\text{bubble}} = \frac{4T}{R} \tag{A.73}$$

$$h = \frac{2T\cos\theta}{r\rho g} = \frac{2T}{R_c \rho g} \qquad \text{[Jurin's Law]} \tag{A.74}$$

Fluid Traps: Terminal Velocity & Insufficient Capillary Length

- **Terminal Velocity (r^2 vs r):** Viscous force $F_v \propto r$, but terminal velocity $v_t \propto r^2$. If two drops of radii $1:2$ coalesce, the radius becomes $R = 2^{1/3} r$; new terminal velocity increases by a factor of $2^{2/3}$.
- **Insufficient Capillary Tube Length ($l < h$):** Liquid **never overflows** from a capillary tube of insufficient height l. Instead, the liquid reaches the upper edge and increases its meniscus radius of curvature R_c' such that $hr = lR_c'$ $(R_c' > R_c)$.

A.7 Thermal Physics & Thermodynamics (Units 7, 8 & 9)

A.7.1 Thermal Expansion, Calorimetry & Heat Transfer

Expansion, Calorimetry, Conduction & Radiation

$$\Delta L = L_0\alpha\Delta T, \quad \Delta A = A_0\beta\Delta T, \quad \Delta V = V_0\gamma\Delta T \tag{A.75}$$

$$Q = mc\Delta T, \qquad Q = mL \tag{A.76}$$

$$\frac{dQ}{dt} = \frac{kA(T_1 - T_2)}{L} = \frac{\Delta T}{R_{\text{th}}} \qquad [R_{\text{th}} = L/(kA)] \tag{A.77}$$

$$E = \sigma e A T^4, \qquad P_{\text{net}} = \sigma e A(T^4 - T_0^4) \tag{A.78}$$

$$\lambda_m T = b = 2.898 \times 10^{-3}\,\text{m K} \qquad \text{[Wien's Law]} \tag{A.79}$$

$$\frac{T_1 - T_2}{t} = K\left(\frac{T_1 + T_2}{2} - T_0\right) \qquad \text{[Cooling Law]} \tag{A.80}$$

A.7.2 First Law of Thermodynamics & Process Formulations

First Law & Thermodynamic Processes

$$\Delta Q = \Delta U + \Delta W \quad [\text{Ideal Gas: } \Delta U = nC_v\Delta T] \tag{A.81}$$

Work Done in Standard Thermodynamic Processes:

$$W_{\text{isochoric}} = 0, \qquad \Delta Q = \Delta U = nC_v\Delta T \tag{A.82}$$

$$W_{\text{isobaric}} = P(V_2 - V_1) = nR(T_2 - T_1), \quad \Delta Q = nC_p\Delta T \tag{A.83}$$

$$W_{\text{isothermal}} = nRT\ln\left(\frac{V_2}{V_1}\right) = 2.303\,nRT\log_{10}\left(\frac{P_1}{P_2}\right) \tag{A.84}$$

$$W_{\text{adiabatic}} = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1} = -\Delta U \tag{A.85}$$

High-Yield Thermodynamic Relations & Polytropic Processes

- **Energy Partition in Isobaric Process:**

$$\Delta W : \Delta U : \Delta Q = nR\Delta T : nC_v\Delta T : nC_p\Delta T = (\gamma - 1) : 1 : \gamma$$

Gas Atomicity	DoF (f)	$C_v = \frac{f}{2}R$	$C_p = C_v + R$	$\gamma = C_p/C_v$
Monoatomic (He, Ne)	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} \approx 1.67$
Diatomic (Rigid: O_2, N_2)	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.40$
Polyatomic (H_2O, NH_3)	6	$3R$	$4R$	$\frac{4}{3} \approx 1.33$

Table A.3: Molar Heat Capacities and Adiabatic Exponents for Standard Gases.

For a diatomic gas ($\gamma = 7/5$), fraction of heat converted to work is $\frac{\Delta W}{\Delta Q} = \frac{\gamma-1}{\gamma} = \frac{2}{7} \approx 28.6\%$.

- **General Polytropic Process ($PV^n = \text{const}$):**

$$W = \frac{nR(T_1 - T_2)}{n-1}, \qquad C = C_v + \frac{R}{1-n}$$

- **Comparison of $P-V$ Curves:** $\left(\frac{dP}{dV}\right)_{\text{adiabatic}} = \gamma \left(\frac{dP}{dV}\right)_{\text{isothermal}}$. Adiabatic slope is γ-times steeper!

A.7.3 Kinetic Theory of Gases (KTG)

Pressure, Speeds & Mean Free Path

$$P = \frac{1}{3}\rho v_{\text{rms}}^2 = \frac{1}{3}\frac{M}{V}v_{\text{rms}}^2, \qquad E_{\text{trans}} = \frac{3}{2}nRT = \frac{3}{2}Nk_BT \tag{A.86}$$

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}, \quad v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}, \quad v_{\text{mp}} = \sqrt{\frac{2RT}{M}} \tag{A.87}$$

$$v_{\text{mp}} : v_{\text{avg}} : v_{\text{rms}} = 1 : 1.128 : 1.224 \tag{A.88}$$

$$\lambda = \frac{1}{\sqrt{2}\pi n d^2} = \frac{k_BT}{\sqrt{2}\pi d^2 P} \quad [\lambda \propto T/P] \tag{A.89}$$

A.8 Oscillations & Waves (Unit 10)

A.8.1 Simple Harmonic Motion (SHM) Kinematics & Energy

Governing Differential Equation, Velocity & Acceleration in SHM

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \implies x(t) = A\sin(\omega t + \phi) \tag{A.90}$$

$$v(t) = \frac{dx}{dt} = \omega\sqrt{A^2 - x^2}, \qquad a(t) = \frac{dv}{dt} = -\omega^2 x \tag{A.91}$$

$$v_{\max} = \omega A \quad (\text{at mean } x = 0), \qquad a_{\max} = \omega^2 A \quad (\text{at extreme } x = \pm A) \tag{A.92}$$

Energy Spectrum of Particle Executing SHM:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2), \qquad U = \frac{1}{2}m\omega^2 x^2 \tag{A.93}$$

$$E_{\text{total}} = K + U = \frac{1}{2}m\omega^2 A^2 \tag{A.94}$$

$$\langle K\rangle_{\text{time}} = \langle U\rangle_{\text{time}} = \frac{1}{4}m\omega^2 A^2 = \frac{1}{2}E_{\text{total}} \tag{A.95}$$

$$\langle K\rangle_{\text{pos}} = \frac{2}{3}E_{\text{total}}, \qquad \langle U\rangle_{\text{pos}} = \frac{1}{3}E_{\text{total}} \tag{A.96}$$

A.8.2 Standard Oscillating Systems & Combinations

Time Period of Springs, Pendulums & Liquid Columns

$$T_{\text{spring}} = 2\pi\sqrt{\frac{m}{k}}, \quad k_s = \frac{k_1 k_2}{k_1 + k_2}, \quad k_p = k_1 + k_2 \tag{A.97}$$

$$T_{\text{cut}} \implies k' = nk \implies T' = \frac{T}{\sqrt{n}} \quad [\text{Cut into } n \text{ pieces}] \tag{A.98}$$

$$T_{\text{pendulum}} = 2\pi\sqrt{\frac{L}{g_{\text{eff}}}}, \qquad g_{\text{eff}} = |\mathbf{g} - \mathbf{a}| \tag{A.99}$$

$$T_{\text{two block}} = 2\pi\sqrt{\frac{\mu}{k}}, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2} \tag{A.100}$$

$$T_{\text{U-tube}} = 2\pi\sqrt{\frac{h}{g}} = 2\pi\sqrt{\frac{L}{2g}} \tag{A.101}$$

A.8.3 Wave Mechanics, Speed of Sound & Organ Pipes

Wave Speed, Harmonics in Strings & Pipes

$$v = \frac{\omega}{k} = \nu\lambda, \qquad y(x,t) = A\sin(kx \mp \omega t + \phi) \tag{A.102}$$

$$v_{\text{string}} = \sqrt{\frac{T}{\mu}}, \qquad v_{\text{gas}} = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}} \tag{A.103}$$

$$\nu_{\text{string},\,n} = \frac{nv}{2L} = \frac{n}{2L}\sqrt{\frac{T}{\mu}} \quad (n = 1, 2, 3, \dots) \tag{A.104}$$

$$\nu_{\text{closed},\,n} = (2n-1)\frac{v}{4L} \implies \nu_1 : \nu_2 : \nu_3 = 1 : 3 : 5 \tag{A.105}$$

$$\nu_{\text{open},\,n} = \frac{nv}{2L} \implies \nu_1 : \nu_2 : \nu_3 = 1 : 2 : 3 \tag{A.106}$$

Resonance Tube Experiment & Beat Frequency:

$$\lambda = 2(l_2 - l_1), \quad e = \frac{l_2 - 3l_1}{2} \approx 0.6r, \quad v = 2\nu(l_2 - l_1) \tag{A.107}$$

$$\nu_{\text{beat}} = |\nu_1 - \nu_2| \quad [\text{Waxing: } \nu\downarrow;\ \text{Filing: } \nu\uparrow] \tag{A.108}$$

Wave Traps: Organ Pipes & Speed of Sound vs Pressure

- **Sound Speed Independence from Pressure:** $v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$. At constant temperature, doubling pressure doubles density, leaving the ratio P/ρ and sound speed v **completely unchanged**.
- **Closed Pipe Odd Harmonics:** In a closed organ pipe of length L, the 3rd harmonic is $\nu = 3\left(\frac{v}{4L}\right)$ which is the **1st overtone**. The n-th overtone is the $(2n+1)$-th harmonic!

APPENDIX B

High-Yield Formula Compendium: Electrodynamics, Magnetism, Optics & Modern Physics

This chapter compiles all core formulas, vector relations, circuit equations, optical relations, quantum constants, and logic gate truth tables spanning Class 12 Physics (Units 11 through 19).

B.1 Electrostatics & Capacitance (Unit 11)

B.1.1 Coulomb's Law, Electric Field & Dipoles

Coulomb's Law, Medium Effect & Dipole Dynamics

$$\mathbf{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{\mathbf{r}}_{12}, \qquad F_{\text{med}} = \frac{F_0}{\kappa} \tag{B.1}$$

$$\mathbf{p} = q(2\mathbf{a}), \qquad \boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}, \qquad U = -\mathbf{p}\cdot\mathbf{E} \tag{B.2}$$

$$E_{\text{axial}} = \frac{1}{4\pi\varepsilon_0}\frac{2pr}{(r^2 - a^2)^2} \approx \frac{2kp}{r^3} \qquad [r \gg a] \tag{B.3}$$

$$E_{\text{equat}} = \frac{1}{4\pi\varepsilon_0}\frac{p}{(r^2 + a^2)^{3/2}} \approx \frac{kp}{r^3} \qquad [r \gg a] \tag{B.4}$$

$$E(r,\theta) = \frac{kp}{r^3}\sqrt{1 + 3\cos^2\theta}, \qquad \tan\alpha = \frac{1}{2}\tan\theta \tag{B.5}$$

Work Done in Rotating an Electric Dipole from θ_1 to θ_2:

$$W = U_f - U_i = pE(\cos\theta_1 - \cos\theta_2) \tag{B.6}$$

$$\implies W_{0\to 180^\circ} = 2pE \quad \text{[Maximum work]} \tag{B.7}$$

B.1.2 Gauss's Law & Symmetrical Charge Distributions

Gauss's Theorem & Canonical Field Formulations

$$\Phi_E = \oint \mathbf{E}\cdot d\mathbf{A} = \frac{q_{\text{enclosed}}}{\varepsilon_0} \tag{B.8}$$

Symmetrical Electric Field Solutions:

$$E_{\text{line}} = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2k\lambda}{r} \qquad \text{[Line Charge]} \tag{B.9}$$

$$E_{\text{sheet}} = \frac{\sigma}{2\varepsilon_0} \qquad \text{[Infinite Sheet]} \tag{B.10}$$

$$E_{\text{conductor}} = \frac{\sigma}{\varepsilon_0} \qquad \text{[Conductor Surface]} \tag{B.11}$$

$$E_{\text{shell}}(r) = \begin{cases} 0 & (r < R) \\ \frac{kq}{R^2} & (r \geq R) \end{cases} \qquad \text{[Shell Field]} \tag{B.12}$$

$$V_{\text{shell}}(r) = \begin{cases} \frac{kq}{R} & (r \leq R) \\ \frac{kq}{r} & (r > R) \end{cases} \qquad \text{[Shell Potential]} \tag{B.13}$$

Uniformly Charged Solid Non-Conducting Sphere (q, R):

$$E_{\text{in}}(r) = \frac{kqr}{R^3} = \frac{\rho r}{3\varepsilon_0} \tag{B.14}$$

$$V_{\text{in}}(r) = \frac{kq}{2R^3}(3R^2 - r^2) \implies V_{\text{center}} = 1.5\, V_{\text{surface}} \tag{B.15}$$

B.1.3 Capacitance, Dielectric Insertion & Stored Energy

Capacitance, Energy Density & Force between Plates

$$C_0 = \frac{\varepsilon_0 A}{d}, \qquad C_{\text{dielectric}} = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{\kappa}\right)}, \qquad C_{\text{conducting slab}} = \frac{\varepsilon_0 A}{d - t} \tag{B.16}$$

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV, \qquad u_E = \frac{U}{\text{Volume}} = \frac{1}{2}\varepsilon_0 E^2 \quad [\text{J m}^{-3}] \tag{B.17}$$

$$F_{\text{attraction}} = \frac{Q^2}{2\varepsilon_0 A} = \frac{1}{2}QE = \frac{1}{2}\varepsilon_0 A E^2, \qquad P_{\text{electrostatic}} = \frac{F}{A} = \frac{1}{2}\varepsilon_0 E^2 \tag{B.18}$$

Common Potential & Heat Dissipation on Connecting Two Capacitors:

$$V_{\text{common}} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}, \qquad \Delta U_{\text{loss}} = \frac{C_1 C_2}{2(C_1 + C_2)}(V_1 - V_2)^2 \tag{B.19}$$

Parameter	Battery CONNECTED	Battery DISCONNECTED
Capacitance (C)	Increases (κC_0)	Increases (κC_0)
Voltage (V)	Constant (V_0)	Decreases (V_0/κ)
Electric Field (E)	Constant (E_0)	Decreases (E_0/κ)
Charge (Q)	Increases (κQ_0)	Constant (Q_0)
Stored Energy (U)	Increases (κU_0)	Decreases (U_0/κ)

Table B.1: Effect of Inserting a Dielectric Slab (κ) into a Parallel Plate Capacitor.

Capacitor Traps: Force between Plates & Common Potential

- **Factor of $1/2$ in Plate Force:** $F = \frac{1}{2}QE$, NOT QE. One plate experiences the field produced strictly by the **other** plate ($E/2 = \frac{\sigma}{2\varepsilon_0}$), not the total field E.
- **Opposite Polarity Connection:** If positive plate of C_1 is connected to negative plate of C_2, the numerator of common potential becomes $(C_1V_1 - C_2V_2)$ and energy loss is $\frac{C_1C_2}{2(C_1+C_2)}(V_1 + V_2)^2$.

B.2 Current Electricity (Unit 12)

B.2.1 Microscopic Conduction, Drift Velocity & Resistance

Microscopic Drift, Ohm's Law & Temperature Dependence

$$I = neAv_d, \qquad v_d = \frac{eE\tau}{m} = \frac{eV\tau}{mL} \tag{B.20}$$

$$J = \frac{I}{A} = nev_d = \sigma E = \frac{E}{\rho} \tag{B.21}$$

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m}, \qquad \rho = \frac{m}{ne^2\tau}, \qquad R = \rho\frac{L}{A} \tag{B.22}$$

$$R_T = R_0(1+\alpha\Delta T), \qquad \rho_T = \rho_0(1+\alpha\Delta T) \tag{B.23}$$

Stretching of Wires (Constant Volume $V = A \cdot L = \text{const}$):

$$R \propto L^2 \propto \frac{1}{A^2} \propto \frac{1}{r^4} \implies \text{If } L \to nL, \quad R' = n^2R \tag{B.24}$$

$$\frac{\Delta R}{R} \approx 2\frac{\Delta L}{L} \quad (\text{for small } \%\Delta L < 5\%) \tag{B.25}$$

B.2.2 Cells, Internal Resistance & Power Transfer

Terminal EMF, Cell Combinations & Max Power Theorem

$$V = \mathcal{E} - Ir \quad [\text{Discharging}], \qquad V = \mathcal{E} + Ir \quad [\text{Charging}] \tag{B.26}$$

$$r = R\left(\frac{\mathcal{E}}{V} - 1\right) = R\left(\frac{l_1 - l_2}{l_2}\right) \tag{B.27}$$

$$\mathcal{E}_{\text{eq}} = \frac{\mathcal{E}_1 r_2 + \mathcal{E}_2 r_1}{r_1 + r_2}, \qquad r_{\text{eq}} = \frac{r_1 r_2}{r_1 + r_2} \quad [\text{Parallel Cells}] \tag{B.28}$$

Maximum Power Transfer Theorem (Load R connected to cell $\mathcal{E}, r$):

$$P_{\text{load}} = \frac{\mathcal{E}^2 R}{(R+r)^2} \implies P = P_{\max} \iff R = r \implies P_{\max} = \frac{\mathcal{E}^2}{4r} \tag{B.29}$$

B.2.3 Bridge Circuits & Electrical Instruments

Wheatstone Bridge & Metre Bridge Balance Equations

$$\frac{P}{Q} = \frac{R}{S} \iff I_G = 0 \quad \text{[Wheatstone Balance Condition: } I_G = 0\text{]} \tag{B.30}$$

Metre Bridge Unknown Resistance S (Balance length l from left):

$$\frac{R}{S} = \frac{l}{100 - l} \implies S = R\left(\frac{100 - l}{l}\right) \tag{B.31}$$

Current Electricity Traps: Cell Grouping & Terminal Voltage

- **Reversed Cell in Series Battery:** In a combination of n identical cells each of EMF $\mathcal{E}$ and internal resistance r, if m cells are connected with reversed polarity:

$$\mathcal{E}_{\text{net}} = (n - 2m)\mathcal{E}, \qquad r_{\text{net}} = nr \quad \text{(Internal resistances always add!)}$$

- **Short Circuit Terminal Potential:** When cell is short-circuited ($R = 0$), current is maximum $I_{\text{sc}} = \mathcal{E}/r$, and terminal voltage drops to **zero** ($V = \mathcal{E} - I_{\text{sc}}r = 0$).

B.3 Magnetic Effects of Current & Magnetism (Unit 13)

B.3.1 Biot-Savart Law & Ampere's Circuital Law

Biot-Savart Law & Symmetrical Field Equations

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{Id\mathbf{l}\times\hat{\mathbf{r}}}{r^2} \qquad [\mu_0/(4\pi)=10^{-7}] \tag{B.32}$$

$$B_{\text{inf}} = \frac{\mu_0 I}{2\pi d}, \qquad B_{\text{semi}} = \frac{\mu_0 I}{4\pi d} \tag{B.33}$$

$$B_{\text{axis}} = \frac{\mu_0 N I R^2}{2(R^2+x^2)^{3/2}} \implies B_{\text{center}} = \frac{\mu_0 N I}{2R} \tag{B.34}$$

$$B_{\text{arc}} = \frac{\mu_0 I\theta}{4\pi R}, \qquad B_{\text{solenoid}} = \mu_0 n I = \frac{\mu_0 N I}{L} \tag{B.35}$$

$$\oint \mathbf{B}\cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}} \qquad [\text{Ampere's Law}] \tag{B.36}$$

B.3.2 Lorentz Force, Particle Trajectories & Force on Conductors

Lorentz Equation & Charged Particle Dynamics in Magnetic Fields

$$\mathbf{F}_B = q(\mathbf{v}\times\mathbf{B}) \implies W_B = 0 \quad [v=\text{const}] \tag{B.37}$$

$$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB} = \frac{\sqrt{2mqV}}{qB} \qquad [\mathbf{v}\perp\mathbf{B}] \tag{B.38}$$

$$T = \frac{2\pi m}{qB}, \qquad \nu = \frac{qB}{2\pi m} \tag{B.39}$$

$$p_{\text{pitch}} = v_{\parallel}T = \frac{2\pi m v\cos\theta}{qB} \qquad [\text{Pitch}] \tag{B.40}$$

$$\mathbf{F} = I(\mathbf{L}\times\mathbf{B}), \qquad \frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} \tag{B.41}$$

B.3.3 Galvanometer Conversions & Magnetic Materials

Moving Coil Galvanometer, Ammeter, Voltmeter & Magnetic Matter

$$\tau = NIAB = C\theta \implies I = \left(\frac{C}{NAB}\right)\theta, \quad I_s = \frac{NAB}{C} \tag{B.42}$$

$$S = \frac{I_g R_g}{I - I_g} \quad \text{[Shunt]}, \qquad R_s = \frac{V}{I_g} - R_g \quad \text{[Multiplier]} \tag{B.43}$$

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu H, \qquad \chi = \frac{M}{H}, \qquad \mu_r = 1 + \chi \tag{B.44}$$

Curie's Law for Paramagnetic Materials & Curie-Weiss Law for Ferromagnets:

$$\chi_{\text{para}} = \frac{C}{T} \implies \chi \propto \frac{1}{T}, \qquad \chi_{\text{ferro}} = \frac{C'}{T - T_c} \quad (T > T_c) \tag{B.45}$$

Magnetism Traps: Work Done by Magnetic Force & Galvanometer Sensitivity

- **Work Done by Magnetic Force is Zero:** Because $\mathbf{F}_B \perp \mathbf{v}$ at every instant, $\mathbf{F}_B \cdot \mathbf{v} = 0$. **A static magnetic field does zero work on a charged particle**. It changes the direction of motion, never the kinetic energy or speed.
- **Current Sensitivity vs Voltage Sensitivity:** Doubling the number of turns N doubles current sensitivity ($I_s \propto N$), but resistance also doubles ($R \propto N$), leaving voltage sensitivity $V_s = I_s/R_G$ **completely unchanged**!

B.4 Electromagnetic Induction & Alternating Current (Unit 14)

B.4.1 Faraday's Laws, Motional EMF & Inductance

Faraday-Lenz Law, Motional EMF & Self/Mutual Inductance

$$\mathcal{E} = -\frac{d\Phi_B}{dt}, \qquad q_{\text{induced}} = \frac{\Delta\Phi_B}{R} \tag{B.46}$$

$$\mathcal{E}_{\text{mot}} = Bvl, \qquad \mathcal{E}_{\text{rot}} = \frac{1}{2}B\omega L^2 \tag{B.47}$$

$$\mathcal{E} = -L\frac{dI}{dt}, \qquad U = \frac{1}{2}LI^2, \qquad L = \frac{\mu_0 N^2 A}{l} \tag{B.48}$$

$$\mathcal{E}_2 = -M\frac{dI_1}{dt}, \qquad M = k\sqrt{L_1 L_2} \quad (0 \le k \le 1) \tag{B.49}$$

B.4.2 Alternating Current Circuits, Impedance & Resonance

AC Values, Reactance, Series LCR Circuit & Resonance

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0, \qquad V_{\text{rms}} = \frac{V_0}{\sqrt{2}} \approx 0.707 V_0 \tag{B.50}$$

$$X_L = \omega L = 2\pi\nu L, \qquad X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C} \tag{B.51}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}, \qquad \tan\phi = \frac{X_L - X_C}{R} \tag{B.52}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \iff \nu_0 = \frac{1}{2\pi\sqrt{LC}} \quad [X_L = X_C] \tag{B.53}$$

$$Z_{\min} = R, \quad I_{\max} = \frac{V_{\text{rms}}}{R}, \quad Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{\nu_0}{\Delta\nu} \tag{B.54}$$

Average Power, Power Factor & Ideal Transformer:

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi = I_{\text{rms}}^2 R, \qquad \cos\phi = \frac{R}{Z} \quad [I_w = I_{\text{rms}} \sin\phi] \tag{B.55}$$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = K, \qquad \eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V_s I_s}{V_p I_p} \tag{B.56}$$

B.5 Electromagnetic Waves (Unit 15)

Displacement Current, Wave Speed, Energy & Radiation Pressure

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}, \qquad \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left(I_c + I_d\right) \tag{B.57}$$

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E_0}{B_0} \approx 3 \times 10^8 \,\mathrm{m\,s^{-1}}, \qquad v = \frac{c}{\sqrt{\mu_r \varepsilon_r}} \tag{B.58}$$

$$u_{\text{total}} = \frac{1}{2}\varepsilon_0 E_{\text{rms}}^2 + \frac{B_{\text{rms}}^2}{2\mu_0} = \varepsilon_0 E_{\text{rms}}^2 = \frac{B_{\text{rms}}^2}{\mu_0} \tag{B.59}$$

$$I = \varepsilon_0 c E_{\text{rms}}^2, \qquad P_{\text{rad}} = \begin{cases} \frac{I}{c} & \text{(Absorption)} \\ \frac{2I}{c} & \text{(Reflection)} \end{cases} \tag{B.60}$$

B.6 Ray Optics & Optical Instruments (Unit 16)

B.6.1 Spherical Mirrors, Snell's Law & Critical Angle

Mirror Formula, Magnification & Normal Shift

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R} \qquad \text{[Mirror Formula; Pole is Origin]} \tag{B.61}$$

$$m = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f} \qquad \text{[Transverse Magnification]} \tag{B.62}$$

$$m_L = \frac{dv}{du} = -m^2 = -\left(\frac{v}{u}\right)^2 \qquad \text{[Longitudinal Magnification]} \tag{B.63}$$

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2} \qquad \text{[Snell's Law; Frequency } \nu = \text{const]} \tag{B.64}$$

$$d_{\text{app}} = \frac{d}{\mu_{\text{rel}}}, \qquad \Delta t = t\left(1 - \frac{1}{\mu}\right) \qquad \text{[Apparent Depth \& Normal Shift]} \tag{B.65}$$

Total Internal Reflection (TIR) & Underwater Illuminance Cone:

$$\sin i_c = \frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}} = \frac{1}{\mu} \qquad \text{[Critical Angle Condition]} \tag{B.66}$$

$$r_{\text{circle}} = \frac{h}{\sqrt{\mu^2 - 1}}, \qquad \text{Area} = \frac{\pi h^2}{\mu^2 - 1} \qquad \text{[Depth } h\text{]} \tag{B.67}$$

Wave Type	Wavelength (λ)	Frequency (ν)	Primary Source & High-Yield Application
γ-Rays	$< 10^{-3}$ nm	$> 10^{20}$ Hz	Radioactive decay; cancer radiotherapy
X-Rays	$1\,\text{nm} - 10^{-3}$ nm	$10^{18} - 10^{20}$ Hz	Fast electrons hitting heavy metal target; bone imaging
UV Rays	$400\,\text{nm} - 1\,\text{nm}$	$10^{15} - 10^{17}$ Hz	Very hot bodies (Sun), arcs; LASIK eye surgery, germicidal
Visible	$700\,\text{nm} - 400\,\text{nm}$	$4 \times 10^{14} - 7.5 \times 10^{14}$ Hz	Atomic electron transitions; human optical vision
Infrared	$1\,\text{mm} - 700\,\text{nm}$	$10^{11} - 4 \times 10^{14}$ Hz	Hot bodies, molecules; thermal therapy, remote controls
Microwaves	$0.1\,\text{m} - 1\,\text{mm}$	$10^{9} - 10^{11}$ Hz	Klystrons, magnetrons; RADAR systems, microwave ovens
Radio Waves	$> 0.1\,\text{m}$	$< 10^{9}$ Hz	Accelerated charges in antennas; cellular/radio broadcasting

Table B.2: Electromagnetic Spectrum in Decreasing Order of Frequency / Increasing Wavelength.

B.6.2 Prisms, Dispersion & Spherical Refracting Surfaces

Prism Deviation, Dispersive Power & Curved Surface Refraction

$$\delta = (i_1 + i_2) - A, \quad r_1 + r_2 = A \quad \text{[Prism Angle } A\text{]} \tag{B.68}$$

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \quad \text{[Min. Dev.; } i = \tfrac{A+\delta_m}{2},\ r = \tfrac{A}{2}\text{]} \tag{B.69}$$

$$\delta_{\text{thin}} = (\mu - 1)A, \quad \omega = \frac{\mu_V - \mu_R}{\mu_Y - 1} \quad \text{[Thin Prism } A \leq 10^\circ\text{]} \tag{B.70}$$

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}, \quad m = \frac{\mu_1 v}{\mu_2 u} \quad \text{[Spherical Interface]} \tag{B.71}$$

B.6.3 Thin Lenses, Combinations & Silvered Systems

Lens Maker's Equation, Power & Silvering Formulations

$$\frac{1}{f} = \left(\frac{\mu_{\text{lens}}}{\mu_{\text{med}}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{[Lens-Maker's Formula]} \tag{B.72}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, \quad m = \frac{v}{u} = \frac{f}{f+u} \quad \text{[Thin Lens Equation]} \tag{B.73}$$

$$P = \frac{1}{f\ [\text{m}]} = \frac{100}{f\ [\text{cm}]}\ [\text{D}], \quad P_{\text{contact}} = P_1 + P_2 \tag{B.74}$$

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \quad \text{[Separation } d\text{]} \tag{B.75}$$

Silvering of a Lens Surface (Acts as an Equivalent Spherical Mirror):

$$P_{\text{eq}} = 2P_{\text{lens}} + P_{\text{mirror}} = \frac{2}{f_L} + \frac{1}{f_M} \tag{B.76}$$

$$F_{\text{eq}} = -\frac{1}{P_{\text{eq}}} \quad \text{[Always acts as equivalent mirror]}$$

Lens Cutting Rules & Medium Immersion Effects

- **Vertical Cut (Perpendicular to Principal Axis):** Cutting a convex lens of focal length f into two halves produces two plano-convex lenses: $f' = 2f, P' = P/2$. Intensity remains **unchanged**.

- **Horizontal Cut (Along Principal Axis):** Each half retains identical curvature: $f' = f, P' = P$. Intensity is halved because light-gathering aperture area is reduced by 50%.
- **Immersion in Liquid (μ_{med}):**

$$\frac{f_{\text{liquid}}}{f_{\text{air}}} = \frac{\mu_g - 1}{\frac{\mu_g}{\mu_l} - 1}$$

$$\text{For glass } (\mu_g = 1.5) \text{ in water } (\mu_l = 4/3): \quad f_{\text{water}} = 4f_{\text{air}}$$

If $\mu_l > \mu_g$, focal length reverses sign: a converging convex lens behaves as a diverging concave lens!

B.6.4 Optical Instruments

Instrument	Normal Adjustment ($v = \infty$)	Near Point Adjustment ($v = D$)	Tube Length (L)
Simple Microscope	$m = \frac{D}{f}$	$m = 1 + \frac{D}{f}$	Eye at lens
Compound Microscope	$m = -\frac{v_o}{u_o}\frac{D}{f_e} \approx -\frac{L}{f_o}\frac{D}{f_e}$	$m = -\frac{v_o}{u_o}\left(1 + \frac{D}{f_e}\right)$	$L = v_o + u_e \approx v_o + f_e$
Astronomical Telescope	$m = -\frac{f_o}{f_e}$	$m = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$	$L = f_o + f_e$ or $f_o + u_e$

Table B.3: Magnifying Powers and Tube Lengths of Canonical Optical Instruments.

B.7 Wave Optics (Unit 16)

B.7.1 Interference & Young's Double Slit Experiment (YDSE)

Interference Intensity, Fringe Position & Fringe Width

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi = 4I_0\cos^2(\phi/2) \quad [I_1 = I_2 = I_0] \tag{B.77}$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2 = \left(\frac{a_1+a_2}{a_1-a_2}\right)^2, \quad \Delta\phi = \frac{2\pi}{\lambda}\Delta x \tag{B.78}$$

$$y_n = n\frac{\lambda D}{d} \quad [\text{Bright: } \Delta x = n\lambda,\ n = 0, 1, 2, \dots] \tag{B.79}$$

$$y'_n = (2n-1)\frac{\lambda D}{2d} \quad [\text{Dark: } \Delta x = (2n-1)\tfrac{\lambda}{2}] \tag{B.80}$$

$$\beta = \frac{\lambda D}{d}, \quad \theta = \frac{\beta}{D} = \frac{\lambda}{d} \quad [\text{Fringe Width } \beta;\ \text{Angular } \theta] \tag{B.81}$$

Shift Produced by Thin Dielectric Slab of Thickness t and Refractive Index μ:

$$\Delta y = \frac{D}{d}(\mu-1)t, \qquad N_{\text{shifted}} = \frac{\Delta y}{\beta} = \frac{(\mu-1)t}{\lambda} \tag{B.82}$$

B.7.2 Diffraction due to a Single Slit

Single Slit Diffraction Maxima, Minima & Fresnel Distance

$$a\sin\theta = n\lambda \implies y_n = n\frac{\lambda D}{a} \quad [\text{Minima: } n = 1, 2, \dots] \tag{B.83}$$

$$a\sin\theta \approx (2n+1)\frac{\lambda}{2} \implies y'_n = (2n+1)\frac{\lambda D}{2a} \quad [\text{Maxima: } n \geq 1] \tag{B.84}$$

$$2\theta_0 = \frac{2\lambda}{a}, \quad W_{\text{central}} = \frac{2\lambda D}{a} = 2\beta_{\text{diff}} \quad [\text{Central Width}] \tag{B.85}$$

$$Z_F = \frac{a^2}{\lambda} \quad [\text{Fresnel Distance; Ray optics for } z \ll Z_F] \tag{B.86}$$

Wave Optics Traps: Interference vs Diffraction Central Fringe

- **Central Maximum Width in Diffraction is Doubled:** In single-slit diffraction, the central maximum spreads between the first minima on both sides ($-\lambda/a$ to $+\lambda/a$). Its linear width is $\frac{2\lambda D}{a}$, exactly **twice** the width of secondary

maxima ($\frac{\lambda D}{a}$). In YDSE, all fringes have identical width $\beta = \frac{\lambda D}{d}$.

- **YDSE Immersed in Liquid:** When apparatus is dipped in liquid of refractive index μ, wavelength shrinks ($\lambda' = \lambda/\mu$). Hence fringe width decreases: $\beta' = \beta/\mu$. Angular width also shrinks: $\theta' = \theta/\mu$.

B.8 Dual Nature of Matter & Radiation (Unit 17)

B.8.1 Photoelectric Effect & Einstein's Equation

Einstein's Photoelectric Governing Equations

$$E = h\nu = \Phi_0 + K_{\max} = \Phi_0 + eV_0 \quad \text{[Photoelectric Law]} \tag{B.87}$$

$$\Phi_0 = h\nu_0 = \frac{hc}{\lambda_0} \approx \frac{1240}{\lambda_0\ [\text{nm}]}\ \text{eV} \quad \text{[Work Function } \Phi_0\text{]} \tag{B.88}$$

$$eV_0 = h\nu - \Phi_0 \implies V_0 = \left(\frac{h}{e}\right)\nu - \frac{\Phi_0}{e} \quad \text{[Slope } = h/e\text{]} \tag{B.89}$$

$$K_{\max} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right) = 1240\left(\frac{1}{\lambda\ [\text{nm}]} - \frac{1}{\lambda_0\ [\text{nm}]}\right)\ \text{eV} \tag{B.90}$$

Photon Dynamics (Rest Mass $m_0 = 0$):

$$E = h\nu = \frac{hc}{\lambda}, \qquad p = \frac{E}{c} = \frac{h}{\lambda} \tag{B.91}$$

$$m_{\text{eff}} = \frac{h\nu}{c^2}, \qquad N_{\text{photons/sec}} = \frac{P}{h\nu} = \frac{P\lambda}{hc} \tag{B.92}$$

B.8.2 de Broglie Matter Wave Formulations

de Broglie Wavelength of Particles & Accelerating Potentials

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} \quad \text{[Matter Wave]} \tag{B.93}$$

$$\lambda_e = \frac{h}{\sqrt{2m_e eV}} = \sqrt{\frac{150}{V}}\ \text{Å} = \frac{1.227}{\sqrt{V}}\ \text{nm} \quad [V \text{ in Volts}] \tag{B.94}$$

$$\lambda_p = \frac{0.286}{\sqrt{V}}\ \text{Å}, \quad \lambda_d = \frac{0.202}{\sqrt{V}}\ \text{Å}, \quad \lambda_\alpha = \frac{0.101}{\sqrt{V}}\ \text{Å} \tag{B.95}$$

$$\lambda_{\text{gas}} = \frac{h}{\sqrt{3mk_BT}} = \frac{h}{\sqrt{2m(\frac{3}{2}k_BT)}} \quad \text{[Thermal Wave at } T\text{]} \tag{B.96}$$

Photoelectric Traps: Intensity vs Frequency Effects

- **Intensity Modifies Current, NOT Kinetic Energy:** Doubling the intensity of incident light doubles the number of photons per second, thereby doubling the **saturation photoelectric current** ($I_{\text{sat}} \propto$ Intensity). The maximum kinetic energy $K_{\max}$ and stopping potential V_0 remain **strictly unchanged**.
- **Frequency Modifies Kinetic Energy, NOT Current:** Increasing frequency ν increases $K_{\max}$ and V_0 linearly ($V_0 = \frac{h}{e}\nu - \frac{\Phi_0}{e}$). Saturation current is independent of frequency above ν_0.

B.9 Atoms & Nuclei (Unit 18)

B.9.1 Bohr's Hydrogen-Like Atom Model

Bohr Postulates, Radius, Velocity, Energy & Rydberg Formula

$$L = mv_n r_n = \frac{nh}{2\pi} \quad [\text{Quantization: } n = 1, 2, \dots] \tag{B.97}$$

$$r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m Z e^2} = 0.529 \left(\frac{n^2}{Z}\right) \text{ Å} = r_1 \left(\frac{n^2}{Z}\right) \tag{B.98}$$

$$v_n = \frac{Ze^2}{2\varepsilon_0 nh} = \left(\frac{c}{137}\right)\frac{Z}{n} \approx 2.18 \times 10^6 \left(\frac{Z}{n}\right) \text{ m s}^{-1} \tag{B.99}$$

$$E_n = -13.6 \left(\frac{Z^2}{n^2}\right) \text{ eV} \quad [K = -E_n,\ U = 2E_n] \tag{B.100}$$

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \quad [R \approx 1.097 \times 10^7 \text{ m}^{-1},\ 1/R \approx 912 \text{ Å}] \tag{B.101}$$

Total Number of Emission Lines from Excited State n to Ground State ($n = 1$):

$$N = \frac{n(n-1)}{2}, \qquad N_{\text{between } n_2 \text{ and } n_1} = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2} \tag{B.102}$$

Series	n_1	n_2	Region	$\lambda_{\min}$ (Series Limit)
Lyman	1	$2, 3, \dots$	UV	$1/R \approx 912$ Å
Balmer	2	$3, 4, \dots$	Visible / Near UV	$4/R \approx 3646$ Å
Paschen	3	$4, 5, \dots$	IR	$9/R \approx 8205$ Å
Brackett	4	$5, 6, \dots$	IR	$16/R \approx 14586$ Å
Pfund	5	$6, 7, \dots$	Far IR	$25/R \approx 22790$ Å

Table B.4: Hydrogen Spectral Series Classifications and Spectral Regimes.

B.9.2 Nuclear Size, Mass Defect & Binding Energy

Nuclear Radius, Density, Mass Defect & Fission/Fusion Energies

$$R = R_0 A^{1/3} \quad [R_0 \approx 1.2 \text{ fm} = 1.2 \times 10^{-15} \text{ m}] \tag{B.103}$$

$$\rho_{\text{nuc}} = \frac{M}{\frac{4}{3}\pi R^3} \approx 2.3 \times 10^{17}\,\text{kg m}^{-3} \quad \text{[Constant for all } A\text{]} \tag{B.104}$$

$$\Delta m = [Zm_p + (A-Z)m_n] - M_{\text{nucleus}} \quad \text{[Mass Defect]} \tag{B.105}$$

$$E_b = \Delta m\, c^2 = \Delta m\ [\text{u}] \times 931.5\,\text{MeV} \quad \text{[Binding Energy]} \tag{B.106}$$

$$Q = [m_{\text{reactants}} - m_{\text{products}}]c^2 = [BE_{\text{products}} - BE_{\text{reactants}}] \tag{B.107}$$

Nuclear Traps: Invariance of Nuclear Density

- **Nuclear Density is Constant:** Because $R \propto A^{1/3}$, nuclear volume $V \propto R^3 \propto A$. Thus nuclear density $\rho = \frac{mA}{V} \propto \frac{A}{A} =$ constant. A Uranium nucleus ($A = 238$) has approximately the **same density** as a Helium nucleus ($A = 4$).
- **Radioactivity Excluded by NMC:** Note that α, β, γ decay rate equations and half-life decay laws have been removed from NEET UG under current NMC guidelines; focus strictly on binding energy per nucleon curves, mass defects, and fission/fusion energy calculations.

B.10 Electronic Devices: Semiconductors (Unit 19)

B.10.1 Intrinsic & Extrinsic Semiconductors

Carrier Densities, Mass Action Law & Electrical Conductivity

$$n_e n_h = n_i^2 \qquad \text{[Law of Mass Action]} \tag{B.108}$$

$$n_e = n_h = n_i \qquad \text{[Intrinsic Semiconductor]} \tag{B.109}$$

$$n_e \approx N_d \gg n_h \qquad \text{[n-Type: Pentavalent Doping]} \tag{B.110}$$

$$n_h \approx N_a \gg n_e \qquad \text{[p-Type: Trivalent Doping]} \tag{B.111}$$

$$\sigma = e(n_e\mu_e + n_h\mu_h), \qquad I = eAE(n_e\mu_e + n_h\mu_h) \tag{B.112}$$

B.10.2 p-n Junction Diodes, Rectifiers & Optoelectronic Devices

Diode Dynamic Resistance & Rectifier Efficiencies

$$r_d = \frac{\Delta V}{\Delta I} \quad [r_f \sim 10\text{–}100\,\Omega,\ r_r > 1\,\text{M}\Omega] \tag{B.113}$$

$$\eta_{\text{HW}} = \frac{0.406 R_L}{r_f + R_L} \implies \eta_{\max} = 40.6\%, \quad \nu_{\text{rip}} = \nu_{\text{in}} \tag{B.114}$$

$$\eta_{\text{FW}} = \frac{0.812 R_L}{r_f + R_L} \implies \eta_{\max} = 81.2\%, \quad \nu_{\text{rip}} = 2\nu_{\text{in}} \tag{B.115}$$

B.10.3 Digital Electronics: Logic Gates & Boolean Algebra

A	B	**AND** $(A \cdot B)$	**OR** $(A + B)$	**NAND** $(\overline{A \cdot B})$	**NOR** $(\overline{A + B})$	**XOR** $(A \oplus B)$	**XNOR** $(\overline{A \oplus B})$
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Table B.5: Comprehensive Truth Tables for Fundamental, Universal and Derived Logic Gates.

De Morgan's Theorems & Universal Logic Construction

- **De Morgan's Laws:**

$$\overline{A+B} = \overline{A} \cdot \overline{B}, \qquad \overline{A \cdot B} = \overline{A} + \overline{B}$$

- **Universal Gates Minimal Realizations:**
  - Number of NAND gates to make: NOT = 1, AND = 2, OR = 3, NOR = 4, XOR = 4.
  - Number of NOR gates to make: NOT = 1, OR = 2, AND = 3, NAND = 4, XNOR = 4.

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